A COGNITIVE-TAXONOMY HANDBOOK

Bloom's Taxonomy for Rasashastra & Bhaishajya Kalpana

Module-by-module domain and level mapping, objective design and usage guide for AYPG-RB — Rasashastra and Bhaishajya Kalpana
MD/MS Ayurveda, Semesters III to VI

Curriculum	NCISM, Board of Ayurveda — four semesters, four papers, applicable from the 2024–25 batch
Coverage	32 modules · 124 units · 577 session learning objectives · 141 practical training and 163 experiential learning activities
Framework	Bloom (1956) and the Anderson & Krathwohl revision (2001) carried in parallel, with the psychomotor and affective domains, read against NCISM Table 3 Column 3E

577
OBJECTIVES CODED

277
THINKING

294
DOING

64
CREDITS · 1920 HOURS

Created by Dr Milind Chatrabhuji using Claude Opus. · Prepared July 2026 · For departmental use in curriculum delivery, objective writing, assessment design and paper setting.

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PART A

Understanding the taxonomy

Nine short chapters. The first five explain what Bloom's taxonomy is, what its two versions are, what its three domains cover, where it already sits inside this curriculum, and how verbs turn it into practice. Chapter 6 covers the fixed one-to-two-to-three split between lecture, practical training and experiential learning. Chapter 8 covers both kinds of assessment this course carries — the modular assessment in every module, and the 800-mark university examination at the end of Semester VI. Read straight through once; after that use the module modules and come back only when something needs settling.

1. **What Bloom's taxonomy is, and why this book carries two versions**
2. **The three domains** — thinking, doing, and how a student goes about the work
3. **The second axis** — what kind of knowledge an objective works on
4. **Where Bloom already sits in your curriculum** — reading Column 3E
5. **Verbs** — the working end of the taxonomy
6. **The one-to-two-to-three split** — lecture, practical training, experiential learning
7. **Choosing how to teach**, level by level
8. **Choosing how to check** — the modular assessment and the university examination
9. **How to use a module in Part B**

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PART A · CHAPTER 1

What Bloom's taxonomy is, and why this book carries two versions

In 1956 a committee of American examiners led by Benjamin Bloom published a way of sorting examination questions. They had noticed what every teacher notices sooner or later: two questions on the same content can be worlds apart, not because one covers more ground, but because one asks the student to *do* something harder with what they know.

What they produced was a ladder of six levels, each asking more than the one below: **Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation**. The ladder is cumulative. You cannot apply what you do not understand, and you cannot understand what you cannot recall. That the levels stack is what makes the taxonomy useful for planning rather than merely for labelling.

THE 2001 REVISION

- The levels became **verbs**: *Remember, Understand, Apply, Analyse, Evaluate, Create.*
- **The top two swapped.** Bloom put Synthesis at level 5 and Evaluation at level 6. The revision reversed them: Evaluate is 5, Create is 6, on the reasoning that making something new is a larger act than judging something that exists.
- A **second axis** was added for the kind of knowledge involved. Chapter 3 covers it.

> **The one thing to carry away.** When a colleague says a question is "at the top level", ask which scheme they mean. In 1956 the top is Evaluation — judging. In 2001 it is Create — producing. Different questions, different teaching, different marking. Most disagreement between two well-informed teachers about Bloom comes down to this.

WHY BOTH, IN THIS BOOK

Your curriculum uses the **1956 scheme**: the codes in Column 3E are CK, CC, CAP, CAN, CS and CE. If this book used only the 2001 names you could not match it against your syllabus document. But most faculty development now teaches the 2001 scheme. So every level here is named twice, 1956 first: *Comprehension / Understand.* Where the two schemes disagree about position — only at the top two — the book says so on the page.

> **This curriculum reaches the top of both ladders.** Synthesis appears five times and Origination four, across Semesters III, V and VI. Every one of the four semesters reaches Evaluation on the thinking ladder. If you have taught only the undergraduate course, where Synthesis is never used at all, the ceiling here is genuinely higher and the teaching has to change with it.

PART A · CHAPTER 2

The three domains — thinking, doing, and how a student goes about the work

Bloom's six levels describe thinking only. The original committee planned three volumes. The **cognitive** volume appeared in 1956; the **affective** volume — attitudes and values — in 1964; the **psychomotor** volume was never written by that committee, and the version your curriculum uses comes from later work by others.

THINKING — SIX LEVELS

Knowledge, Comprehension, Application, Analysis, Synthesis, Evaluation. Across the four semesters, 277 of the 577 objectives are cognitive — 48%.

DOING — FIVE LEVELS

- **Set** — ready to begin; knows what is needed and can lay it out.
- **Guided response** — performs while being shown or guided.
- **Mechanism** — performs reliably alone, without prompting.
- **Adaptation** — changes the method when conditions change.
- **Origination** — devises a procedure that was not given.

Two hundred and ninety-four of the 577 objectives are here — half the course. That is the single most important fact about this curriculum, and it is more pronounced than in any other year of the programme: 12% of the undergraduate syllabus is psychomotor, 38% of Semester II, and 50% here.

The distribution inside the ladder matters as much as the total. **One hundred and ninety-four objectives — a third of the entire course — sit at *Guided response***, the level at which a student performs while being shown. Sixty-five sit at *Mechanism*, where they perform unaided. That is worth pausing on: the curriculum concentrates its largest single block of objectives one rung *below* independent competence. Moving students from the first to the second is not a side effect of the practical hours; in this course it is the main work of them.

GOING ABOUT IT — FIVE LEVELS

Receiving, Responding, Valuing, Organisation, Characterisation — from paying attention when it matters, through treating something as worth doing unprompted, to it having become how the student works.

> **This course is nearly silent here too.** Six objectives in 577 carry an affective code. That is worth knowing rather than working around: professional habits — verifying material, recording the reading actually taken, declining to sign off on what was not checked — are being formed across these four semesters whether or not the curriculum codes them, and a postgraduate is precisely the person in whom they matter. If you want evidence of them you will have to plan for it, because the syllabus does not ask. Part B therefore shows two ladders per module; the six affective objectives appear in their own bands in the units that hold them.

THE DOMAINS DO NOT MOVE TOGETHER

An objective at Mechanism may need only level 1 thinking if the procedure is followed by rote. Another at Set may need level 4 thinking if choosing the right equipment requires judgement. The two ceilings move independently across this course, and the modules print both for that reason. Read both before planning, and before choosing a modular assessment.

PART A · CHAPTER 3

The second axis — what kind of knowledge an objective works on

The 2001 revision added a second question. The first is: what thinking must the student do? The second is: *what kind of knowledge is it about?*

- **Factual** — terminology, names, specific details. Learned by rehearsal and self-testing.
- **Conceptual** — classifications, principles, relationships, the reason a thing is so. Learned by explaining it to somebody.
- **Procedural** — methods, techniques, criteria for choosing between them. Learned by carrying them out.
- **Metacognitive** — knowing about your own learning. Learned by being made to look back at how you worked.

> **This course is overwhelmingly Procedural.** Procedural knowledge dominates, conceptual follows, and factual knowledge is a small minority. That is the profile of a laboratory and workshop course rather than a reading course, and it lines up with the 50% psychomotor figure. The practical consequence is blunt: **reading a procedure is not learning it**, and a student who has only read will look identical to one who has practised right up until the moment they are asked to perform.

WHY IT CHANGES THE TEACHING

Two objectives at the same level can need entirely different sessions. One asks for the names in a group; another asks for the steps of a procedure in order. Rehearsal fixes the first. The second will not hold until the student has done it, because a remembered list of steps and a performable procedure are not the same possession. This is why “they know it but they cannot do it” is such a common complaint — usually the objective was Procedural and the teaching was pitched as though it were Factual.

In Part B every objective is labelled with its kind of knowledge, read from the words of the objective itself. Where an objective works on two kinds, both are named. Where the wording supports a reading without settling it, the label carries a small circle (°), so you can see a judgement was made and overrule it.

PART A · CHAPTER 4

Where Bloom already sits in your curriculum

Bloom is not something this book adds. It has been in the curriculum since publication, in a column most teachers have never had pointed out.

COLUMN 3E

Open Table 3. The fifth column is headed *Domain/Sub Domain*. Every one of the 577 session learning objectives carries a code there: CK, CC, CAP, CAN, CS, CE for thinking; PSY-SET, PSY-GUD, PSY-MEC, PSY-ADT, PSY-ORG for doing; AFT-REC through AFT-CHR for how the student goes about the work. The full key is Appendix C1.

Table 3 in this curriculum is organised by **paper first, then semester** — all four semesters of Paper 1, then all four of Paper 2, and so on. Part B reorders it by semester, which is how a department actually teaches, so Module 1 and Module 2 sit together in Semester III rather than being separated by three semesters of Paper 1.

THE COLUMNS AROUND IT

- **3A — Course outcome.** Which of the course-level outcomes the objective serves.
- **3C — Notional learning hours.** How long the objective is expected to take.
- **3D — Lecture, practical training or experiential learning.** The mode, and for the last two a reference into the activity tables. Chapter 6 is about this column.
- **3G — Teaching and learning methods.**

> **A word about Column 3F, which this book does not use.** Column 3F is headed *Level*, but it carries a different scheme — Knows, Knows how, Shows how, Does — which is Miller's pyramid, not Bloom. It answers a different question: not what kind of thinking is needed, but how authentic the evidence has to be. It has its own companion volume. In *this* book, "level" always means the Bloom level in Column 3E. Where you read "level 5" here, it never means "Shows how".

MODULE LEARNING OBJECTIVES SIT ABOVE SESSION OBJECTIVES

Each module states its own learning objectives, and these are what the modular assessment tests. They are pitched at their own levels, which do not always match the session objectives beneath them. Part B shows each module's objectives with the level of their verb, so you can see at a glance whether the module assessment is aimed higher or lower than the sessions that feed it.

FOUR PAPERS, AND WHAT THEY COVER

- **Paper 1** — Rasashastra (Iatrochemistry): Principles and Practices
- **Paper 2** — Exposition of Ayurvedic Pharmaceutics
- **Paper 3** — Standardization and clinical Utility of Ayurvedic formulations
- **Paper 4** — Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

Each paper carries two modules per semester, so eight modules per paper across the four semesters, and thirty-two in all.

PART A · CHAPTER 5

Verbs — the working end of the taxonomy

A level is an idea; a verb is what a student meets. **An objective is taught and examined at the level of its verb**, whatever else is written beside it.

- **Level 1, Knowledge / Remember** — define, enlist, list, name, recall, recite, state, identify, recognise, observe, record.
- **Level 2, Comprehension / Understand** — describe, explain, discuss, illustrate, classify, differentiate, outline, present.
- **Level 3, Application / Apply** — apply, demonstrate, perform, prepare, calculate, measure, conduct, practise, operate, handle.
- **Level 4, Analysis / Analyse** — analyse, compare, examine, determine, categorise, correlate, review, explore, investigate, search.
- **Level 5 in 1956, Synthesis** — design, formulate, develop, construct, devise, synthesise, integrate, optimise. (Level 6, Create, in the 2001 scheme.)
- **Level 6 in 1956, Evaluation** — evaluate, justify, appraise, critique, assess, judge, defend, interpret. (Level 5 in the 2001 scheme.)

The affective domain has its own verbs, and they are not thinking verbs: *appreciate, value, accept, comply, respond, defend, advocate, adhere, display, practise.*

> **Where a verb and a code disagree, follow the verb — and know you are choosing.** The verb is what appears on the paper, so it wins by default. That is fine provided it is a decision rather than an accident. If the verb asks less than the code, you have room to teach higher than the syllabus requires. If it asks more, plan for the verb. Part B tells you, objective by objective, which case you are in.

WRITING YOUR OWN

Decide the level first, then take a verb from that level, then check the verb can be assessed by the tool you have in mind. One verb, one level, one objective. Where you find yourself writing “identify, demonstrate and evaluate” on a single line, you have three objectives at three levels and they will need three different checks.

PART A · CHAPTER 6

The one-to-two-to-three split — lecture, practical training, experiential learning

This is the feature of postgraduate study that most decides what students actually reach, and it has no equivalent in the undergraduate course. Every credit is thirty hours, and the curriculum fixes how those hours are spent: **five hours of lecture, ten of practical training, thirteen of experiential learning, and two of modular assessment** — a ratio of roughly one to two to three.

Across four semesters and 64 credits that comes to **1920 notional hours**: 1792 of teaching and 128 of modular assessment. Two thirds of this course happens with the teacher not talking, and it stays that way for two full years.

THE THREE KINDS OF HOUR DO NOT REACH THE SAME LEVELS

- **Lecture** puts material in front of students and lets them retrieve it. It builds level 1 well and level 2 poorly, and on its own it builds nothing above level 2. **273 of the 577 objectives are set as lecture** — the largest single mode by count, though the smallest by hours.
- **Practical training** moves a student from performing with help to performing unaided. That single transition is what these hours are for, and it is slow. **141 objectives** are set here, each pointing at one of the 141 practical training activities.
- **Experiential learning** hands over an unfamiliar situation and requires a judgement, a decision or a product. This is where the upper levels are built and where the modular assessment lives. **163 objectives** are set here, against 163 activities.

> **The commonest failure is silent demotion.** An objective set as experiential learning and then delivered as a lecture has been quietly moved down two levels, and nothing in the timetable will show it. The students will have been told about the thing instead of doing it. Part B prints the kind of hour against every objective for this reason: it is the clearest statement the curriculum makes about how a thing is meant to be learned.

THE ACTIVITY TABLES

Every practical and experiential objective carries a reference — 1.1, 8.2 and so on — into the curriculum's activity tables, which describe what the student is to do. There are **141 practical training activities and 163 experiential ones**, 304 in all. Appendices C6 and C7 reproduce every one with the level it is cited to serve, so you can see what a reference points at without turning back to the syllabus.

PART A · CHAPTER 7

Choosing how to teach, level by level

The commonest planning error is to choose the method by the hour rather than by the level. What follows is what each level needs.

LEVEL 1 — KNOWLEDGE / REMEMBER

Exposure, then rehearsal, then retrieval practice. Make students retrieve without looking, then check. In this course level 1 is very small — seven objectives in 577 — so it should be taking almost none of your contact time. If it is taking more, something pitched higher is being taught as recall.

LEVEL 2 — COMPREHENSION / UNDERSTAND

Students must handle the material, not receive it again: sort it, contrast it, put it in their own order, explain it to a classmate. **152 objectives sit here**, the largest cognitive group in the course, and lecture alone will not deliver them.

LEVEL 3 — APPLICATION / APPLY

Work examples through together, then hand over a situation the student has not seen. The hand-over is the teaching.

LEVEL 4 — ANALYSIS / ANALYSE

Give students two things and ask how they differ and why. Analysis needs something to set against something else.

LEVEL 5 IN 1956, SYNTHESIS

Used five times across the four semesters. It means the student assembles something that was not handed to them — a protocol, a specification, a procedure. Set the criteria in advance and mark against them.

LEVEL 6 — EVALUATION

Give something to judge and criteria to judge it against, then require the criteria to be stated. Small-group critique works well. This is the level most often taught as though it were level 2 — explaining the accepted view rather than reaching one.

THE DOING LADDER

- **Set** — show the setup, name every item, then have students lay it out themselves.
- **Guided response** — demonstrate once at speed, once slowly, then let them copy while you correct in the moment.
- **Mechanism** — repeated supervised practice with your help withdrawn a little each time. **Most practical hours belong here.**
- **Adaptation** — change one condition and let them work out the adjustment.
- **Origination** — used four times, in Semesters III, V and VI. The student devises a procedure that was never demonstrated. Reaching it needs the earlier levels genuinely secure, which is why it appears late.

PART A · CHAPTER 8

Choosing how to check — the modular assessment and the university examination

A check is fair when it can only be passed by a student who has reached the level, and can be passed by any student who has. Most unfair checks fail the first half.

- **Level 1** — short questions asking for a list, a name, a definition. Many of them, spread out.
- **Level 2** — questions requiring the material rearranged. If it can be answered word-for-word from notes, it has slipped to level 1.
- **Level 3** — an unseen case. A familiar case measures memory.
- **Level 4** — questions asking for the parts and how they relate.
- **Level 5** — something the student produced, marked against criteria set in advance.
- **Level 6** — give the marks for the reasoning and the criteria, not for the facts repeated inside the answer.

DOING OBJECTIVES CAN ONLY BE CHECKED BY WATCHING

A viva shows whether a student knows a procedure and can explain it. It cannot show whether they can carry it out, and a fluent student who has never handled the material will pass it comfortably. With **294 doing objectives** — half this course — this matters more here than anywhere else in the programme. The evidence has to be seen: a checklist ticked while the student works, an OSPE station, an observed performance. A checklist of the steps you will look for, written before the session, is enough.

THE TWO ASSESSMENTS THIS COURSE CARRIES

Formative — the modular assessment. Every module carries one, worth 25 marks per credit and using two of its hours per credit. That is 400 marks a semester and **1600 marks across the four**. The curriculum requires a different method in each module, with a record of the pattern used.

Summative — the university examination, at the end of Semester VI: four papers of 100 marks and a practical examination of 400, **800 marks in all**. It examines two years of work at once, which has one consequence worth planning for from Semester III onward: **anything certified only by observation during a practical session will not be re-examinable at the end.** If a student's competence at Mechanism or Adaptation was seen but never recorded, there is nothing to carry forward.

> **Choose the modular method against the module's ceiling, not by rotation.** A module whose ceiling is Origination or Synthesis needs an assessment that lets a student produce something. One that stops at Application does not, and an elaborate instrument there wastes everyone's time. The modules in Part B print both ceilings for exactly this decision.

> **Keep the record as you go, and keep it usable.** Across 124 units and 304 activities, a log book and a set of checklists are the only things that will still exist in Semester VI. Decide at the start of each module what artefact each doing objective will leave behind — a signed checklist, a log entry, a dated record — and the summative examination becomes a matter of drawing on evidence rather than of memory.

PART A · CHAPTER 9

How to use a module in Part B

Part B has thirty-two modules, ordered by semester and then by paper, so the two modules a department teaches together sit together. Each begins on its own page and carries its units inside it.

WHAT IS ON THE MODULE

1. **The heading** — module number, semester, paper, and the paper's title.
2. **The strip** — credits, units, thinking and doing counts, formative marks.
3. **Where the hours go** — the lecture, practical and experiential split for this module, with what each kind of hour must deliver here.
4. **What the module asks students to do** — both ladders across all its units. Empty levels are shown as empty, because knowing what a module does *not* ask for is as useful as knowing what it does.
5. **Module learning objectives**, each with the level of its verb. These are what the modular assessment tests.
6. **The units**, each as a strip: its title, objective count, hours and both ceilings, then its syllabus points as the curriculum numbers them, then its objectives grouped by level.
7. **Two panels** — one for the teacher, one to give to students.

READING A UNIT STRIP

Inside each strip the objectives are gathered under coloured bands, one band per level, thinking before doing. Every objective shows its verb, its kind of knowledge, and its kind of hour. Where an objective's verb sits at a different level from its code, a note in the same cell says which to teach to — and the verb is what a student will meet in the examination.

THREE WAYS TO USE IT

Planning a session. Find the unit, take one band at a time, and teach upward. Check the method against Chapter 7 and the kind of hour against Chapter 6.

Designing the modular assessment. Start from the module's two ceilings and its module learning objectives, then choose an instrument that can reach them.

Planning for Semester VI. Use the doing ladder to see which objectives will need a recorded artefact, and decide what that artefact is before the module starts.

> **What this book does not do.** It does not rewrite the curriculum and it does not tell you what to teach — every objective in Part B appears in the curriculum's own words. What it adds is the taxonomy layer: what kind of learning each objective asks for, and what follows for teaching and checking.

PART B

The modules

Thirty-two modules, ordered by semester and then by paper so the modules a department teaches together sit together. Each carries its units inside it, and every objective appears in the curriculum's own words, gathered under its level so a session can be planned around one level at a time, with the kind of hour printed against each. Chapter 9 explains the layout; after that the units are meant to be opened at whichever module you are about to teach.

1. **Semester III** — Modules 1, 2, 9, 10, 17, 18, 25, 26 across the four papers
2. **Semester IV** — Modules 3, 4, 11, 12, 19, 20, 27, 28
3. **Semester V** — Modules 5, 6, 13, 14, 21, 22, 29, 30
4. **Semester VI** — Modules 7, 8, 15, 16, 23, 24, 31, 32

↑ Contents

MODULE 1 · SEMESTER 3 · PAPER 1

Chronological Development and Taxonomy of Rasa Dravyas

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	11	8	50
60 notional hours	19 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

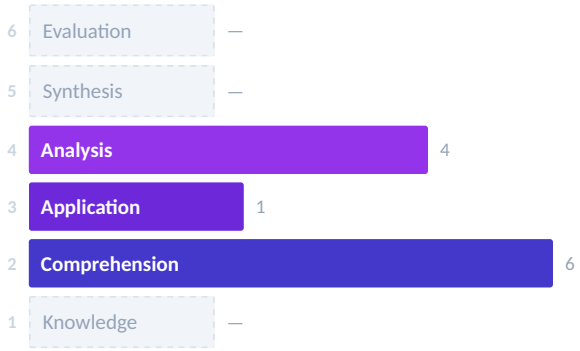
WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Discuss the philosophical relevance and chronological development of Rasashastra,	*discuss*	L2 Comprehension / Understand
2	Integrate the development of Iatrochemistry with modern chemistry	*integrate*	L5 Synthesis / Create
3	Differentiate Rasadravyas as per different ancient classics, and Mendeleev Periodic Table	*differentiate*	L4 Analysis / Analyse
4	Discuss the Sandigdha, Anupalabdha Rasadravyas, and authenticate ideal substitutes	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Chronological Development and Overview of Rasashastra

5 obj · 9 h · Analysis / Adaptation

1.1 Philosophical Relevance of Rasashastra
1.2 Chronological Development of Rasashastra
1.3 Integration of the development of Iatrochemistry with modern chemistry

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Discuss the philosophical relevance of Rasashastra and historical evidence till the 6th century AD.	*discuss*	Conceptual	Lecture 1h
Describe the chronological development of Rasashastra from the 6th Century AD to the medieval period.	*describe*	Conceptual ◦	Lecture 1h
Integrate the development of Iatrochemistry with modern chemistry. — teach at level 5, the verb *integrate* is what will be asked	*integrate*	Procedural ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss the chronological development of Rasashastra after the medieval period. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Construct a chronological tree of Rasashastra and its integration with other medical traditions.	*construct*	Procedural	Experiential 5h

Unit 2 · Taxonomy of Rasadravyas

5 obj · 13 h · Analysis / Guided response

2.1 Taxonomy of Rasadravyas
2.2 Concepts and justification about division of Rasadravyas
2.3 Mendeleeve periodic table

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Justify the division of Rasadravyas. — teach at level 6, the verb *justify* is what will be asked	*justify*	Conceptual ◦	Lecture 1h
Discuss the critics of the classification of Rasadravya.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2			
Elaborate classification of Rasadravyas as per different ancient classics. — teach at level 2, the verb *elaborate* is what will be asked	*elaborate*	Conceptual	Lecture 1h
Identify and analyse selected metals and minerals with the periodic table elements. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Experiential 5h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the application of the Mendeleev Periodic table in mapping Rasadravyas.	*demonstrate*	Procedural	Practical 5h

Unit 3 · Geological/mineralogical identification of Rasadravyas

6 obj · 22 h · Analysis / Mechanism

3.1 Grahya-Agrahyata of Rasadravyas
3.2 Geological/Mineralogical identification of Rasadravyas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss classical parameters for differentiating acceptable and non-acceptable Rasadravya. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Procedural	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Apply geological/mineralogical characters for the identification of Rasadravya. — teach at level 3, the verb *apply* is what will be asked	*apply*	Procedural	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the comparison and analyze the classification of Rasa Dravyas.	*demonstrate*	Procedural	Practical 5h
Demonstrate Geological/Mineralogical identification of Rasadravyas.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 2			
Recognise Rasadravya based on their Graahyatva-Agraahyatva.	*recognise*	Procedural	Experiential 5h
Identify Rasadravya using geological/mineralogical characters.	*identify*	Procedural	Experiential 5h

Unit 4 · Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya

3 obj · 12 h · Comprehension / Guided response

4.1 Sandigdha (Controversial)
4.3 Pratinidhi (Substitute) Rasadravya
4.2 Anupalabdha (Not Available) and Krutrima (Artificial) Rasadravya

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 1 · SET — 1			
Identify and analyse Sandigdha and Anuplabhdha Rasadravyas and propose suitable alternatives.	*identify*	Procedural	Experiential 6h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstratie the Krutrima and Pratinidhi Rasadravyas.	*demonstrate*	Procedural	Practical 5h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 4 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 2 · SEMESTER 3 · PAPER 1

Puta and Samskara (Transformative processes of drug manufacturing)

Rasashastra (Iatrochemistry) — Principles and Practices

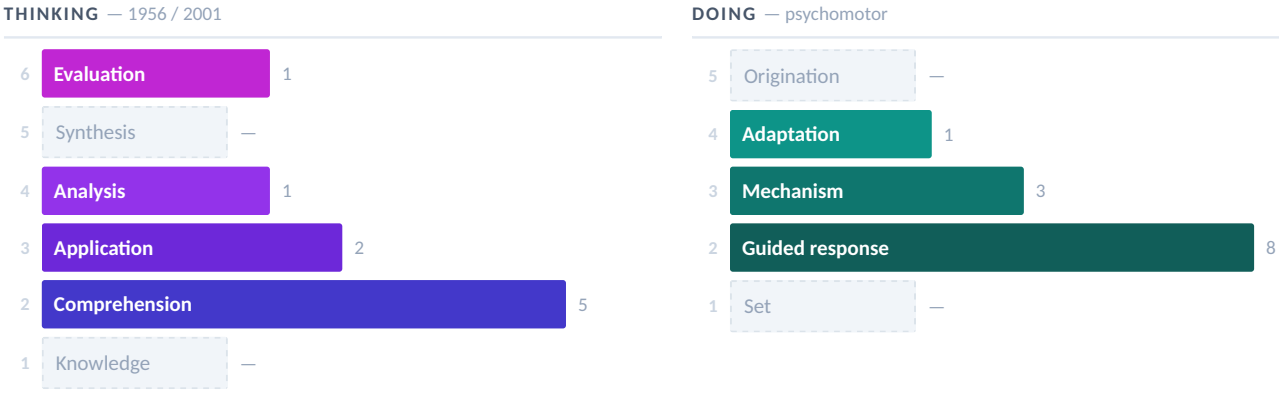
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	6	9	12	50
60 notional hours	21 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	9	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	8	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the definition, significance, construction, and design of traditional Puta	*describe*	L2 Comprehension / Understand
2	Elaborate on the different theories and laws of thermodynamics with traditional Puta	*elaborate*	L2 Comprehension / Understand
3	Describe the principles and significance of Jarana, Amritikarana, Lohitikarana, Satva, and Druti procedures	*describe*	L2 Comprehension / Understand
4	Integrate the principles of classical drug manufacturing (Samskara-transformative process) with modern techniques	*integrate*	L5 Synthesis / Create

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Puta and Law of thermodynamics

6 obj · 18 h · Evaluation / Mechanism

1.1 Definition, Significance and Classification of Puta

1.2 Laws of thermodynamics, Thermochemistry, Griffith theory, Collision theory, etc

1.3 Heating devices and fuels, with detailed knowledge about temperature, duration, designing and construction of different Puta

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Explore the traditional concept of Puta, discussing its definition, significance, and classification in the context of Ayurvedic pharmaceutical practices. — teach at level 4, the verb *explore* is what will be asked	*explore*	Conceptual	Lecture 1h
Explain the laws of thermodynamics (Zeroth, First, Second, Third), the law of thermochemistry, Griffith's theory, and Collision theory.	*explain*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Examine the modern applications and relevance of Puta in contemporary pharmaceutical processing, integrating traditional knowledge with modern technology. — teach at level 4, the verb *examine* is what will be asked	*examine*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate and compare the temperature pattern and total duration of different traditional Puta with EMF.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 2			
Perform comparison of Traditional and Modern Puta Methods: Observing and Analyzing Temperature Patterns and Duration.	*perform*	Procedural	Experiential 5h
Explore Thermal Dynamics in Traditional Puta Systems: An of Fuel Variations, Temperature Patterns, and Thermodynamic Principles.	*explore*	Conceptual + Procedural ◦	Experiential 5h

Unit 2 · Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana

4 obj · 16 h · Analysis / Guided response

2.1 Principle and Significance of

2.2 Application of Amritikarana and Lohitikarana procedures in bhasma preparation

2.3 Standard Operating Procedures (SOP) and Chemical Changes of Jarana, Amritikarana, and Lohitikarana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss the fundamental principles and significance of Jarana, Amritikarana, and Lohitikarana procedures in the preparation of bhasmas. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual + Procedural ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the art of Gandhaka Jarana: A Process, Understanding its Role and Significance in Rasashastra.	*demonstrate*	Procedural	Practical 5h
Demonstrate Amritikarana and Lohitikarana: Traditional Methods for Mica Processing in Rasashastra.	*demonstrate*	Procedural	Practical 5h
Perform an experiential Investigation of Amritikarana and Lohitikarana Procedures Exploring Safety, Efficacy, and Principles of Transformation.	*perform*	Procedural	Experiential 4h

Unit 3 · Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures

4 obj · 9 h · Application / Guided response

3.2 Application and analysis of Satva and Druti procedures

3.3 SOP and chemical changes of Satva and Druti

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the principles, significance, and applications of Satva procedures within the context of Ayurvedic pharmacy and modern pharmaceutical practices. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Describe the principles, significance, and apply Druti procedures within the context of Ayurvedic pharmacy. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the procedure of Satvapatana.	*demonstrate*	Procedural	Practical 5h
Perform and analyse the Satva Patana procedure.	*perform*	Procedural	Experiential 2h

Unit 4 · Samskara (Principles of Transformative processes of drug manufacturing)

2 obj · 3 h · Comprehension / Adaptation

4.1 Application of Toya Sannikarsha (Processing with water) and Agni

4.2 Shaucha and Manthana (churning)

4.3 Vasana (process of infusing aroma), and Bhavana (Effect of Levigation / Trituration)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Integrate and apply classical Ayurvedic manufacturing principles, including water, heat and combined processing, cleaning, churning, proximity, and trituration effects. — teach at level 5, the verb *integrate* is what will be asked	*integrate*	Conceptual	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Apply Time-Tested Principles of Classical Drug Manufacturing: by integrating Toya Sannikarsha, Agni Sannikarsha, Toyagni Sannikarsha, Shaucha, and Manthana to Optimize Pharmaceutical Quality, Safety, and Efficacy.	*apply*	Procedural	Experiential 2h

Unit 5 · Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities

2 obj · 3 h · Comprehension / Mechanism

5.1 Influence of Desha (geographical area and place of

5.2 Influence of Kala (processing time, season, and maturity of herbs) on drug quality

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Description of the influence of Desha (geographical area) and Kala (time and seasonal factors) on the quality of Ayurvedic drugs, including the impact of geographical location, climate, and seasonal variations on drug efficacy and safety.	*description*	—	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Perform A Hands-on Investigation of Geographical and Temporal Influences on drug 7, CO 8 quality: Desha–Kala in Pharmaceutical Manufacturing.	*perform*	Procedural	Experiential 2h

Unit 6 · Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel)

3 obj · 7 h · Comprehension / Guided response

5.1 Knowledge of Kalaprakarsha (Time taken

5.2 Knowledge of Bhajana (vessels used for manufacturing)

5.3 Environmental and Temporal Influences on Drug Quality and Processing

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the concepts of Kalaprakarsha (processing with specific time exposure) and Bhajana (processing vessels) in Ayurvedic pharmaceutical processing, including the significance of processing time, vessel material, and other temporal factors on drug quality and efficacy.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate and analyze the role of Vasana and Bhavana to Optimize Pharmaceutical Quality, Safety, and Efficacy.	*demonstrate*	Procedural	Experiential 4h
Demonstrate the effect of Kalaprakarsha and Bhajana in Pharmaceutical Processing Analyzing Processes and Findings to Optimize Quality, and stability.	*demonstrate*	Procedural	Experiential 2h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 9 lecture, 4 practical, 8 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 9 · SEMESTER 3 · PAPER 2

Scientific fundamentals of Ayurvedic Pharmaceutical preparations

Exposition of Ayurvedic Pharmaceutics

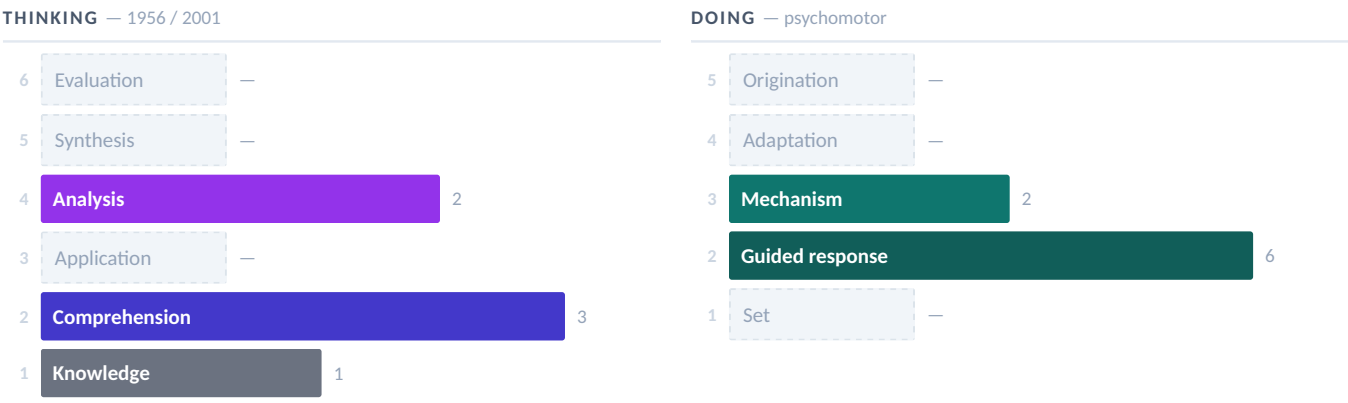
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	6	8	50
60 notional hours	15 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	5	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the classical Principles of drug processing and the scientific basis of Panchavidha Kashaya Kalpana	*describe*	L2 Comprehension / Understand
2	Evaluate Identity, Purity, and Quality of Raw Materials Used in Ayurveda Pharmaceutics	*evaluate*	L6 Evaluation / Evaluate
3	Conduct pharmaceutical processing of drugs with rationality	*conduct*	L3 Application / Apply
4	Analyze Ayurvedic Classical Dosage Forms and their integration with modern scientific validation	*analyze*	L4 Analysis / Analyse

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Classical Principles of drug processing and scientific basis of Panchavidha Kashaya Kalpana

3 obj · 6 h · Comprehension / Mechanism

1.1 Bhaishajya Synonyms and significance of its
1.2 Classical principles of processing
1.3 Scientific review of Panchavidha Kashaya Kalpana and Upakalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the rationality behind the selection of Panchavidha Kashaya Kalpana and compare classical and modern drug extraction techniques.	*discuss*	Procedural	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Prepare unique formulation from the category of Panchavidha Kashaya Kalpana.	*prepare*	Procedural	Experiential 3h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate processing of unique Panchavidha Kashaya Kalpana with their ingredients proportion of water and procedural details.	*demonstrate*	Procedural	Practical 2h

Unit 2 · Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics

3 obj · 12 h · Comprehension / Guided response

2.1 Classical knowledge about Identity, Purity, Quality of Raw
2.2 Collection of Raw material- GACP, QAQC parameters
2.3 Adulteration and Contamination
2.4 Substitute drugs (Pratinidhi Dravya)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe classical and contemporary parameters for quality assurance of raw drugs, as well as adulteration and contamination of the raw material.	*describe*	Procedural	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate application of QAQC parameters used for quality control of herbal drugs.	*demonstrate*	Procedural	Practical 4h
Identify and apply various tools and techniques used for quality control of raw materials and appraise the classical rules of the collection of raw material.	*identify*	Procedural	Experiential 6h

Unit 3 · Pharmaceutical processing of drugs with its rationality

3 obj · 13 h · Analysis / Mechanism

3.1 Processing of Drugs- Classical methodology
3.2 Simple and unique remedies from classical texts prepared with various processing methods
3.3 Rationality behind the Processing Method of Classical Dosage Forms
3.4 Siddhilakshanani (End-point assessment of pharmaceutical process) and its significance

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the classical and contemporary methods used for the processing of drugs including the assessment of the End-Point(Siddhi Lakshanani) of the dosage form, C_L.	*describe*	Procedural	Lecture 2h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Interpret significance of understanding the proportion of ingredients of formulations quoted in the classics. — teach at level 6, the verb *interpret* is what will be asked	*interpret*	Conceptual	Experiential 6h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate various formulations by describing their ingredients and understanding the uniqueness of their proportion.	*demonstrate*	Procedural	Practical 5h

Unit 4 · Ayurvedic Classical Dosage Forms and their integration with modern scientific validation

6 obj · 25 h · Analysis / Guided response

4.1 Classification of the source of raw material and the
4.2 Multiple factors responsible for Process optimization

4.3 Key qualities of good pharmaceutical products to ensure safety, efficacy, and authenticity

4.4 Research updates about the Integration of traditional wisdom with modern scientific validation

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Recognize the factors responsible for a change in the concentration of the prepared drug and assess the extracted phytochemicals with the help of TLC and other methods of analysis.	*recognize*	Procedural	Lecture 3h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe multiple factors responsible for the process optimization of Ayurvedic pharmaceutical preparations and illustrate key qualities of good pharmaceutical products to ensure safety, efficacy, and authenticity. — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the extraction of phytochemicals in the prepared dosage form.	*demonstrate*	Procedural	Practical 4h
Perform the process of Extraction of Phytochemicals in Ayurvedic formulations.	*perform*	Procedural	Experiential 5h
Demonstrate the kwatha preparation considering its process optimization.	*demonstrate*	Procedural	Practical 5h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Review Research Updates in the field of Ayurvedic pharmaceutics along with modern scientific validation methods.	*review*	Conceptual + Procedural ◦	Experiential 6h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 5 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 10 · SEMESTER 3 · PAPER 2

Principles of pharmaceutical formulation

Exposition of Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	5	6	50
60 notional hours	12 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Principles of Samskara (transformative processes of drug manufacturing) in relation to drug development

2 obj · 6 h · Application / Guided response

1.1 Qualities of drugs/Procedures
1.2 Classical concept of Samskara and expected alterations in qualities
1.3 Key role of Samskara in pharmacological and therapeutic modification of the drug
1.4 Modifications in samskaras in current practices and their consequences

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the applied principles of samskara involved in the preparation of various kalpana. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual + Procedural °	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Illustrate the role of samskaras in the changes observed during the preparation of various dosage forms according to classical texts.	*illustrate*	Procedural	Practical 4h

Unit 2 · Rationality behind Composition of Formulations

2 obj · 6 h · Analysis / Guided response

2.1 Combinations of ingredients (desirable/ inadmissible)
2.2 Rational integrity behind the combination of ingredients
2.3 Aushadhi Gana (therapeutic groups) or Varga (categories) of medicinal herbs and their role in formulations
2.4 Combinations based on predominant pharmacological actions, qualities, or therapeutic usage of the ingredients

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Analyze the purpose of composition and combinations of ingredients in classical formulations.	*analyze*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate combinations of various classical formulations and justify the rationale behind the ingredients.	*demonstrate*	Procedural	Practical 4h

Unit 3 · Principles and designing of various dosage forms

2 obj · 8 h · — / Guided response

3.1 Principles of dosage form design
3.2 Pharmaceutical Considerations
3.3 Formulation Considerations
3.4 Biopharmaceutical Considerations
3.5 Pharmacokinetic Considerations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate various designs of dosage forms, considering all fundamentals of pharmacology.	*demonstrate*	Procedural	Practical 6h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Justify the Principles of dosage form design as per modern pharmaceutical science.	*justify*	Conceptual	Lecture 2h

Unit 4 · Principles of Prakshep dravyas (Pharmaceutical Additives)

3 obj · 14 h · Comprehension / Mechanism

4.1 Prakshepa Dravya- Need, Importance, Proportion (matra)
4.2 Pharmaceutical Excipients

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify the selection and relevance of Prakshepa dravya added to various dosage forms and describe Pharmaceutical excipients. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate and analyze the importance of pharmaceutical excipients.	*demonstrate*	Procedural	Practical 6h
DOING — LEVEL 3 · MECHANISM — 1			
Illustrate various Prakshepa Dravyas and their proportions stated in the classics.	*illustrate*	Procedural	Experiential 6h

Unit 5 · Principles of Potency and Efficacy enhancement

3 obj · 22 h · Synthesis / Mechanism

5.1 Classical principles of Potency and efficacy enhancement (Shaktyutkarsha)
5.2 Methods of Potency and Efficacy Enhancement in Ayurvedic Classics
5.3 Methods of Potency and Efficacy Enhancement in Modern Pharmaceutics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe and compare various techniques used in the classics for Shaktyutkarsha (Potency enhancement). — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural	Lecture 2h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Investigate the principles for improving potency and efficacy in modern pharmaceutics. — teach at level 4, the verb *investigate* is what will be asked	*investigate*	Conceptual	Experiential 10h
DOING — LEVEL 3 · MECHANISM — 1			
Justify the need for Shaktyutkarsha (Potency enhancement)in drug preparation and Advocate techniques of potency enhancement used in Ayurvedic pharmaceutics.	*justify*	Procedural	Experiential 10h

FOR THE TEACHER

- **This module tops out at Synthesis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 17 · SEMESTER 3 · PAPER 3

Guidelines for administration of Rasoushadhi in clinical Practice

Standardization and clinical Utility of Ayurvedic formulations

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	11	11	50
60 notional hours	22 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	7	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Discuss the concept of Rasa Chikitsa and explain the clinical superiority of Rasoushadhi.	*discuss*	L2 Comprehension / Understand
2	Illustrate the concept, need, and procedure of Kshetrikarana and its significance in the safe administration of Rasoushadhi.	*illustrate*	L2 Comprehension / Understand
3	Discuss appropriate Pathya apathya during Rasa Yoga Sevana.	*discuss*	L2 Comprehension / Understand
4	Elaborate on the toxicological profile of Rasoushadhi, including Ashuddha, Apakwa Bhasma Sevana Janya Dosha, and its management with safe prescription	*elaborate*	L2 Comprehension / Understand
5	Discuss heavy metal toxicity and its management with chelation therapy.	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Concept of Rasa chikitsa and significance of Rasoushadhi

4 obj · 11 h · Application / Guided response

1.1 Perspectives on the Notion of Rasa
1.2 Significance of Rasa Chikitsa
1.3 Various treatment modalities of Ayurveda

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Analyse the term Rasa from the Sanskrit and Ayurvedic literatures and discuss the significance of Rasoushadhi. — teach at level 4, the verb *analyse* is what will be asked	*analyse*	Factual + Conceptual °	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss various treatment modalities used in the Ayurvedic system of medicine. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual °	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the rational prescription for a selected disease.	*demonstrate*	Procedural	Practical 3h
Apply the principles of Rasa Chikitsa in the clinical practice/ alloted case.	*apply*	Procedural	Experiential 6h

Unit 2 · Concept of Kshetrikarana and its significance prior to administration of Rasoushadhi

4 obj · 11 h · Application / Guided response

2.1 Etymology and Concept of Kshetrikarana
2.2 Kshetrikarana for Optimising Medicine Efficacy
2.3 Procedure of Kshetrikarana (Panchakarma & Pancha Shodhana)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define kshetrikarana and discuss the concept of kshetrikarana. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual °	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the need and benefits of Kshetrikarana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual °	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the need and role of Kshetrikarana.	*demonstrate*	Procedural	Practical 3h
Demonstrate the procedure and benefits of kshetrikarana.	*demonstrate*	Procedural	Experiential 6h

Unit 3 · Pathya -Apathya during Rasayoga Sevana

5 obj · 12 h · Application / Mechanism

3.1 Pathya and Apathya Ahara, Vihara, and Manasika Vikara during Rasayoga Sevana

3.2 Importance of Pathya-Apathya during Rasayoga Sevana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define Pathya and Apathya ahara and describe its importance. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the application of the concept of what pathya and apathya vihara with respect to the Manasika vikara. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Enlist and demonstrate common pathya and apathya Ahara during rasa yoga sevana.	*enlist*	Procedural	Practical 2h
Enlist and demonstrate the common pathya-apathya Vihara during rasa yoga sevana.	*enlist*	Procedural	Practical 2h
DOING — LEVEL 3 · MECHANISM — 1			
Generate document depicting the outcome of compliance and non-compliance with the pathya apathya guidelines.	*generate*	Procedural	Experiential 6h

Unit 4 · Safety and toxicity of Rasoushadhi

5 obj · 14 h · Evaluation / Guided response

4.1 Safety considerations of Rasoushadhi

4.2 Ashuddha and Apakwa Bhasma Sevana Janya Vikara and their management

4.3 Safety and prescription trends of Rasoushadhi

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the safety considerations of Rasa oushadhies in detail. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Analyse the consequences of Ashuddha or Apakwa Bhasma sevana Dosha, along with the treatment.	*analyse*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evaluate the prescription errors containing Rasoushadhis.	*evaluate*	Conceptual ◦	Experiential 4h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate a case study of signs and symptoms of rasa vikaras and their management.	*demonstrate*	Procedural	Practical 6h
Assess and comapre Ashuddha and Shuddha Bhasma effects in the clinical practice.	*assess*	Procedural	Practical 2h

Unit 5 · Heavy metal toxicity and chelation therapy

4 obj · 8 h · Comprehension / Mechanism

5.1 The mechanism of heavy metal toxicity
5.2 Symptoms of heavy metal toxicity
5.3 Factors influencing heavy metal toxicity
5.4 Concept of chelation therapy and risks, and benefits

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss in detail the sources, pathways, and factors influencing heavy metal toxicity.	*discuss*	Conceptual	Lecture 1h
Discuss chelating agents and their role in managing heavy metal toxicity.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate identification of heavy metal toxicity.	*demonstrate*	Procedural	Practical 2h
DOING — LEVEL 3 · MECHANISM — 1			
Evalute the effect of your chelation therapy in the management of our heavy metal toxicity.	*evaluate*	Procedural	Experiential 4h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 7 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 18 · SEMESTER 3 · PAPER 3

Prayogika Bhaishajya Kalpana(Clinical application of Pharmaceutical products)

Standardization and clinical Utility of Ayurvedic formulations

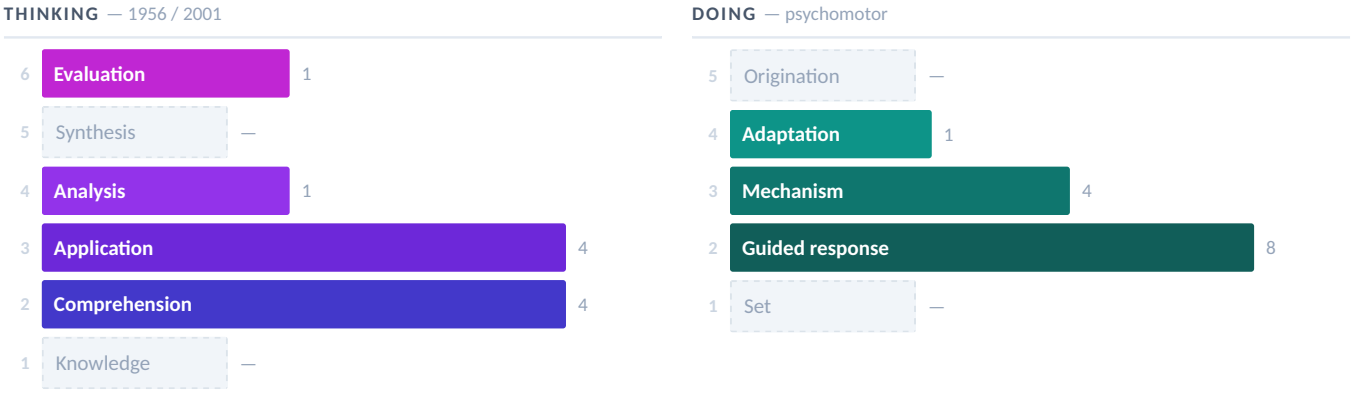
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	10	13	50
60 notional hours	24 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	8	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Aushadha Sevana Kala (Time of administration of drug)

7 obj · 15 h · Application / Adaptation

1.1 Classical concept of time of administration
1.2 Biological rhythms and chronobiology
1.3 Chrono-pharmaceutics and chronopharmaceutical drug delivery systems
1.4 Principles of chronopharmacology
1.5 Benefits of chronotherapeutics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss in detail the concept of Oushadha Sevana Kala in Ayurveda.	*discuss*	Conceptual	Lecture 1h
Elaborate the principles of chrono pharmaceutical drug delivery systems and their benefits.	*elaborate*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss biological rhythms and chronobiology and their significance in drug delivery. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics.	*demonstrate*	Procedural	Practical 3h
Demonstrate chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dynamics.	*demonstrate*	Procedural	Experiential 3h
Illustrate the modulation of the action of the dosage form by chronotherapeutic administration in Ayurveda.	*illustrate*	Procedural	Experiential 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Demonstrate Modulation of dosage form action by chronotherapeutic administration in Ayurveda.	*demonstrate*	Procedural	Practical 3h

Unit 2 · Concept and scientific basis of Anupana

6 obj · 16 h · Evaluation / Mechanism

2.1 Classical concepts of Anupana in Ayurveda
2.2 Role of Anupana in bioavailability
2.3 Anupana for targeted drug delivery and distribution
2.4 Role of Anupana in the mitigation of adverse effects and toxicity
2.5 Anupana for synergistic effect

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss in detail the concept of anupana in Ayurveda. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Explain the role of Anupana in bioavailability and targeted drug delivery. — teach at level 2, the verb *explain* is what will be asked	*explain*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate appropriate Anupana for the selected dosage form in different clinical conditions.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 3 · MECHANISM — 2			
Enlist appropriate Anupanas for various formulations in different clinical conditions.	*enlist*	Procedural	Experiential 3h
Describe the role of Anupana in the modulation of drug tolerance and safety.	*describe*	Conceptual + Procedural	Experiential 4h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Demonstrate drug response through synergistic Anupana integration.	*demonstrate*	Conceptual	Experiential 4h

Unit 3 · Matra (Posology)

5 obj · 12 h · Application / Guided response

3.1 Definition, importance, and clinical significance
3.2 Factors affecting drug dose
3.3 Posology according to modern pharmacology
3.4 Calculation of paediatric dose

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Elaborate on the concept, importance, and clinical significance of Matra.	*elaborate*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the factors affecting drug dose in detail. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the calculation of paediatric dose using different formulae.	*demonstrate*	Procedural	Practical 3h
Demonstrate the Patient-centric evaluation of Matra.	*demonstrate*	Procedural	Practical 3h
Demonstrate the utility of posology principles from an ayurvedic and modern pharmacology perspective.	*demonstrate*	Procedural	Experiential 4h

Unit 4 · Concept of drug repurposing

6 obj · 13 h · Application / Mechanism

4.1 Concept of drug repurposing
4.2 Systematic approaches to drug repurposing
4.3 Strategies of drug repurposing
4.4 Methodologies of drug repurposing

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the concept, need, and key aspects of drug repurposing.	*describe*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Describe systematic approaches to drug repurposing. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
Discuss strategies and methodologies of drug repurposing. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the analysis of classical ayurvedic drugs for novel indications.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 2			
Evaluate novel indications for classical Ayurvedic formulations.	*evaluate*	Procedural	Experiential 3h
Demonstrate the Reverse pharmacological approach for drug repurposing in Ayurveda.	*demonstrate*	Procedural	Experiential 2h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 8 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 25 · SEMESTER 3 · PAPER 4

Artificial intelligence in drug development and Pharmacogenetics; Pharmacogenomics

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	10	10	50
60 notional hours	20 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	5	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the role of Artificial Intelligence in drug development	*describe*	L2 Comprehension / Understand
2	Demonstrate the basics of machine learning and its applications	*demonstrate*	L3 Application / Apply
3	Discuss 3D-Printed Drug Delivery Systems Based on Natural Products	*discuss*	L2 Comprehension / Understand
4	Justify clinical application of pharmacogenetics	*justify*	L6 Evaluation / Evaluate
5	Describe the Current status and future perspective of Pharmacogenomics	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Role of Artificial intelligence in drug development

4 obj · 11 h · Application / Mechanism

1.1 Introduction to AI in drug development
1.2 AI, machine-based intelligence learning
1.3 Role of AI in Drug discovery and Drug development

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define AI and discuss its importance in drug development. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the Role of AI in Drug Discovery and Drug development with a suitable example. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the basics of machine learning and its application.	*demonstrate*	Procedural	Practical 4h
DOING — LEVEL 3 · MECHANISM — 1			
Perform De Novo Drug. Design Using Generative Models.	*perform*	Procedural	Experiential 5h

Unit 2 · Types of software used in Artificial intelligence

3 obj · 6 h · Application / Origination

2.1 Different types of software used in AI
2.2 Data mining and analysing medical records, medication management
2.3 Recent updates

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Describe different types of software used in AI. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss Data mining and analysing medical records, medication management, with recent updates. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 5 · ORIGINATION — 1			
Demonstrate prediction of drug likeliness using machine learning.	*demonstrate*	Procedural	Practical 4h

Unit 3 · 3D printed drug delivery system

5 obj · 16 h · Application / Adaptation

3.1 Approaches of 3D printing in current drug delivery
3D printing in personalized drug delivery
3D-Printed Drug Delivery Systems Based on Natural Products

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the fundamentals and explore the materials and technology used in 3D printing.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the concept of personalised drug delivery, formulations, and natural products. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate different 3D printers used in the drug delivery system.	*demonstrate*	Procedural	Practical 4h
DOING — LEVEL 4 · ADAPTATION — 2			
Demonstrate the comparison of traditional dosage forms versus 3D printed ones regarding the structure and dissolution.	*demonstrate*	Procedural	Practical 4h
Analyse ADMET properties using AI tools.	*analyse*	Procedural	Experiential 6h

Unit 4 · Pharmacogenetics and Pharmacogenomics

4 obj · 11 h · Application / Guided response

4.1 Introduction to Pharmacogenetics and Pharmacogenomics along with its importance
4.2 Factors influencing drug response and ADE (Adverse drug events)
4.3 Methods of pharmacogenetics testing

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Appraise key concepts and terminologies and explore the role of Pharmacogenetics and Pharmacogenomics. — teach at level 6, the verb *appraise* is what will be asked	*appraise*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Justify clinical application of pharmacogenetics, benefits, and challenges. — teach at level 6, the verb *justify* is what will be asked	*justify*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate to predict Drug-Target Interaction.	*demonstrate*	Procedural	Practical 4h
Identify potential adverse effects of drug candidates using dataset and tools.	*identify*	Procedural	Experiential 5h

Unit 5 · Pharmacogenomics in drug discovery

4 obj · 12 h · Comprehension / Mechanism

5.1 Benefits of Pharmacogenomics
5.2 Drug Repurposing
5.3 Basis of gene-drug association
5.4 Current status and future perspective of Pharmacogenomics
5.5 Ethical concern

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss pharmacogenomics data contributes to the identification and validation of drug targets.	*discuss*	Conceptual ◦	Lecture 1h
Discuss the current status and future perspectives of pharmacogenomics, including ethical concerns.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 2			
Extract potential drug-disease relationship by using through Literature /data Mining.	*extract*	Conceptual + Procedural ◦	Experiential 5h
Interpret Pharmacokinetics and Toxicology Modelling using appropriate AI tools and methods.	*interpret*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Origination on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 5 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Origination.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 26 · SEMESTER 3 · PAPER 4

Pharmacology

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	11	5	50
60 notional hours	16 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	2	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the basics of pharmacology, including Pharmacodynamics and pharmacokinetics	*describe*	L2 Comprehension / Understand
2	Describe Pharmacokinetics, therapeutic index, and drug interactions	*describe*	L2 Comprehension / Understand
3	Discuss the pharmacology of drugs acting on different systems, like the Central and Peripheral nervous systems	*discuss*	L2 Comprehension / Understand
4	Describe Adverse Drug Reactions and Emergency Pharmacology	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Introduction to Pharmacology

3 obj · 12 h · Comprehension / Guided response

1.1 Introduction to Pharmacology
1.2 Pharmacodynamics
1.3 Dose-response relationship and factors influencing drug action

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define and classify pharmacology and describe the principles and mechanisms of drug action, receptor theories, and classification of rece. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual	Lecture 1h
Discuss dose response relationship, and factors influencing drug action.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate pharmacological experiments using videos or simulation software.	*demonstrate*	Procedural	Practical 10h

Unit 2 · Pharmacokinetics, therapeutic index, and drug interactions

3 obj · 12 h · Application / Adaptation

2.1 Pharmacokinetics (absorption, distribution, metabolism, excretion)
2.2 Therapeutic index and combined effects of drugs
2.3 Adverse drug reaction
2.4 Drug interactions (pharmacokinetic and pharmacodynamic)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss therapeutic index, drug interactions, and adverse drug reactions.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the basic principles of pharmacokinetics (absorption, distribution, metabolism 7, CO 8 excretion). — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Demonstrate the effect of drugs on animals by simulated experiments.	*demonstrate*	Procedural	Practical 10h

Unit 3 · Pharmacology of drugs that act on the peripheral nervous system

3 obj · 9 h · Synthesis / —

3.1 Organization and function of ANS (Autonomic Nervous System)
3.2 Neurohumoral transmission, co-transmission, and classification of neurotransmitters
3.3 Parasympathomimetics, Parasympatholytics, Sympathomimetics, Sympatholytics
3.4 Neuromuscular blocking agents and skeletal muscle relaxants (peripheral)
3.5 Local anesthetic agents

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the organisation and function of ANS (Autonomic Nervous System) and describe Neurohumoral transmission, co-transmission, and classification of neurotransmitters.	*discuss*	Conceptual	Lecture 1h
Enlist Parasympathomimetics, Parasympatholytics, Sympathomimetics, sympatholytics Neuromuscular blocking agents, and skeletal muscle relaxants (peripheral) and Local anaesthetic agents. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Factual ◦	Lecture 1h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Analyse the mechanism of action of drugs acting on the Cardiovascular system through a case study scenario. — teach at level 4, the verb *analyse* is what will be asked	*analyse*	Conceptual ◦	Experiential 7h

Unit 4 · Pharmacology of drugs that act on the central nervous system I

3 obj · 9 h · Application / Mechanism

4.1 Neurohumoral transmission in the C.N.S. special emphasis on the importance of
4.2 General anaesthetics and pre-anaesthetics
4.3 Sedatives, hypnotics, and centrally acting muscle relaxants
4.4 Anti-epileptics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Describe Neurohumoral transmission in the C.N.S and illustrate General anaesthetics and pre-anaesthetics. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Categorize drugs into classes such as Sedatives, hypnotics, and centrally acting muscle relaxants, and discuss anti-epileptics. — teach at level 4, the verb *categorize* is what will be asked	*categorize*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Analyse the mechanism of action of drugs acting on the respiratory system through a case study scenario.	*analyse*	Procedural	Experiential 7h

Unit 5 · Pharmacology of drugs that act on the central nervous system II

4 obj · 14 h · Comprehension / Mechanism

5.1 Psychopharmacological agents: Antipsychotics, antidepressants, anti-anxiety
5.2 Drugs used in Parkinson's disease and Alzheimer's disease
5.3 CNS stimulants and nootropics
5.4 Drug addiction, drug abuse, tolerance, and dependence

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss Psychopharmacological agents and enlist drugs used in Parkinson's disease and Alzheimer's disease.	*discuss*	Conceptual ◦	Lecture 1h
Differentiate CNS stimulants and nootropics, and discuss drug addiction, drug abuse tolerance, and dependence. — teach at level 4, the verb *differentiate* is what will be asked	*differentiate*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 2			
Analyse Adverse Drug Reaction in a polypharmacy patient through a case scenario.	*analyse*	Procedural	Experiential 6h
Discuss Emergency Pharmacology through a case scenario of anaphylaxis management.	*discuss*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Synthesis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 2 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 3 · SEMESTER 4 · PAPER 1

Parada (Mercury) and Its Processing: Ancient wisdom and Scientific advances

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	8	9	50
60 notional hours	17 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	9	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the ancient and contemporary perspectives of Parada (Mercury)	*describe*	L2 Comprehension / Understand
2	Perform and analyse the Samanya Shodhana, Vishesha Shodhana, and procedures of Parada processing (Ashta Samskara)	*perform*	L3 Application / Apply
3	Discuss the Physicochemical properties and pharmacology of mercury and mercurial compounds	*discuss*	L2 Comprehension / Understand
4	Describe the mercury poisoning and its management by the ancient and contemporary approaches	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Parada: Ancient and Contemporary Perspectives on Its Shodhana and Underlying Principles

8 obj · 22 h · Comprehension / Mechanism

1.1 Ancient & Contemporary knowledge of Parada/
1.2 Parada Dosha, Shodhana(Purification) of Parada and its significance
1.3 Principles of Parada Jarana and Parada Murcchana
1.4 Utility of Parada Bandha

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Describe the ancient and contemporary perspectives of Parada (Mercury).	*describe*	Conceptual ◦	Lecture 1h
Discuss the processes related to Parada and their significance.	*discuss*	Conceptual + Procedural ◦	Lecture 1h
Describe the scientific basis of Parada Jarana, Murcchana, and Bandha.	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Discuss the rationale behind Samanya and Vishesha Shodhana.	*discuss*	Conceptual + Procedural ◦	Lecture 1h
Demonstrate Samanya and Vishesha Shodhana procedures of Parada.	*demonstrate*	Procedural	Practical 7h
Perform the Parada Jarana, Parada Murchana and Bandhana processes.	*perform*	Procedural	Practical 5h
Perform the Samanya Shodhana of Parada.	*perform*	Procedural	Experiential 3h
DOING — LEVEL 3 · MECHANISM — 1			
Perform the Vishesha Shodhana procedures of Parada.	*perform*	Procedural	Experiential 3h

Unit 2 · Parada Ashta Samskara and Ashtadasha Samskara

4 obj · 21 h · Comprehension / Guided response

2.1 Parada Samskara with research updates
2.2 Pharmaceutical equipment design for Parada Ashta Samskara
2.3 Hands-on experience of Parada Ashta Samskara

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe and discuss the utility of Ashtadasha Samskara.	*describe*	Conceptual ◦	Lecture 2h
Describe the design of pharmaceutical equipment to carry out Parada Ashta Samskara.	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the Parada Samskara procedures- Swedana, Mardana, Murchana, and Utthapana.	*demonstrate*	Procedural	Practical 8h
Elaborate and perform the Parada Ashta Samskara procedures- Patana, Bodhana Niyamana and Deepana.	*elaborate*	Procedural	Experiential 10h

Unit 3 · Physico-chemical proprties and pharmacology of mercury and mercurial compounds

2 obj · 6 h · Analysis / Mechanism

3.1 Mercury metallurgy
3.2 Stoichiometry of mercury
3.3 Pharmacodynamics and pharmacokinetics of mercury and mercurial products

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe and elaborate on the metallurgy and stoichiometry of mercury. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Compare and analyze the Pharmacodynamics and pharmacokinetics of mercury and mercurial products.	*compare*	Procedural	Experiential 5h

Unit 4 · Mercury toxicity, ADR and its management

3 obj · 7 h · Analysis / Guided response

4.1 Mercury poisoning and its types

4.2 Ancient and Contemporary knowledge about the Treatment of Mercury poisoning

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the antidotes for mercury poisoning and its possible treatment by ancient and contemporary knowledge. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe the mercury poisoning and analyze its mechanism. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Analyse mercury toxicity and ADR incidences.	*analyse*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 9 lecture, 3 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 4 · SEMESTER 4 · PAPER 1

Maharasa Dravya, its processing and utility

Rasashastra (Iatrochemistry) — Principles and Practices

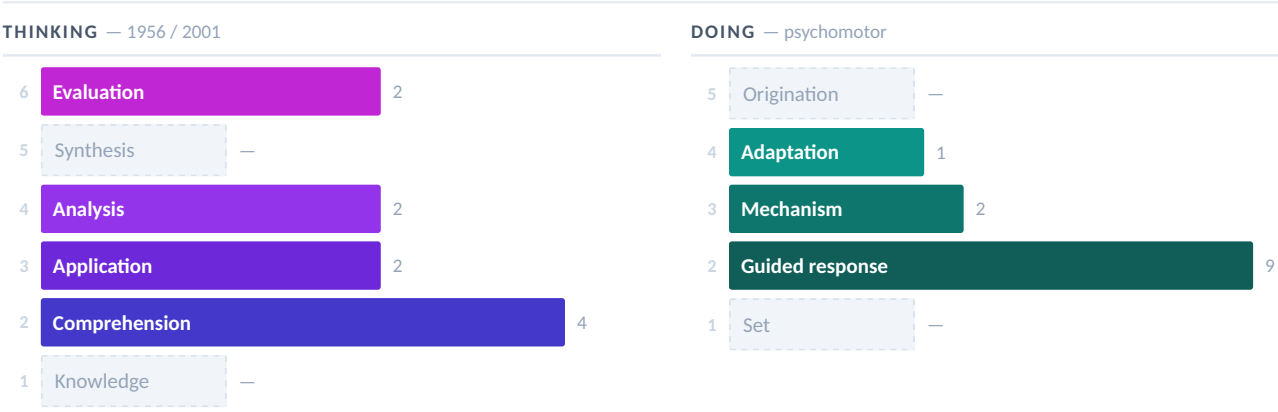
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	10	12	50
60 notional hours	22 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	6	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Discuss the characterization of the samples of Maharasa Dravyas	*discuss*	L2 Comprehension / Understand
2	Perform various pharmaceutical procedures like Shodhana, Marana, etc.	*perform*	L3 Application / Apply
3	Perform the physicochemical analysis with the current scientific updates	*perform*	L3 Application / Apply
4	Discuss the pharmacology and therapeutic application of Maharasa Dravya	*discuss*	L2 Comprehension / Understand
5	Describe the new metal, minerals, procedures, and principles related to Maharasa Dravyas	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Maharasa Dravyas with special reference to Abhraka and Vaikranta

8 obj · 18 h · Analysis / Adaptation

1.1 Comparative analysis of Maharasa Dravyas mentioned in various texts
1.2 Characterization of Maharasa dravyas
1.3 Abhraka
1.4 Vaikranta

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe and critically discuss the different Maharasa Dravyas mentioned in the classics.	*describe*	Conceptual ◦	Lecture 1h
Describe the significance of Abhraka Pancha Samskara.	*describe*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the mineralogical identity of Vaikranta. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Analyze the technique of characterization for Maharasa Dravyas.	*analyze*	Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Analyze and Verify Maharasa Dravya Samples.	*analyze*	Procedural	Practical 2h
Perform Shodhana, Marana, and other pharmaceutical procedures related to Abhraka and Vaikranta.	*perform*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Analyze the pharmacotherapeutic attributes of Abhraka Bhasma and Vaikranta Bhasma.	*analyze*	Procedural	Experiential 5h
DOING — LEVEL 4 · ADAPTATION — 1			
Analyse the pharmacology and clinical applications of Abhraka and Vaikranta.	*analyse*	Procedural	Experiential 2h

Unit 2 · Principles and Processing of Makshika and Vimala

3 obj · 11 h · Comprehension / Guided response

2.1 Critical understanding of Makshika (Copper Pyrite)
2.2 Detailed understanding of Vimala (Iron Pyrite)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the therapeutic applications of Makshika and Vimala Bhasma, along with the antidotes for toxicity/ADR caused by them.	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Makshika and Vimala.	*perform*	Procedural	Practical 5h
Perform the Shodhana of Makshika Bhasma and Vimala Bhasma.	*perform*	Procedural	Experiential 5h

Unit 3 · Principles and Processing of Shilajit and Sasyaka

3 obj · 10 h · Analysis / Guided response

3.1 Shilajatu (Black Bitumen)
3.2 Sasyaka(Bornite)/Tuttha(Blue Vitriol)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe and analyze the therapeutic applications of Shilajatu and Sasyaka/Tuttha, along with the antidotes for toxicity/ADR caused by them. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Shilajatu and Sasyaka/Tuttha.	*perform*	Procedural	Practical 3h
Perform the Shodhana of Shilajit and Sasyaka/Tuttha.	*perform*	Procedural	Experiential 6h

Unit 4 · Principles and Processing of Chapala and Kharpara, along with physico-chemical analysis, with research updates of all Maharasa Dravyas

4 obj · 9 h · Application / Mechanism

4.2 Kharpara

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the therapeutic applications of Chapala and Kharpara Bhasma, along with the antidotes for toxicity/ADR caused by them. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Perform Shodhana, Marana, and other pharmaceutical procedures related to Chapala/Kharpara.	*perform*	Procedural	Practical 2h
Demonstrate and perform physico-chemical analysis of Bhasma / Shodhita products of Maharasa Dravya category.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 3 · MECHANISM — 1			
Explore and perform the pharmaceutical procedure, like Shodhana, Marana for Chapala/ Kharpara, using authentic or substitute drugs.	*explore*	Procedural	Experiential 3h

Unit 5 · Utility of new metals and minerals

4 obj · 8 h · Evaluation / Guided response

5.1 New metals/ minerals that align with Maharasa Dravyas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the utility of new metals and minerals that align with Maharasa Dravyas.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 2			
Explore Shodhana process for the new metal/ minerals that align with Maharasa Dravyas. — teach at level 4, the verb *explore* is what will be asked	*explore*	Procedural ◦	Lecture 1h
Describe Marana process for the new metal/ minerals that align with Maharasa Dravyas. — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Identify and analyze New Metallic/Mineral Resources that align with Maharasa Dravyas Principles.	*identify*	Conceptual + Procedural ◦	Experiential 5h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 6 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 11 · SEMESTER 4 · PAPER 2

Mana Paribhasha (Metrology in Ayurvedic Pharmaceutical preparation)

Exposition of Ayurvedic Pharmaceutics

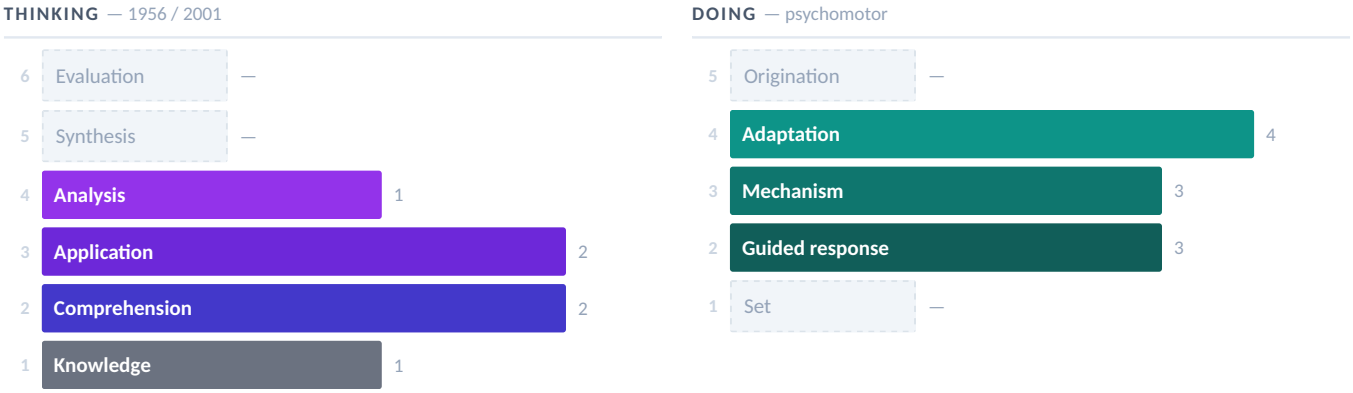
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	6	10	50
60 notional hours	16 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	6	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	5	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Mana Paribhasha (Classical system of measurement)

4 obj · 12 h · Analysis / Adaptation

1.1 Importance of accurate measurements in proportions of ingredients

1.2 Importance of accurate measurements in dosage and administration of drugs

1.3 Types of Mana and details of classical units used for measurement (Weight/ Volume/ Length/ Time)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Interpret the application of Mana in classical formulations with a variety of examples. — teach at level 6, the verb *interpret* is what will be asked	*interpret*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Recall classical knowledge of Mana Paribhasha with a comparison of various classical texts. EC, LS, S. — teach at level 1, the verb *recall* is what will be asked	*recall*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Apply the Knowledge of mana Paribhasha in the preparation of various formulations.	*apply*	Procedural	Practical 4h
DOING — LEVEL 4 · ADAPTATION — 1			
Compare and judge the differences in the Mana mentioned by different classics.	*compare*	Procedural	Experiential 6h

Unit 2 · Standard Units of Measurement

3 obj · 6 h · Knowledge / Adaptation

2.1 AYUSH Unit Converter

2.2 Utility of converted units in the preparation of Classical formulations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Describe standard units of weights and measures and AYUSH Unit Converter. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Preparation of charts of unit converter (Ayurvedic Mana- Metric).	*preparation*	Procedural	Practical 2h
DOING — LEVEL 4 · ADAPTATION — 1			
Demonstrate preparation of classical formulations prepared with converted Units.	*demonstrate*	Procedural	Experiential 3h

Unit 3 · Volumetric Vs Gravimetric Methods

3 obj · 6 h · Application / Guided response

3.1 Key differences between Volumetric and gravimetric methods of measurement for a variety of drugs

3.2 Application of volumetric method for preparation of various classical formulations

3.3 Variation in proportion by using both methods and alteration in outcome

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Define the terms Volumetric methods and Gravimetric methods and apply the methods of both measurements to Ayurvedic Pharmaceutical Preparations. — teach at level 1, the verb *define* is what will be asked	*define*	Factual + Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Apply and demonstrate the volumetric method for the preparation of classical formulations.	*apply*	Procedural	Practical 2h
Justify and interpret differences occurring in the final product prepared by varying the weight of the ingredients once taken by volume and once taken by mass(Gravimetric Method).	*justify*	Procedural	Experiential 3h

Unit 4 · Rationality of Proportion of Ingredients in Ayurvedic formulations

3 obj · 16 h · Comprehension / Adaptation

4.1 Factors accountable for deciding the proportion of ingredients of the

4.2 Profile of Raw Material (Fresh/ dry, Soft/ hard, part used, therapeutic properties, compatibility, etc.)

4.3 Rules for Duplication of ingredients in the same formulation

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the classical rules and principles for the proportion of ingredients based on raw material and its nature/ Qualities.	*describe*	Conceptual	Lecture 3h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate the skill of preparing formulations with a diverse range of raw materials.	*demonstrate*	Procedural	Practical 6h
DOING — LEVEL 4 · ADAPTATION — 1			
Analyze and justify the proportions of ingredients of classical formulations.	*analyze*	Procedural	Experiential 7h

Unit 5 · Bulk production of Ayurvedic Formulations

3 obj · 16 h · Comprehension / Adaptation

5.1 Modifications expected for bulk production (in Material, Machine, Method, Mechanism)

5.2 Material- Type, proportion, effect of temperature pattern

5.3 Machine used for bulk manufacturing and automated drug manufacturing

5.4 Mechanism of machines used for bulk production and its merits and demerits

5.5 Methods used for bulk production in Ayurvedic Pharmacies and probable alterations in the qualities of the end products

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Enlist the machines, heating devices for bulk production. and describe their methods currently used by Ayurvedic pharmacies f. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Procedural	Lecture 3h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate the use of machinery used for large-scale production in Ayurvedic pharmacies, with their mechanism.	*demonstrate*	Procedural	Practical 6h
DOING — LEVEL 4 · ADAPTATION — 1			
Debate on merits and demerits of bulk production and the adaptation of advanced machines in Ayurvedic Pharmacies.	*debate*	Procedural	Experiential 7h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 6 lecture, 5 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 12 · SEMESTER 4 · PAPER 2

Process validation Protocol

Exposition of Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	3	4	6	50
60 notional hours	10 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	4	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	2	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Pharmaceutical Preparation process: Classical Dosage forms

6 obj · 41 h · Evaluation / Guided response

1.1 Primary Dosage forms- Panchavidha Kashaya Kalpana and their Upakalpana

1.2 Secondary Dosage Forms- Avaleha Kalpana, Lavana Kalpana, Sneha Kalpana, Guti-vati-guggulu kalpana, Sandhana Kalpana, Arka-Kalpana, Kshara-kalpana

1.3 Topical Dosage forms- Lepa kalpana, Malahara Kalpana, Upanaha Kalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the SOP of Primary dosage forms, Secondary dosage forms, and topical dosage forms.	*describe*	Procedural ◦	Lecture 3h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Assess the significance of the Process Validation Protocol.	*assess*	Conceptual + Procedural ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate the development of a process validation protocol for Ayurvedic dosage forms.	*demonstrate*	Procedural	Practical 10h
Develop a process validation protocol for Ayurvedic dosage forms.	*develop*	Procedural	Practical 10h
Interpret the documentation related to Standard Operating Procedures (SOPs) used in pharmacies.	*interpret*	Procedural	Experiential 6h
Prepare unique classical formulations following constructed SOP.	*prepare*	Procedural	Experiential 10h

Unit 2 · Pharmaceutical Preparation process: Recent advances of Dosage forms

3 obj · 10 h · Comprehension / Guided response

2.1 Principles of processes viz. Expression, Extraction, Maceration,

2.2 Solid Dosage forms- Powders, Pills, Tablets, Capsules, Soft gel capsules, Gummies, Lozenges, Granules

2.3 Liquid Dosage Forms- Emulsions, Suspensions, Lotions, Liniments, Elixir, Syrup, Nasal Drops, Gargle

2.4 Semi-solid dosage forms- Ointment, Gel, Paste, Cream, Plasters, Suppositories

2.5 Gaseous dosage forms- Aerosols, Nasal Spray, Inhalers

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe principles of different dosage forms, viz. Extraction, Maceration, Percolation Infusion, Distillation, Active principle isolation, Particle size reduction, Separation techniques, and Mixing techniques.	*describe*	Conceptual	Lecture 2h
Discuss Pharmaceutical Preparation Process of Solid, semisolid, liquid, and gaseous dosage forms.	*discuss*	Procedural	Lecture 3h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Prepare different novel modern dosage forms of the given drug.	*prepare*	Procedural	Experiential 5h

Unit 3 · Novel Dosage and drug delivery Forms

1 obj · 5 h · — / Guided response

3.1 Importance and Need of Different Dosage Forms in Drug Delivery Systems and Treatment Modalities

3.2 Definition, significance, technologies driving Novel dosage forms, applications, and challenges

3.3 Modification of different dosage forms

3.4 Potential areas for future research and development

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Analyze, justify, and demonstrate the importance and need of the preparation of Novel dosage forms.	*analyze*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Guided response on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 4 lecture, 2 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Guided response.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 19 · SEMESTER 4 · PAPER 3

SOP and clinical application of Chaturvidha Rasayana

Standardization and clinical Utility of Ayurvedic formulations

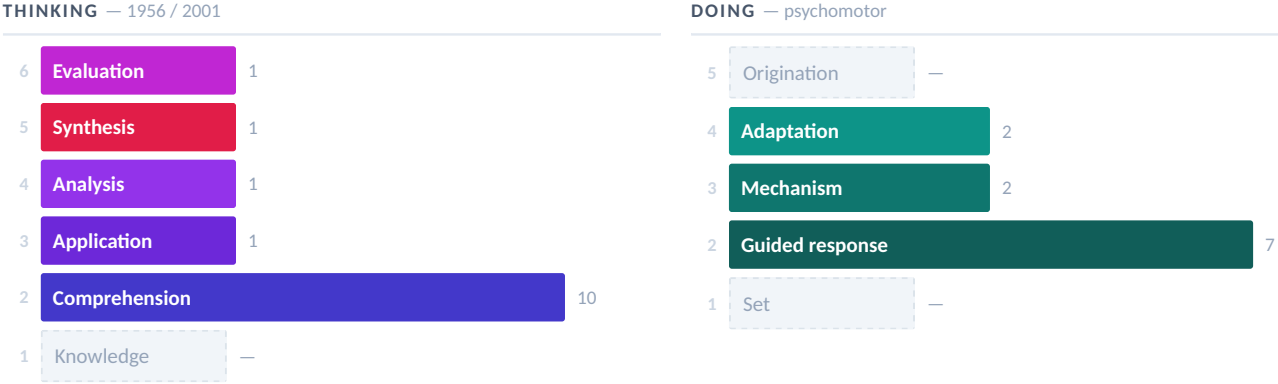
CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	4	14	11	75
90 notional hours	25 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	14	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	7	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe SOP, significance, and clinical application of Khalveeya Rasayana	*describe*	L2 Comprehension / Understand
2	Discuss SOP, significance, mechanism of action, clinical applications with Pathya-Apathya of Parpati Kalpana	*discuss*	L2 Comprehension / Understand
3	Describe SOP, role of Gandhaka Drava paka, significance, and clinical attributes of Pottali Rasayana	*describe*	L2 Comprehension / Understand
4	Discuss SOP, Significance, role of Kramagni Paka, and clinical application of Kupipakwa Rasayana	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Standard Operating Procedure (SOP) and Clinical Application of Khalveeya Rasayana

5 obj · 17 h · Application / Adaptation

1.1 Definition and significance of Khalveeya Rasayana
1.2 Standard Operating Procedure for the preparation of Khalveeya Rasayana
1.3 Role of bhavana in potentiating Khalveeya rasayana
1.4 Clinical application of Khalveeya Rasayana with examples

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the unique features, types, and significance of parpati kalpana.	*describe*	Conceptual	Lecture 2h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the role of bhavana (pharmacodynamics of trituration) in potentiating kharaleeya rasayana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the method of preparation and SOP of Kharaleeya Rasayana.	*demonstrate*	Procedural	Practical 7h
Demonstrate the impact of Bhavana (A bioactivation tool) in Kharaleeya Rasayana.	*demonstrate*	Procedural	Experiential 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Appraise the utility of kharaleeya rasayana to clinical case practice.	*appraise*	Procedural	Experiential 4h

Unit 2 · SOP and clinical application of Parpati Rasayana

6 obj · 15 h · Analysis / Adaptation

2.1 Definition and significance of Parpati
2.2 Standard Operating Procedure for the preparation of Parpati with process validation
2.3 Mechanism of action of Parpati kalpana
2.4 Pathyapathya in Parpati Prayoga and its significance
2.5 Clinical application of Parpati Kalpa with examples

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Describe the unique features, types, and significance of parpati kalpana.	*describe*	Conceptual	Lecture 1h
Discuss SOP and quality control parameters of Parpati Kalpa.	*discuss*	Procedural	Lecture 1h
Discuss in detail the pathya-apathya, matra, prayoga of parpati kalpana.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Review and analyze research updates on Parpati Kalpana.	*review*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the SOP of parpati kalpana.	*demonstrate*	Procedural	Practical 7h
DOING — LEVEL 4 · ADAPTATION — 1			
Evaulate the pharmacodynamic rationale and clinical application of parpati kalpana.	*evaluate*	Conceptual + Procedural	Experiential 4h

Unit 3 · SOP and clinical application of Pottali Rasayana

7 obj · 25 h · Evaluation / Mechanism

3.1 Definition and significance of Pottali rasayana
3.2 Standard Operating Procedure for the preparation of Pottali Kalpa with process validation
3.3 Role of Gandhaka drava paka in the preparation Pottali rasayana
3.4 Special clinical attributes of Pottali with examples

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define pottali and discuss in detail the concept, types, and significance of pottali rasayana. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual	Lecture 1h
Discuss the standardization and quality control parameters for Pottali Kalpa.	*discuss*	Procedural ◦	Lecture 1h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Discuss in detail the various classical methods of preparation of pottali kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Procedural	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evalaute the safety and efficacy of Pottali Rasayana.	*evaluate*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the method of preparation and SOP of pottali kalpana.	*demonstrate*	Procedural	Practical 8h
DOING — LEVEL 3 · MECHANISM — 2			
Prepare Pottali following SOP.	*prepare*	Procedural	Experiential 8h
Analyse the pharmacological profile of the ingredients, bhavana dravya, and the pharmaceutical procedure adopted in achieving therapeutic precision.	*analyse*	Procedural	Experiential 5h

Unit 4 · SOP and clinical application of Kupipakwa Rasayana

7 obj · 27 h · Comprehension / Guided response

4.1 Definition and significance of Kupipakwa rasayana
4.2 Standard Operative Procedure for the preparation of Kupipakwa Rasayana with process validation
4.3 Role of Kramagni paka in preparation of Kupipakwa Rasayana
4.4 Chemistry of Kupipakwa rasayana
4.5 Clinical application of Kupipakwa Rasayana Kalpa with examples

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Discuss the concept and significance of Kupipakwa Rasayana.	*discuss*	Conceptual	Lecture 1h
Describe in detail the role of Kramagni Paka in the preparation of Kupipakwa Rasayana.	*describe*	Conceptual + Procedural ◦	Lecture 1h
Describe the role of gandhaka jarana in the context of sagandha kupipakwa rasayana.	*describe*	Conceptual	Lecture 1h
Discuss the thermo–chemistry of kupipakwa rasayana.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the method of preparation and SOP of Kupipakwa Rasayana.	*demonstrate*	Procedural	Practical 8h
Prepare the selected Kupipakwa Rasayana.	*prepare*	Procedural	Experiential 10h
Demonstrate the scientific rationality for the selection of specific Kupipakwa Rasayana in the given scenario.	*demonstrate*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 14 lecture, 4 practical, 7 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 20 · SEMESTER 4 · PAPER 3

SOP and Clinical application of Ayaskriti, Louha, and Mandura Kalpa

Standardization and clinical Utility of Ayurvedic formulations

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	3	5	7	25
30 notional hours	12 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the concept of Ayaskriti, with SOP, clinical applications, and safety concerns of Ayaskriti	*describe*	L2 Comprehension / Understand
2	Discuss the concept, SOP, clinical applications, and key chemical considerations of Louha Kalpa	*discuss*	L2 Comprehension / Understand
3	Discuss the concept, SOP, and clinical applications of Mandura Kalpa	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · SOP and clinical application of Ayaskriti

3 obj · 8 h · Comprehension / Mechanism

1.1 Ayaskriti
1.2 Standard Operating Procedure for the preparation of Ayaskriti
1.3 Types (forms) of ayaskriti
1.4 Method of administration and clinical utility of Ayaskriti
1.5 Safety concerns of Ayaskriti

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss in detail the concept of Ayaskriti, its types, method of administration, and safety concerns.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the SOP and preparation of Ayaskriti.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 3 · MECHANISM — 1			
Evaluate the role of Ayaskriti.as a multi-target formulation.	*evaluate*	Conceptual + Procedural ◦	Experiential 4h

Unit 2 · SOP and clinical application of Louha kalpa

4 obj · 9 h · Comprehension / Adaptation

2.1 Pharmaceutical vistas of Louha Kalpas
2.2 Key chemical considerations in Louha Kalpa
2.3 Pharmacology and clinical utility of Louha kalpa

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the concept of Louha Kalpa.	*discuss*	Conceptual	Lecture 1h
Discuss the key chemical considerations in Louha Kalpa.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the SOP and preparation of Louha Kalpa.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Apply Louha Kalpa by integrating through textual insights with real word cases for the therapeutic purpose.	*apply*	Procedural	Experiential 4h

Unit 3 · SOP and Clinical application of Mandura Kalpa

5 obj · 11 h · Comprehension / Mechanism

3.1 Concept of mandura kalpa and its SOP
3.2 Types of mandura kalpa based on the method of preparation
3.3 Significance of go mutra in mandura kalpa
3.4 Clinical application of mandura kalpa

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the concept and pharmaceutical vistas of Mandura Kalpa.	*discuss*	Conceptual	Lecture 1h
Describe chemistry and pharmacology associated with structure and preparation of Mandura Kalpas.	*describe*	Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the SOP and preparation of Mandura Kalpa.	*demonstrate*	Procedural	Practical 2h
Demonstrate the clinical application of Mandura Kalpa.	*demonstrate*	Procedural	Practical 2h
DOING — LEVEL 3 · MECHANISM — 1			
Evaluate the role of Gomutra in pharmaceutical processing and as a bio enhancer.	*evaluate*	Conceptual + Procedural ◦	Experiential 5h

FOR THE TEACHER

- **This module tops out at Comprehension on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 27 · SEMESTER 4 · PAPER 4

Network pharmacology & Reverse Pharmacology

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	2	10	6	50
60 notional hours	16 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	2	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Netwok Pharmacology

8 obj · 28 h · Analysis / Mechanism

1.1 Fundamental knowledge of Network pharmacology
1.2 Network pharmacology concepts
1.3 Computational methods
1.4 Applications
1.5 Research updates on Network pharmacology

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Define network pharmacology and differentiate it from traditional pharmacology.	*define*	Factual ◦	Lecture 1h
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Recognize key network concepts like nodes (genes/proteins/drugs) and edges(interactions). — teach at level 1, the verb *recognize* is what will be asked	*recognize*	Factual ◦	Lecture 1h
Describe network construction from herbal compounds.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss various applications and research updates on Network pharmacology. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss and analyze Research updates on Reverse pharmacology. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate network pharmacology applications with the help of a video.	*demonstrate*	Procedural	Practical 10h
DOING — LEVEL 3 · MECHANISM — 2			
Practice Network Pharmacology.	*practice*	Procedural	Experiential 10h
Carry out a mini-project on network pharmacology.	*carry*	Procedural	Experiential 3h

Unit 2 · Reverse Pharmacology

8 obj · 28 h · Analysis / Mechanism

2.1 Introduction, Definition of Reverse Pharmacology, History, and Scope
2.2 Drug Discovery Current Scenario, Challenges, and Future Perspective
2.3 Reverse pharmacology for herbal drug-based modern medicines
2.4 Research updates on Reverse pharmacology

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Describe the definition and history of reverse pharmacology and discuss its Scope.	*describe*	Factual ◦	Lecture 1h
Elaborate current scenario, challenges, and future perspectives of reverse pharmacology.	*elaborate*	Conceptual ◦	Lecture 1h
Assess reverse pharmacology leverages traditional medicine and clinical experience to accelerate drug development. — teach at level 6, the verb *assess* is what will be asked	*assess*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Apply Reverse pharmacology for herbal-based modern medicines.	*apply*	Procedural ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss and analyze Research updates on Reverse pharmacology. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate reverse pharmacology applications with the help of a video.	*demonstrate*	Procedural	Practical 10h
DOING — LEVEL 3 · MECHANISM — 2			
Present observations of analysis of Reverse Pharmacology in selected cases.	*present*	Procedural	Experiential 10h
Validate Reverse pharmacology with a systems biology approach.	*validate*	Procedural	Experiential 3h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 2 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 28 · SEMESTER 4 · PAPER 4

Pharmaceutical Quality Assurance

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

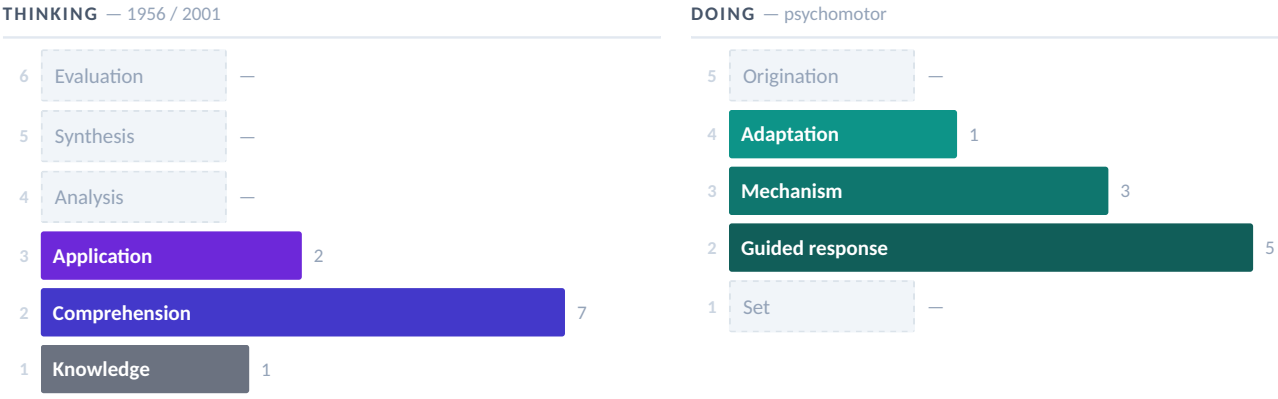
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	10	9	50
60 notional hours	19 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe Quality Assurance and Quality Management concepts	*describe*	L2 Comprehension / Understand
2	Explain Personnel Responsibilities	*explain*	L2 Comprehension / Understand
3	Discuss types of Containers, Packing materials	*discuss*	L2 Comprehension / Understand
4	Discuss Good Laboratory Practices	*discuss*	L2 Comprehension / Understand
5	Justify document maintenance in the pharmaceutical industry	*justify*	L6 Evaluation / Evaluate

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Quality Assurance and Quality Management concepts

6 obj · 19 h · Application / Guided response

1.1 Introduction to Quality Assurance and Quality Management concepts
1.2 Definition and concept of Quality control,
1.3 Quality assurance and GMP
1.4 Total Quality Management (TQM)
1.5 ICH Guidelines
1.6 Brief overview of Pharmaceutical industry guidelines for Quality, Safety, Efficacy and Multidisciplinary (QSEM)
1.7 Overview of ISO 9000 & ISO14000
1.8 Steps for registration of NABL accreditation

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define and elaborate the concept of Quality Assurance and Quality Management as per. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
Describe Total Quality Management (TQM) and ICH Guidelines, QSEM, and ISO 9000.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Illustrate the Steps for registration of NABL accreditation. — teach at level 2, the verb *illustrate* is what will be asked	*illustrate*	Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the process of mock GMP Inspection.	*demonstrate*	Procedural	Practical 5h
Demonstrate GMP Hierarchy and Responsibilities.	*demonstrate*	Procedural	Practical 5h
Identify the Root cause, affected personnel roles, and suggest corrective and preventive measures through Case Study Discussion on GMP Violation.	*identify*	Procedural	Experiential 6h

Unit 2 · Personnel Responsibilities

3 obj · 8 h · Comprehension / Adaptation

2.1 Organization and personnel
2.2 Personnel responsibilities, training, hygiene, and personal records
2.3 Premises: Design, construction, and plant layout, maintenance, sanitation, environmental control, utilities, and maintenance of sterile areas, control of contamination
2.4 Equipment and raw materials: Equipment selection, purchase specifications, maintenance, and purchase specifications and maintenance of stores for raw materials

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe Organizational structure and personnel management.	*describe*	Conceptual ◦	Lecture 1h
Discuss design, construction, and maintenance of Premises.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Document a personal record of pharmacy personnel.	*document*	Procedural	Experiential 6h

Unit 3 · Containers, Packing materials

4 obj · 13 h · Comprehension / Mechanism

3.1 Types of Packaging
3.2 Packaging materials
3.3 Container types and closures
3.4 Packaging and stability
3.5 Regulatory aspects
3.6 Quality Control
3.7 Smart Packaging

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss selection and maintenance of equipment, purchase specifications of raw materials, and maintenance of stores.	*discuss*	Conceptual ◦	Lecture 1h
Discuss Types of containers and packaging, Regulatory aspects, their Quality Control and Smart Packaging.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the identification and Classification of packaging materials.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Inspect Packaging Materials to check their compliance with quality specifications.	*inspect*	Procedural	Experiential 6h

Unit 4 · Good Laboratory Practices

3 obj · 8 h · Knowledge / Mechanism

4.1 GLP Principles and Requirements
4.2 GLP documentation
4.3 Regulatory compliance
4.4 Audit procedures

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Elaborate GLP with its Principles and Requirements, documentation, Regulatory compliance, Audit procedures. — teach at level 2, the verb *elaborate* is what will be asked	*elaborate*	Conceptual + Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate SOP Writing for different equipment.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Perform calibration of Analytical Instruments.	*perform*	Procedural	Experiential 2h

Unit 5 · Document maintenance in pharmaceutical industry

3 obj · 8 h · Application / Mechanism

4.1 Documentation related to Grievances and Redressal
4.2 Complaints and evaluation of complaints, Handling of returned goods, recalling, and waste disposal
4.3 Document maintenance in the pharmaceutical industry: Batch Formula Record, Master Formula Record, SOP, Quality audit, Quality Review, Quality documentation,

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define Complaints, their evaluation, product recall, and waste disposal. — teach at level 1, the verb *define* is what will be asked	*define*	Factual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe document maintenance in the pharmaceutical industry. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Conduct QA Documentation audit.	*conduct*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 4 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 5 · SEMESTER 5 · PAPER 1

Critical appraisal of Uparasa Dravya and Sadharana Rasa Dravya

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	2	10	14	50
60 notional hours	24 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	8	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe Uparasa and Sadharana Rasa Dravyas with their classical references.	*describe*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Principle and Processing of Uparasa Dravya

12 obj · 28 h · Application / Adaptation

1.1 Uparasa Dravyas mentioned in various texts
1.2 Characterization of Uparasa Dravyas
1.3 Pharmaceutical processes of Uparasa Dravya
1.4 Pharmacology and clinical application of Uparasa Dravya

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Describe he different Uparasa Dravyas mentioned in the classics.	*describe*	Conceptual ◦	Lecture 1h
Discuss the controversial drugs under Uparasa category.	*discuss*	Conceptual ◦	Lecture 1h
Discuss the mineralogical characterization of Uparasa.	*discuss*	Conceptual ◦	Lecture 1h
Discuss research updates on Uparasa drugs.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Recognise the therapeutic applications and toxico-kinetics of Uparasa Dravyas. — teach at level 1, the verb *recognise* is what will be asked	*recognise*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 5			
Demonstrate the authentication of Uparasa Dravyas.	*demonstrate*	Procedural	Practical 2h
Demonstrate Shodhana, Marana and other pharmaceutical procedures for Uparasa.	*demonstrate*	Procedural	Practical 4h
Demonstrate Quality Control testing for the products of Uparasa Dravya.	*demonstrate*	Procedural	Practical 4h
Perform Quality Control tests for the products of Uparasa Dravya.	*perform*	Procedural	Experiential 3h
Prepare formulations of Uparasa Dravya.	*prepare*	Procedural	Experiential 5h
DOING — LEVEL 3 · MECHANISM — 1			
Review articles on the toxicity and ADRs of Haratala and Manahshila and interpret.	*review*	Procedural	Experiential 2h
DOING — LEVEL 4 · ADAPTATION — 1			
Identify and authenticate Uparasa Dravyas.	*identify*	Procedural	Experiential 3h

Unit 2 · Principle and Processing of Sadharana Rasa

12 obj · 28 h · Evaluation / Adaptation

2.1 Sadharana rasa Dravyas mentioned in various texts
2.2 Characterization of Sadharana Rasa Dravyas
2.3 Pharmaceutical processes of Sadharana rasa Dravya
2.4 Pharmacology and clinical application of Sadharana rasa Dravya

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe and critically discuss the different Sadharana Rasa Dravyas mentioned in the classics.	*describe*	Conceptual ◦	Lecture 1h
Recognise the therapeutic applications and toxico-kinetics of Sadharana Rasa Dravyas. — teach at level 1, the verb *recognise* is what will be asked	*recognise*	Factual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Analyze the controversial drugs under Sadharana Rasa category. — teach at level 4, the verb *analyze* is what will be asked	*analyze*	Conceptual ◦	Lecture 1h
Discuss the mineralogical characterization of Sadharana Rasa. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Discuss research updates on Sadharana Rasa drugs. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 5			
Demonstrate authentication of Sadharana Rasa Dravya.	*demonstrate*	Procedural	Practical 2h
Demonstratie Shodhana, Marana and other pharmaceutical procedures for Sadharana.	*demonstrate*	Procedural	Practical 4h
Demonstrate Quality Control testing for the products of Sadharana Rasa Dravya.	*demonstrate*	Procedural	Practical 4h
Identify and authenticate Sadharana Rasa Dravyas.	*identify*	Procedural	Experiential 3h
Assemble the utility of Sadharana Dravyas in the clinical practice through a Field Survey of manufacturers and clinicians.	*assemble*	Procedural	Experiential 4h
DOING — LEVEL 4 · ADAPTATION — 2			
Perform Quality Control tests for the products of Sadharana Rasa Dravya.	*perform*	Procedural	Experiential 3h
Prepare and present the formulations of Sadharana Rasa Dravya.	*prepare*	Procedural	Experiential 3h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 8 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 6 · SEMESTER 5 · PAPER 1

Critical appraisal of Dhatu and Updhatu

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	2	10	14	50
60 notional hours	24 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

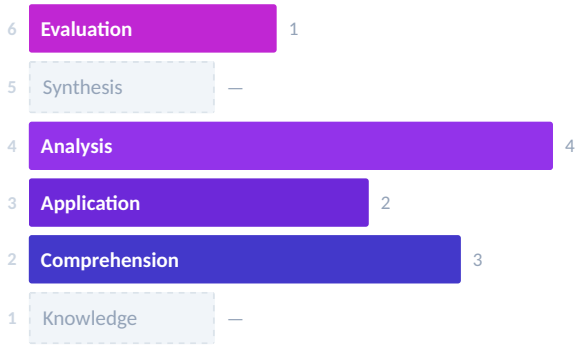
WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	8	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe classical references, properties, and classifications of Dhatu and Upadhatu.	*describe*	L2 Comprehension / Understand
2	Perform Shodhana, Marana, and other pharmaceutical procedures specific to Dhatu and Upadhatu Dravyas.	*perform*	L3 Application / Apply
3	Analyze physico-chemical characteristics of processed Dhatu and Upadhatu with scientific and classical correlation.	*analyze*	L4 Analysis / Analyse
4	Evaluate toxicological and therapeutic properties of commonly used Dhatu and Upadhatu formulations.	*evaluate*	L6 Evaluation / Evaluate

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Principle and Processing of Dhatu / Metals

12 obj · 28 h · Analysis / Mechanism

1.1 Dhatus mentioned in Rasa classics
1.2 Classification and characterization of metals
1.3 Pharmaceutical processing (Shodhana, Marana, etc.)
1.4 Therapeutic uses, toxicity, and safety of metallic preparations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe classical Dhatu Dravyas.	*describe*	Conceptual °	Lecture 1h
Discuss the controversial metals in Ayurveda texts.	*discuss*	Conceptual °	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Identify therapeutic applications and toxico-kinetics of Dhatu. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual °	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2			
Characterize Dhatu Dravyas mineralogically and pharmaceutically.	*characterize*	—	Lecture 1h
Compile and interpret research and regulatory updates on metal-based Ayurvedic drugs.	*compile*	Conceptual °	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 6			
Demonstrate the authentication of Dhatu Dravya.	*demonstrate*	Procedural	Practical 2h
Perform pharmaceutical processing (Shodhana & Marana) of metals.	*perform*	Procedural	Practical 4h
Demonstrate Quality control of metallic preparations.	*demonstrate*	Procedural	Practical 4h
Perform Quality assessment of Dhatu Bhasma.	*perform*	Procedural	Experiential 3h
Prepare formuations using Dhatu Bhasma.	*prepare*	Procedural	Experiential 5h
Identify and analyze Adverse drug reactions (ADRs) due to metals.	*identify*	Procedural	Experiential 2h
DOING — LEVEL 3 · MECHANISM — 1			
Identify and document classical Dhatus.	*identify*	Procedural	Experiential 3h

Unit 2 · Principle and Processing of Upadhatu

12 obj · 28 h · Evaluation / Adaptation

2.1 Classical analysis of Upadhatu

2.2 Classification, sources, and characteristics of Upadhatu

2.3 Pharmaceutical processing (Shodhana, Marana, etc.)of Upadhatu

2.4 Pharmacological actions and clinical application of Upadhatu

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe and classify Upadhatu from Rasashastra references.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss controversial and newly interpreted Upadhatu. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2			
Analyze Upadhatu mineralogically and pharmaceutically.	*analyze*	Conceptual ◦	Lecture 1h
Review and interpret findings of research and current practices involving Upadhatu.	*review*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Recognize therapeutic applications and possible ADRs of Upadhatu. — teach at level 1, the verb *recognize* is what will be asked	*recognize*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 6			
Demonstrate the Identification of Upadhatu samples.	*demonstrate*	Procedural	Practical 2h
Demonstrate Shodhana and Marana of Upadhatu.	*demonstrate*	Procedural	Practical 4h
Demonstrate the Quality control tests for sub-metallic formulations.	*demonstrate*	Procedural	Practical 4h
Identify the Upadhatu samples.	*identify*	Procedural	Experiential 3h
Perform Quality Control tests for the products of Upadhatu.	*perform*	Procedural	Experiential 3h
Analyze and interpret findings of the utility of Dhatu and Upadhatu in clinical practice through a Field survey with Manufacturers or Vaidyas.	*analyze*	Procedural	Experiential 4h
DOING — LEVEL 4 · ADAPTATION — 1			
Prepare Upadhatu formulations.	*prepare*	Procedural	Experiential 3h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 8 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 13 · SEMESTER 5 · PAPER 2

Krutanna (Ayurvedic Food Preparations)

Exposition of Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	4	5	10	50
60 notional hours	15 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	5	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Classification of food substances and Formulations

3 obj · 12 h · Comprehension / Guided response

1.1 Ayurvedic dietary principles and Ahara Dravya Varga (Groups of food substance)

1.2 Aahara -mahabhaishajya: Significance of food substances used as medicine

1.3 Unique classical formulations prepared using Food substances belonging to various groups and their utility in treatment

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe Ayurvedic dietary principles and the significance of food substances used as medicine.	*describe*	Conceptual	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the method of preparation of distinctive formulations using Aahara Dravya from different groups (Vargas).	*demonstrate*	Procedural	Practical 4h
Prepare distinctive formulations using Aahara Dravya from different groups (Vargas).	*prepare*	Procedural	Experiential 6h

Unit 2 · Customized food for treatment

6 obj · 24 h · Comprehension / Guided response

2.1 Disease-specific Ayurvedic Aahara- Therapeutic utility

2.2 Samsarjana krama- Manda, Peya, Vilepi, Yavagu, Yusha, Odana, Krushara, Mansarasa- Process validation Protocol for all formulations

2.3 Personalised prescription of Aushada Siddha Aahara for Shamana and during panchakarma

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe disease-specific diet formulations.	*describe*	Conceptual ◦	Lecture 2h
Recite paribhasha and describe the importance of Yusha, Yavagu, Odana, and Mamsarasa. — teach at level 1, the verb *recite* is what will be asked	*recite*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate preparation of therapeutic diet formulations.	*demonstrate*	Procedural	Practical 4h
Identify and prepare disease specific Ayurvedic Ahara.	*identify*	Procedural	Experiential 6h
Demonstrate the method of preparation of various Aahara Kalpana.	*demonstrate*	Procedural	Practical 4h
Develop the disease-specific Personalised prescription of Aushadhi Siddha Ahara.	*develop*	Procedural	Experiential 6h

Unit 3 · Ayurvedic Food Industry

3 obj · 10 h · Knowledge / Guided response

3.1 Market survey about Ayurvedic food products and research updates

3.2 Global needs and designing of modified food formulations

3.3 Semi-processed food/ semi-cooked food, RTC- ready to cook / RTD-Ready to drink Foods and drinks

3.4 Regulation Standards for Ayurvedic Food Preparation (Ayurveda Aahara Regulations- 2021), FSSAI Regulations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 1 · KNOWLEDGE / REMEMBER — 1			
Describe the regulation standards for Ayurvedic food preparations- FSSAI (Ayurveda Aahara regulations). — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of novel Ayurvedic food products.	*demonstrate*	Procedural	Practical 4h
Collect the data and interpret the findings through a market survey to explore the available Ayurvedic food products and assess the current trends in the Ayurvedic food industry.	*collect*	Procedural	Experiential 4h

Unit 4 · Food processing technology traditional and modern methods

3 obj · 10 h · Analysis / Adaptation

4.1 Traditional Methods- Steaming, Boiling, Drying, Salting, Smoking, Frying,

4.2 Modern Methods- Heating, Freeze Drying, Vacuum Drying, Canning, Juicing, Fermentation, Modified atmosphere packaging, Pasteurisation, Smoking, Pulse electric

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Differentiate between classical and contemporary food processing techniques and support the current and emerging trends in the Ayurvedic food industry.	*differentiate*	Procedural ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate various food processing techniques used in current Ayurvedic food Industries.	*demonstrate*	Procedural	Practical 4h
DOING — LEVEL 4 · ADAPTATION — 1			
Analyze Functional Food and Modified Food Formulations in Ayurvedic Food Industry.	*analyze*	Procedural	Experiential 4h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 5 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 14 · SEMESTER 5 · PAPER 2

Ayurvedic Formulations for Holistic Wellbeing, Vitality and Body Care

Exposition of Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	2	4	7	50
60 notional hours	11 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	4	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Rasayana-Personalized Rejuvenation Formulations

5 obj · 24 h · Comprehension / Mechanism

1.1 Key aspects of Rasayana Formulations

1.2 General Rasayana and Organ-specific/Target-specific Rasayana- Varnya, Tvachya, Kanthya, Swarya, Keshya, Hrudya, Medhya- Formulations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe the mode of action of Rasayana Formulations.	*describe*	Conceptual ◦	Lecture 2h
Describe classical Rasayana and Vajikarana Formulations.	*describe*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the role of Ayurvedic Rasayana formulations in the maintenance of health and Vajikarana Formulations for better progeny(Suprajanana).	*demonstrate*	Procedural	Experiential 5h
Select appropriate Ayurvedic Formulations for the fitness of specific body parts and analyze their role in the maintenance of health.	*select*	Conceptual + Procedural ◦	Experiential 7h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate the preparation method of Rasayana Drugs following validated SOP.	*demonstrate*	Procedural	Practical 8h

Unit 2 · Ayurvedic based cosmetic preparation

6 obj · 32 h · Evaluation / Mechanism

2.1 Body/ Hair/ Lip/ Oral/Skin/ Foot care products- Preparation methods and application in Ayurvedic

2.2 Preparation process of Lip balm, lipsticks, shampoos, cold cream and vanishing cream, toothpaste, hair dyes, Sunscreen lotion, Face pack, Mask, Toners, herbal soaps,

2.3 Current and Emerging trends – Ayurvastram, Ayurvedic footwear, Eye mask

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the role of Ayurvedic formulations in Cosmetic preparations.	*discuss*	Conceptual	Lecture 3h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evaluate the Global need for body care products in the healthcare sector.	*evaluate*	Conceptual ◦	Lecture 3h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the preparation of Modified Cosmeceuticals.	*demonstrate*	Procedural	Practical 6h
Analyze different cosmetics available in the market.	*analyze*	Procedural	Experiential 7h
Design and prepare novel Ayurvedic cosmeceuticals.	*design*	Procedural	Experiential 7h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate preparation of Herbal Cosmetics.	*demonstrate*	Procedural	Practical 6h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 4 lecture, 3 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 21 · SEMESTER 5 · PAPER 3

Clinical application of Rasoushadhi according to Srotas and in selected challenging clinical conditions

Standardization and clinical Utility of Ayurvedic formulations

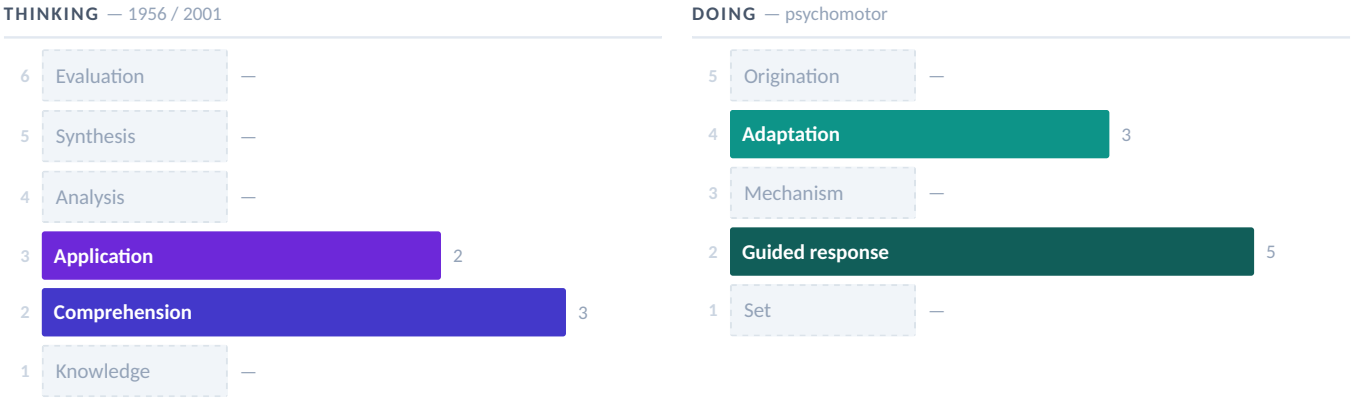
CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	3	5	8	25
30 notional hours	13 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Clinical application of Rasoushadhis specific to Srotas

4 obj · 8 h · Application / Adaptation

1.1 Concept and rationality of selection and application of Rasoushadhi according to Srotas

1.2 Clinical application of Rasouashdhi in disorders of different Srotas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss Srotas-specific Rasoushadhi, tailored Ayurvedic formulations for individual biological channels.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Enlist and discuss Rasoushadhi for different Srotas, explaining their therapeutic efficacy and mechanism of action. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the clinical application of Rasoushadhi in selected Srotas, illustrating their practical utility and therapeutic outcomes in addressing specific health conditions.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Illustrate the therapeutic application of Rasoushadhi in different Srotas.	*illustrate*	Procedural	Experiential 3h

Unit 2 · Clinical application of Rasoushadhi in emergency conditions

4 obj · 9 h · Comprehension / Adaptation

2.1 Role of Rasoushadhis in Managing Medical Emergencies

2.2 Specific Rasa Yogas among Chaturvidha Rasayana kalpas for Managing Medical Emergencies

2.3 Application of Pottali Kalpa in Emergency Management

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the unique properties of Rasoushadhi, enabling its use in clinical emergencies.	*discuss*	Conceptual ◦	Lecture 1h
Enlist specific Rasa Yogas for managing medical emergencies and discuss their rationality, highlighting the therapeutic mechanisms and clinical applications. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the clinical applicability of Rasoushadhi in Medical Emergencies.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Apply the appropriate Rasa Oushadhi in a alloted /selected Medical Emergency conditions.	*apply*	Procedural	Experiential 4h

Unit 3 · Rasoushadhi in specific, clinically challenging conditions

5 obj · 11 h · Application / Adaptation

3.1 Rational application of Rasoushadhi in cancer

3.2 Application of Rasoushadhi in neurodevelopmental conditions

3.3 Utility of Rasoushadhi in neurodegenerative conditions

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the role of Rasoushadhi in clinically challenging conditions like cancers neurodevelopmental and neurodegenerative disorders. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the role of Rasoushadhi in clinically challenging conditions.	*demonstrate*	Procedural	Practical 2h
Demonstrate the role of Rasoushadhi as Rasayana in clinically challenging conditions.	*demonstrate*	Procedural	Practical 2h
Evalaute the role of Rasoushadhi in clinically challenging conditions such as Cancer.	*evaluate*	Conceptual + Procedural ◦	Experiential 3h
DOING — LEVEL 4 · ADAPTATION — 1			
Explore the role of Rasoushadhi in clinically challenging conditions such as neurodevelopmental and neurodegenerative diseases.	*explore*	Conceptual + Procedural ◦	Experiential 3h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 22 · SEMESTER 5 · PAPER 3

Yogas from different Kalpanas

Standardization and clinical Utility of Ayurvedic formulations

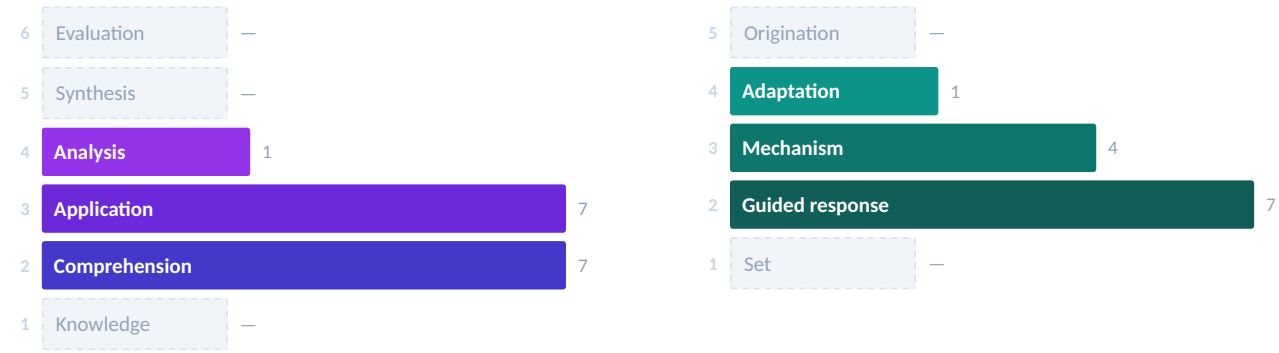
CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	5	15	12	75
90 notional hours	27 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	15	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	6	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · SOP of Vati, Gutika, Guggulu and Varti Kalpas and their therapeutic applications

6 obj · 20 h · Application / Mechanism

1.1 SOP of selected Vati preparation including packing,
1.2 SOP regarding preparation of selected formulations from Guggulu Kalpana, including packing, storage, and therapeutic applications
1.3 Types of Varti and their SOP
1.4 Clinical applications of different Varti
1.5 Suppositories and pharmaceutical modifications of Ayurvedic suppositories

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss modern aspects of solid dosage forms.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Describe the therapeutic significance of vati kalpana. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
Summarize the therapeutic significance of Guggulu Kalpa and Varti Kalpana. — teach at level 2, the verb *summarize* is what will be asked	*summarize*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of Vati, Guggulu Kalpa, and their modified modern dosage forms.	*demonstrate*	Procedural	Practical 5h
Design an S.O.P for the preparation of Ayurvedic suppository specifying its clinical utility based on Varti Kalpana.	*design*	Procedural	Practical 4h
DOING — LEVEL 3 · MECHANISM — 1			
Analyse the role of Gugulu Kalpana in various clinical conditions by selecting a specific Gulggulu Yoga.	*analyse*	Conceptual + Procedural ◦	Experiential 8h

Unit 2 · SOP and Clinical applications of Avaleha and Khanda kalpana

5 obj · 16 h · Application / Mechanism

2.1 SOP of Avaleha preparation with suitable Leha yogas
2.2 Therapeutic application of different varieties of Avaleha yogas
2.3 SOP of Khanda kalpana with suitable Khanda Yogas
2.4 Therapeutic application of different varieties of Khanda yogas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss modern semisolid dosage forms.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the therapeutic significance of Avaleha kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Discuss therapeutic significance of Paka and Khanda kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the role of pharmaceutical additives and Samyak Paka of Avaleha.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Analyse the role of selected leha yoga in different clinical conditions.	*analyse*	Conceptual + Procedural ◦	Experiential 8h

Unit 3 · SOP of Sneha Kalpana and its clinical significance

5 obj · 17 h · Application / Adaptation

3.1 Different categories of Sneha with special reference to therapeutic application
3.2 Relevance of Avarthita Taila, Taila murchana and Taila Paka with special reference to critical endpoints
3.3 SOP of Sneha kalpana in selected formulations

3.4 Therapeutic application of selected Ghrita yogas and Taila yogas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify the similarities of Sneha Kalpa with liposomal drug delivery system. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the concept of Sneha Kalpana with special reference to critical end points (Sneha Paka) and its pharmaceutical and therapeutical significance. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Discuss and compare the significance of Murchana of Katu Taila, Tila Taila, Eranda Taila and Ghrita with special reference to analytical profile and pharmacotherapeutics. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate S.O.P for a selected Sneha Kalpa which is widely used and determine its analytical profile.	*demonstrate*	Procedural	Practical 6h
DOING — LEVEL 4 · ADAPTATION — 1			
Analyze different stages of Sneha Paka.	*analyze*	Procedural	Experiential 8h

Unit 4 · SOP and clinical applications of Arka and Dravakam in various disorders

5 obj · 14 h · Analysis / Mechanism

4.1 SOP for preparing selected Arkas
4.2 Analytical profiling of various yogas from Arka Kalpana
4.3 SOP of preparing various Dravakas
4.4 Analytical Profiling of selected formulations of Dravaka
4.5 Therapeutic applications and method of administration of Dravaka

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Desribe therapeutic significance of Arka Kalpana and Dravaka.	*describe*	Conceptual	Lecture 1h
Discuss different distillation techniques used in pharmacy practice.	*discuss*	Procedural ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Elaborate Dravaka Kalpa with a critical correlation with acidic and alkaline preparations. — teach at level 2, the verb *elaborate* is what will be asked	*elaborate*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate S.O.P for the preparation of a selected Arka and Dravaka and create their analytical profile.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Illusrate the clinical application of selected Arka/Dravaka Kalpa.	*illustrate*	Procedural	Experiential 6h

Unit 5 · SOP and therapeutic applications of Sandhana Kalpana

6 obj · 17 h · Application / Mechanism

5.1 Different and novel fermentation techniques
5.2 SOP of preparing different types of Formulations in Sandhana Kalpana
5.3 Therapeutic applications of selected yogas from Sandhana Kalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe chemistry behind classical fermentation techniques.	*describe*	Procedural	Lecture 1h
Discuss industrial fermentation techniques.	*discuss*	Procedural	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Elaborate therapeutic significance of Sandhana kalpana. — teach at level 2, the verb *elaborate* is what will be asked	*elaborate*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate S.O.P for preparing a selected sandhana Yoga and determine the analytical standards.	*demonstrate*	Procedural	Practical 5h
Record the dose, duration, indications, and clinical outcome of the various formulation i.e. Sandhana Kalpanas in the prescribed patients.	*record*	Procedural	Experiential 4h
DOING — LEVEL 3 · MECHANISM — 1			
Perform Comparative analysis of Asava Arishta Samples having different percentage and types of sugar.	*perform*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 15 lecture, 6 practical, 6 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 29 · SEMESTER 5 · PAPER 4

Pharmacy Practice

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	3	10	8	50
60 notional hours	18 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the Hospital Pharmacy and the Community Pharmacy with their organization	*describe*	L2 Comprehension / Understand
2	Discuss Therapeutic Drug Monitoring	*discuss*	L2 Comprehension / Understand
3	Discuss the Pharmacy and Therapeutic Committee	*discuss*	L2 Comprehension / Understand
4	Enlist the emergency drug list	*enlist*	L1 Knowledge / Remember

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Hospital Pharmacy and Community Pharmacy

7 obj · 26 h · Comprehension / Mechanism

1.1 Hospital and its organization
1.2 Definition, Classification of Hospital
1.3 Hospital pharmacy and its organization
1.4 Functions of hospital pharmacy
1.5 Community Pharmacy
1.6 Organization and structure of retail and wholesale drug stores

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Define and classify Hospital Pharmacy, and discuss its organization. — teach at level 1, the verb *define* is what will be asked	*define*	Factual ◦	Lecture 1h
Describe Community Pharmacy.	*describe*	Conceptual ◦	Lecture 1h
Discuss the Organization and structure of retail and wholesale drug stores.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate Drug Information and Case Analysis.	*demonstrate*	Procedural	Practical 5h
Demonstrate Prescription Handling and Dispensing.	*demonstrate*	Procedural	Practical 5h
Identify Adverse drug reaction in selected case.	*identify*	Procedural	Experiential 7h
DOING — LEVEL 3 · MECHANISM — 1			
Appraise about Basic Health Screening at community setup.	*appraise*	Procedural	Experiential 6h

Unit 2 · Therapeutic Drug Monitoring

7 obj · 17 h · Evaluation / Adaptation

2.1 Drug distribution system in a hospital
2.2 Dispensing of drugs to inpatients, types of drug distribution systems, charging policy and labelling, dispensing of drugs to ambulatory patients, and dispensing of
2.3 Hospital formulary
2.4 Therapeutic drug monitoring
2.5 Factors to be considered during the Therapeutic Drug Monitoring
2.6 Medication adherence
2.7 Patient medication history interview
2.8 Community pharmacy management
2.9 Financial, materials, staff, and infrastructure requirements

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss about Hospital formulary and Therapeutic drug monitoring.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 3			
Describe the Drug distribution system in a hospital. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
Discuss Dispensing of drugs to inpatients, types of drug distribution systems, charging policy, and labelling, dispensing of drugs to ambulatory patients, and dispensing of controlled drugs. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Analyse Financial, materials, staff, and infrastructure requirements. — teach at level 4, the verb *analyse* is what will be asked	*analyse*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evaluate Medication Adherence.	*evaluate*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Perform Data Interpretation & Pharmacokinetic Calculations.	*perform*	Procedural	Practical 5h
DOING — LEVEL 4 · ADAPTATION — 1			
Discuss Multidisciplinary Integration for Therapeutic Drug Monitoring.	*discuss*	Procedural	Experiential 7h

Unit 3 · Pharmacy and therapeutic committee

4 obj · 13 h · Comprehension / Origination

3.1 Introduction to Pharmacy and Therapeutic Committee

3.2 Drug information services

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe Pharmacy and Therapeutic Committee along with Drug Information Services.	*describe*	Conceptual ◦	Lecture 1h
Enlist emergency drugs used in pharmacy. — teach at level 1, the verb *enlist* is what will be asked	*enlist*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate Formulary Review and Management.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 5 · ORIGINATION — 1			
Appraise and. Develop Standard Therapeutic Guidelines.	*appraise*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Origination on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 4 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Origination.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 30 · SEMESTER 5 · PAPER 4

Pharmaceutical Microbiology

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

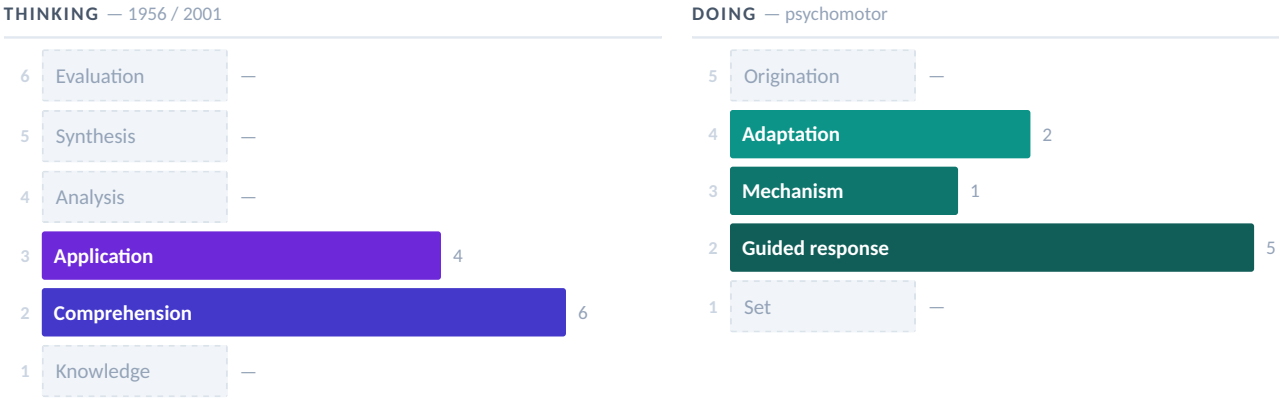
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	3	10	8	50
60 notional hours	18 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Introduction to microbes with applications of sterilization

5 obj · 12 h · Application / Guided response

1.1 Identification of bacteria using staining techniques (simple, Gram's & Acid-fast

1.2 Study of principles, procedures, merits, demerits, and applications of physical, chemical, gaseous, radiation, and mechanical methods of sterilization

1.3 Evaluation of the efficiency of sterilization methods

1.4 Equipment employed in large-scale sterilization and Sterility indicators

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define and classify microorganisms (microbes) and explain their general characteristics. — teach at level 1, the verb *define* is what will be asked	*define*	Factual ◦	Lecture 1h
Describe the concept of sterilization, its importance, and differentiate various sterilization methods.	*describe*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the beneficial and harmful roles of microbes in health, industry, and the environment. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate dry heat sterilization method.	*demonstrate*	Procedural	Practical 3h
Conduct dry heat sterilization.	*conduct*	Procedural	Experiential 6h

Unit 2 · Sources of contamination

6 obj · 12 h · Application / Adaptation

2.1 Designing an aseptic area

2.2 Study of different sources of contamination in an aseptic area and methods of prevention, clean area classification

2.3 Principles and methods of different microbiological assay methods

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define contamination and explain its significance in aseptic and cleanroom environments. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
Discuss the impact of contamination and describe methods used to minimize or prevent contamination.	*discuss*	Procedural ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Recognise common sources of contamination in pharmaceutical and microbiological settings and discuss each source contributes to microbial and particulate contamination. — teach at level 1, the verb *recognise* is what will be asked	*recognise*	Factual ◦	Lecture 1h
Discuss the principles and methods of different microbiological assay methods. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual + Procedural ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the Aseptic Technique Practice for the transfer of microbes.	*demonstrate*	Procedural	Practical 2h
DOING — LEVEL 4 · ADAPTATION — 1			
Perform Simulation for Aseptic Practices.	*perform*	Procedural	Experiential 6h

Unit 3 · Types of spoilage and use of antimicrobial agents

7 obj · 32 h · Application / Adaptation

3.1 Types of spoilage, factors affecting the microbial spoilage of pharmaceutical products,

3.2 Preservation of pharmaceutical products using antimicrobial agents, evaluation of microbial stability of formulations

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Define spoilage, explain its significance in pharmaceuticals, food, and cosmetics, and discuss the factors that contribute to spoilage. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
Describe the signs of spoilage.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Classify different types of antimicrobial agents (e.g., preservatives, disinfectants antibiotics) based on their mode of action and explain their antimicrobial mechanisms. — teach at level 2, the verb *classify* is what will be asked	*classify*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate Personnel Contamination Assessment.	*demonstrate*	Procedural	Practical 7h
Demonstrate Different Types of Spoilage.	*demonstrate*	Procedural	Practical 8h
DOING — LEVEL 3 · MECHANISM — 1			
Conduct Real-life case studies on spoilage outbreaks and preservative failures.	*conduct*	Procedural	Experiential 7h
DOING — LEVEL 4 · ADAPTATION — 1			
Observe and detect Spoilage of Pharmaceutical Product in Real Time.	*observe*	Procedural	Experiential 7h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 4 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 7 · SEMESTER 6 · PAPER 1

Sudha Varga, Sikata Varga, Visha Varga Dravyas

Rasashastra (Iatrochemistry) — Principles and Practices

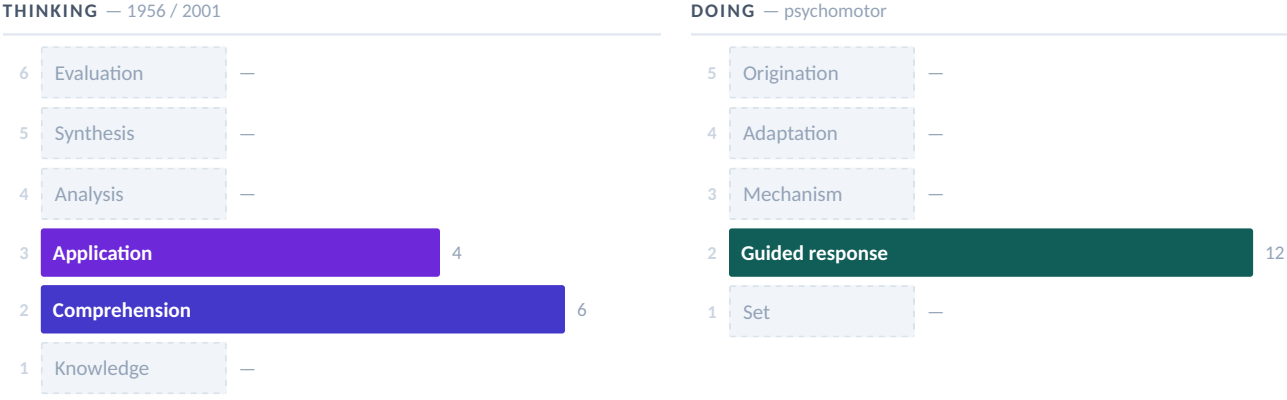
CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	3	10	12	75
90 notional hours	22 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	10	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	6	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Identify and classify the drugs of Sudha Varga, Sikata Varga, and Visha Varga.	*identify*	L1 Knowledge / Remember
2	Perform pharmaceutical processes, including Shodhana and Marana, for Sudha Varga and Sikata Varga drugs, and Shodhana for Visha Varga drugs.	*perform*	L3 Application / Apply
3	Describe the physico-chemical properties of Sudha Varga, Sikata Varga, and Visha Varga drugs.	*describe*	L2 Comprehension / Understand
4	Discuss the pharmacology and therapeutic applications of Sudha Varga, Sikata Varga, and Visha Varga drugs.	*discuss*	L2 Comprehension / Understand

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Characterization, Pharmaceutical Processing, and Clinical Applications of Sudha Varga Dravya

7 obj · 28 h · Application / Guided response

1.1 Classification and Identification of Sudha
1.2 Pharmaceutical Processing of Sudha Varga Dravya
1.3 Quality Control Methods for Sudha Varga Dravya
1.4 Pharmacology, Therapeutic Applications, and Research Updates on Sudha Varga Dravya

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the list of Sudha varga Dravya mentioned in various classics and their mineralogical correlation.	*discuss*	Factual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the pharmacological and therapeutic applications of Sudha Varga Dravya. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 2h
Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sudhavarga Dravya. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya.	*demonstrate*	Procedural	Practical 6h
Demonstrate the quality testing procedures for Bhasma/Pishti of Sudha vargeeya Dravya.	*demonstrate*	Procedural	Practical 4h
Perform Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya.	*perform*	Procedural	Experiential 7h
Perform quality testing procedures for Bhasma/Pishti of Sudha Vargeeya Dravya.	*perform*	Procedural	Experiential 6h

Unit 2 · Characterization, pharmaceutical processing, and clinical applications of Sikata Varga Dravya

7 obj · 28 h · Application / Guided response

2.1 Identification of Sikata Varga dravyas
2.2 Shodhana, marana, and other pharmaceutical processes of Sikata varga Dravya as per the SOP laid down
2.3 Quality control methods for the drugs of Sikata Varga Dravya
2.4 Pharmacology, therapeutic application, and research updates on Sikata Varga Dravya)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss the list of Sikata varga Dravya mentioned in various classics and their mineralogical correlation.	*discuss*	Factual ◦	Lecture 1h
Discuss research updates on characterization, pharmaceutical processing, QC, and pharmacotherapeutics of Sikata Varga Dravya.	*discuss*	Conceptual ◦	Lecture 2h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the pharmacology and therapeutic applications of Sikata Varga Dravya. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate Shodhana, marana, and Pishti Nirmana of Sikata varga Dravya.	*demonstrate*	Procedural	Practical 6h
Demonstrate quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya.	*demonstrate*	Procedural	Practical 4h
Perform Shodhana, marana and Pishti Nirmana of Sikata varga Dravya.	*perform*	Procedural	Experiential 7h
Perform quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya.	*perform*	Procedural	Experiential 6h

Unit 3 · Pharmaceutical processing, phytopharmacology and clinical applications of Visha Varga dravya

8 obj · 28 h · Application / Guided response

3.1 Drugs of Visha and Upavisha categories
3.2 Shodhana of Visha-Upavisha Dravya based on different references
3.3 Phytochemical changes after the process of Shodhana
3.4 Clinical application and therapeutics of Visha Varga

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Discuss and compare the drugs of the Visha and Upavisha categories from different classics.	*discuss*	Conceptual ◦	Lecture 1h
Describe the identification of herbal, animal, and mineral drugs mentioned under the Visha-Upavisha category based on organoleptic and physico-chemical characters.	*describe*	Conceptual ◦	Lecture 1h
Discuss the phytopharmacology of herbal drugs mentioned under the Visha-Upavisha category.	*discuss*	Conceptual ◦	Lecture 2h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe formulations and therapeutic applications of Visha Dravya. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate the Shodhana Process of Visha Dravya.	*demonstrate*	Procedural	Practical 6h
Demonstrate the analytical testing process for Shodhita Visha dravya.	*demonstrate*	Procedural	Practical 4h
Perform Shodhana of Visha Dravyas as per different classics.	*perform*	Procedural	Experiential 7h
Perform analytical testing process for Shodhita Visha dravya.	*perform*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Guided response on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 10 lecture, 6 practical, 6 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Guided response.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 8 · SEMESTER 6 · PAPER 1

Appraisal of Ratna and Uparatna and therapeutic significance of Pishti and Druti Kalpana

Rasashastra (Iatrochemistry) — Principles and Practices

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	2	5	8	25
30 notional hours	13 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	4	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Identify and explore precious and semiprecious stones classified under Ratna and Uparatna Varga.	*identify*	L1 Knowledge / Remember
2	Prepare Pishti Kalpa adhering to Standard Operating Procedures (SOPs).	*prepare*	L3 Application / Apply
3	Prepare and pharmaceutically modify Gandhaka Druti and Gandhaka Taila Kalpas.	*prepare*	L3 Application / Apply
4	Analyze the pharmacology and therapeutic applications of selected Pishti, Gandhaka Druti, and Gandhaka Taila Kalpas.	*analyze*	L4 Analysis / Analyse

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Mineralogy, Crystal chemistry,and Pharmacotherapeutics of Ratna-Uparatna Varga

7 obj · 15 h · Comprehension / Mechanism

1.1 Precious stones
1.2 Uparatna and semiprecious stones
1.3 Crystal chemistry and mineralogical properties of Ratna-Uparatna
1.4 Pharmacology and therapeutic applications of Ratna and Uparatnas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 3			
Discuss the characteristics of precious stones with special reference to the mineralogical identification of Ratnas.	*discuss*	Conceptual ◦	Lecture 1h
Describe different semiprecious stones mentioned in Rasashastra texts under the category of Uparatna.	*describe*	Conceptual ◦	Lecture 1h
Discuss crystal chemistry, pharmacology and therapeutic applications of Ratna and Uparatnas.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate mineralogical identification of precious and semiprecious stones under Ratna and Uparatna varga.	*demonstrate*	Procedural	Practical 3h
Demonstrate SOP of Pishti Kalpana of Ratna-Uparatna.	*demonstrate*	Procedural	Practical 3h
Identify the precious and semiprecious stones under Ratna and Uparatna varga using keys for mineralogical identification.	*identify*	Procedural	Experiential 3h
DOING — LEVEL 3 · MECHANISM — 1			
Identify substitutes for Ratnavarga and Uparatna varga drugs.	*identify*	Procedural	Experiential 3h

Unit 2 · SOP and therapeutic applications of Pishti, Gandhaka Druti and Gandhaka Taila kalpana

6 obj · 13 h · Application / Guided response

2.1 Pishti Kalpana: Concept, SOP, and Therapeutic
2.2 Classical concept of Druti
2.2 Gandhaka Druti and Gandhaka Taila Kalpana: Concept, SOP, and Therapeutic Applications
2.3 Pharmaceutical Modifications of Gandhaka Druti and Gandhaka Taila

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the concept, significance, and therapeutic value of Pishti kalpas. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Discuss the concept of Druti mentioned in Rasa classics and explore the therapeutic value of Gandhaka Druti and Gandhaka Taila. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 4			
Demonstrate different methods of Gandhaka Druti.	*demonstrate*	Procedural	Practical 2h
Demonstrate Pharmaceutical modification of Gandhaka Druti and Gandhaka Taila.	*demonstrate*	Procedural	Practical 2h
Prepare Pishti of Selected Ratna or Uparatna.	*prepare*	Procedural	Experiential 3h
Prepare Gandhaka Druti using different media.	*prepare*	Procedural	Experiential 4h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 4 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 15 · SEMESTER 6 · PAPER 2

Disease condition specific selection of formulation

Exposition of Ayurvedic Pharmaceutics

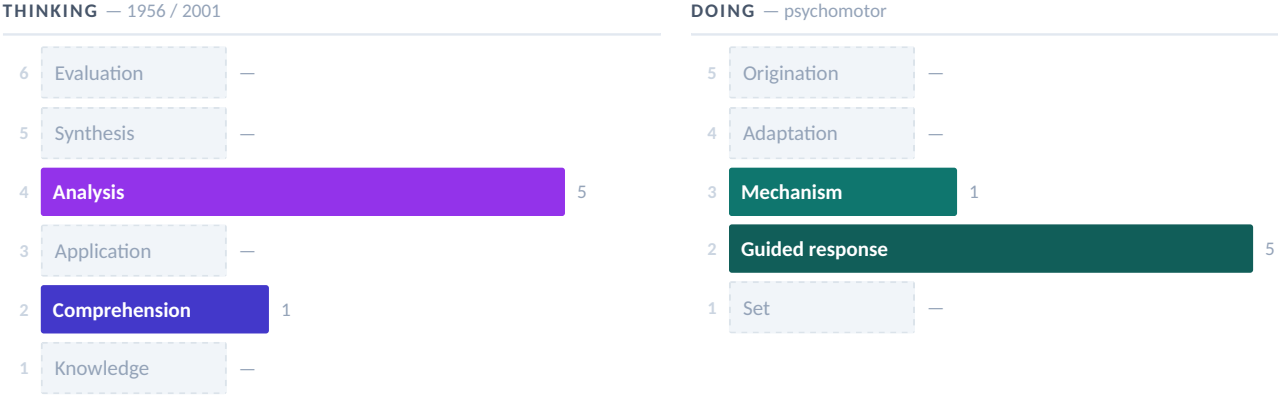
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	5	6	6	50
60 notional hours	12 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	6	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Application of Yukti Pramana in Formulation Selection

2 obj · 2 h · Analysis / —

1.1 Disease Category and Determination of multiple factors involved in diagnosis and

1.2 Strategic selection of drugs and dosage forms

1.3 Stage-specific selection (Rogavastha and Bhaishajya Kalpana)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 2			
Appraise the rational relationship between Roga and Bhaishajya Kalpana based on classical references. — teach at level 6, the verb *appraise* is what will be asked	*appraise*	Conceptual	Lecture 1h
Analyse and justify the selection of formulation according to the Rogavastha (stage of disease).	*analyse*	Conceptual	Lecture 1h

Unit 2 · Selection of formulations for Speciality practice in Ayurveda

3 obj · 12 h · Analysis / Guided response

2.1 Speciality practice in Ayurveda

2.2 Alignment of formulations with the specialty branch of clinical Practice

2.3 Drugs with ideal guna, virya, vipaka, and of appropriate dosage forms for the patient and pathology

2.4 Evidence-based drugs and dosage form selection for specialty practice

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Describe different branches /specialty practices in Ayurveda and justify the need for specific dosage forms for effective clinical outcomes. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the essential formulations needed for the diverse branches of Ayurveda.	*demonstrate*	Procedural	Practical 5h
Identify different dosage forms used by local health practitioners. through filed based survey.	*identify*	Procedural	Practical 5h

Unit 3 · Routes of Drug Administration

2 obj · 12 h · Analysis / Mechanism

3.1 Classical and conventional concepts of Drug Administration

3.2 Routes of Drug Administration – Need and Relevance

3.3 Factors in the selection of drug and dosage form based on route of administration

3.4 Matching Dosage Forms to Pharmacological Needs by Selected Route

3.5 Properties of formulations and their effect on Pharmacodynamics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Analyze the formulations and routes of drug administration, and justify the relevance of each route in pharmacology.	*analyze*	Conceptual ◦	Lecture 2h
DOING — LEVEL 3 · MECHANISM — 1			
Analyze the key characteristics of dosage forms that meet the needs of various routes of drug administration.	*analyze*	Procedural	Experiential 10h

Unit 4 · Designing formulations for Shodhana and Shamana Chikitsa

2 obj · 12 h · Analysis / Guided response

4.1 Ideal characteristics of drugs as per classics (Matravat Bheshaja lakshanani)

4.2 Principles of Shodhana chikitsa (Purificatory/detoxifying therapy) and actions of formulations used for shodhana - Langhana, Doshanirhana, srotahshodhana

4.3 Principles of Shamana chikitsa (palliative treatment) and actions of formulations used for Shamana- Deepana, Pachana, Stambhana, Rasayana, etc

4.4 Evidence-based dosage forms and formulations for common diseases

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Compare dosage forms and formulations used for shodhana and shamana chikitsa.	*compare*	Conceptual ◦	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the method of prepartion of frequently used formulation for Shodhana and Shamana Chikitsa of common diseases.	*demonstrate*	Procedural	Practical 10h

Unit 5 · Innovation in Ayurvedic Formulations

3 obj · 18 h · Comprehension / Guided response

5.1 Need and current trends in the industry of Ayurvedic Pharmaceutics

5.2 Innovative Formulations-Transdermal patches, Nebulizers, Eye masks, etc

5.3 Relevance and consequences of innovation done by the R&D department of Ayurvedic Pharmacies

5.4 Knowledge, Attitude, and Practice (KAP) studies on various aspects related to Ayurvedic Formulations available in the market

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss challenges associated with recent modifications and innovations in Ayurvedic formulations, specifically focusing on transdermal patches and nebulizers.	*discuss*	Conceptual	Lecture 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the method of nnovative formulations that enhance well-being and redefine healthcare.	*demonstrate*	Procedural	Experiential 10h
Demonstrate the essential competencies needed to develop R&D department of Ayurvedic Pharmacy.	*demonstrate*	Procedural	Experiential 6h

FOR THE TEACHER

- **This module tops out at Analysis on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 6 lecture, 3 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 16 · SEMESTER 6 · PAPER 2

Current trends in Ayurvedic Pharmaceutical Industry.

Exposition of Ayurvedic Pharmaceutics

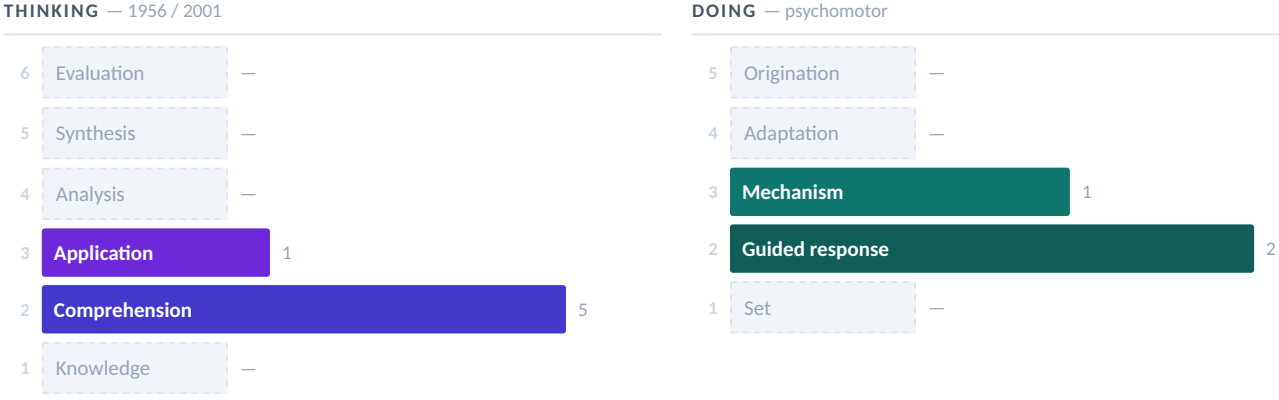
CREDITS	UNITS	THINKING	DOING	FORMATIVE
2	3	6	3	50
60 notional hours	12 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	10 h	6	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	20 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	26 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 4 hours carry the modular assessment, worth 50 marks, bringing the module to its 60 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Key Competencies Required for Employment in Pharmacy

6 obj · 22 h · Comprehension / Mechanism

1.1 Orientation about current trends in the Ayurvedic Pharmaceutical Industry

1.2 Ayurvedic Pharmacy Industry (India & Global Overview): Market size, growth, and key importing countries

1.3 Ayurvedic Products and Market Stakeholders

1.4 Consumer Preferences and Distribution Channels

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Discuss current trends in Ayurvedic pharma industry.	*discuss*	Conceptual ◦	Lecture 1h
Discuss key competencies required for employment in the Ayurvedic pharmacy.	*discuss*	Conceptual ◦	Lecture 1h
Describe career opportunities in Ayurvedic pharma industries.	*describe*	Conceptual ◦	Lecture 1h
Discuss key competencies required for employment in the Ayurvedic pharmacy.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Analyze the Market Status of Ayurvedic Pharmacy Industry.	*analyze*	Procedural	Practical 8h
DOING — LEVEL 3 · MECHANISM — 1			
Assess real-life experience of the market for the growing popularity of Ayurvedic products.	*assess*	Procedural	Experiential 10h

Unit 2 · Internship/ Visits to GMP certified Pharmacies

3 obj · 16 h · Comprehension / —

2.1 Factory set up and daily Work flow

2.2 Hygiene checks & Cleaning

2.3 Inventory Management, material management and Production

2.4 Packaging and labelling

2.5 Documentation (batch records) and QC record

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe the sections and working flow of GMP-certified pharmacy.	*describe*	Conceptual ◦	Lecture 2h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Observe and appraise the operational management of Ayurvedic Pharmacy.	*observe*	Factual ◦	Practical 6h
GOING ABOUT IT — LEVEL 4 · ORGANISATION — 1			
Interpret benefits of field visit and Internship.	*interpret*	Conceptual	Experiential 8h

Unit 3 · Quality Assurance and Quality Control (QAQC) Laboratory Training

3 obj · 18 h · Application / Guided response

3.1 Information about Leading Ayurvedic Research Labs in India
3.2 Regulatory & Certification Bodies for QAQC of Ayurvedic Formulations
3.3 Laboratory set up and daily Work flow
3.4 LIMS (Laboratory Information Management System)
3.5 Instrument Calibration, Sample testing and analysis, Record keeping

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe modern parameters used for quality assurance and quality control of Ayurvedic Drugs. — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural	Lecture 4h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate and assist quality control practical to display quality standards of Ayurvedic formulations.	*demonstrate*	Procedural	Practical 6h
GOING ABOUT IT — LEVEL 3 · VALUING — 1			
Justify the utility of LIMS(Laboratory Information Management System) in Ayurvedic research laboratories and the workflow of the laboratory.	*justify*	Conceptual	Experiential 8h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 6 lecture, 3 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 23 · SEMESTER 6 · PAPER 3

SOP and clinical applications of Churna Kalpana(Powder dosage forms)

Standardization and clinical Utility of Ayurvedic formulations

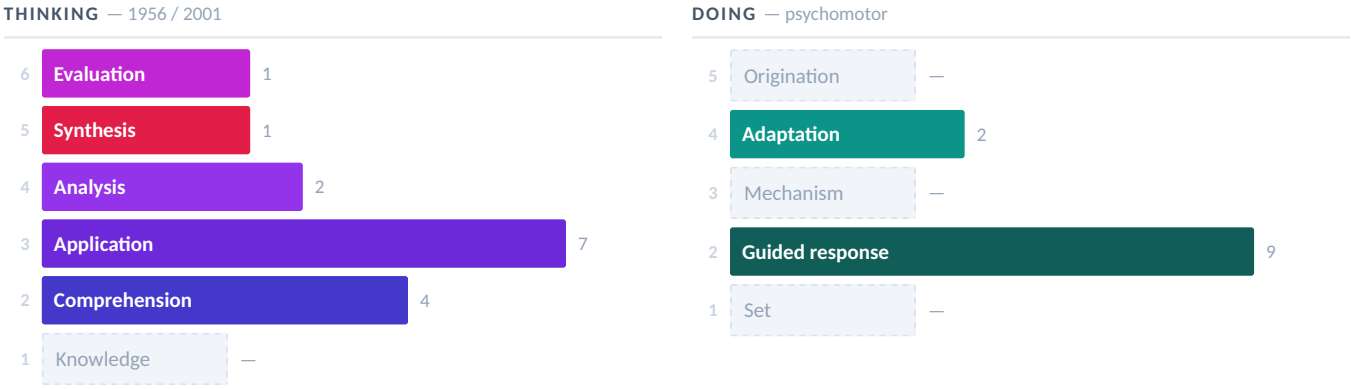
CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	5	15	11	75
90 notional hours	26 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	15	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	6	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · SOP and clinical utility of Churna Kalpana

4 obj · 15 h · Application / Adaptation

1.1 Standardisation of various Churna yogas with Standard Operating Procedure
1.2 Therapeutic application of various Churna yogas
1.3 Relevance of Churna kalpana in therapeutics

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss the importance of particle size of different types of Churna concerning its pharmaceutical and therapeutic significance. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
Describe the pharmacological and therapeutic significance of Churna kalpana. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the SOP and quality analysis of a selected Churna yoga.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 4 · ADAPTATION — 1			
Evaluate the role of a selected Churna in a specific clinical condition.	*evaluate*	Conceptual + Procedural ◦	Experiential 8h

Unit 2 · Standardization, analytical profiling, and clinical application of Kshara

5 obj · 16 h · Application / Guided response

2.1 Relevance of Kshara kalpana and Kshara sutra in therapeutics
2.2 SOP of preparation of Paneeya Kshara and Pratisaraneeya Kshara
2.3 SOP of preparation of Kshara Sutra
2.4 Analytical profiling of Kshara

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Discuss Kshara Kalpana, its sources, and evaluate the chemical composition.	*discuss*	Conceptual ◦	Lecture 1h
Discuss the techniques for the preparation of Kshara Sutra.	*discuss*	Procedural	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe the therapeutic application of Paneeya Kshara and Prathisaraneeya Kshara. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of a selected Paneeya Kshara and Pratisaraneeya Kshara and their quality analysis.	*demonstrate*	Procedural	Practical 5h
Demonstrate the therapeutic applications of Kshara and Ksharasutra in a given selected.	*demonstrate*	Procedural	Experiential 8h

Unit 3 · Lavana Kalpana with special reference to pharmaceutico-analytical aspects and therapeutic attributes

5 obj · 16 h · Application / Guided response

3.1 Standard Operating
3.2 Analytical Profiling of Lavanas
3.3 Therapeutic Applications of Selected Formulations in Lavana Kalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe salt-based formulations (Lavana Kalpa) mentioned in different texts.	*describe*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 2			
Discuss chemical profiling and standardization of Lavana Kalpa. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
Illustrate the probable mechanism of action and clinical significance of Lavana Kalpa. — teach at level 2, the verb *illustrate* is what will be asked	*illustrate*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of a selected Lavana formulation following different methods and analytical profiling.	*demonstrate*	Procedural	Practical 5h
Demonstrate the role of Lavana Kalpana in various clinical conditions.	*demonstrate*	Procedural	Experiential 8h

Unit 4 · Satva Kalpana SOP and clinical significance

5 obj · 15 h · Evaluation / Guided response

4.1 SOP for preparing selected Satvas
4.2 Analytical profiling of various yogas from Satva Kalpana
4.3 Therapeutic application of Satva kalpana

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the uniqueness and pharmaceutical relevance of Satva kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss SOP and analytical profiling of Satva Kalpana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Procedural ◦	Lecture 1h
THINKING — LEVEL 6 · EVALUATION / EVALUATE — 1			
Evaluate the clinical significance of Satva Kalpana.	*evaluate*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of a selected Satva and perform its analytical profiling.	*demonstrate*	Procedural	Practical 5h
Demonstrate the role of a selected satva in a specified disease condition.	*demonstrate*	Procedural	Experiential 7h

Unit 5 · Therapeutic attributes of Masi Kalpana

7 obj · 22 h · Synthesis / Adaptation

5.1 Masi from plant and animal origin
5.2 Pharmacology of Masi
5.3 Activated charcoal in therapeutics
5.4 Clinical applications of Masi

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the therapeutic significance of various Masi Kalpas.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Evaluate the clinical significance of activated charcoal. — teach at level 6, the verb *evaluate* is what will be asked	*evaluate*	Conceptual	Lecture 1h
THINKING — LEVEL 4 · ANALYSIS / ANALYSE — 1			
Discuss the analytical profiling of Mashi Kalpas. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Discuss SOP for the preparation of Masi by following different classical and advanced methods IS. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Procedural	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate the preparation of Mashi and interpret its analytical profiling.	*demonstrate*	Procedural	Practical 5h
Demonstrate pharmaceutical modification of Masi Kalpana.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 4 · ADAPTATION — 1			
Evaluate the Role of Malaharas as a Topical application in Skin disease.	*evaluate*	Conceptual + Procedural ◦	Experiential 8h

FOR THE TEACHER

- **This module tops out at Evaluation on the thinking ladder and Adaptation on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 15 lecture, 6 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Adaptation.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 24 · SEMESTER 6 · PAPER 3

Therapeutic significance of Topical dosage forms

Standardization and clinical Utility of Ayurvedic formulations

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	3	5	6	25
30 notional hours	11 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	3	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	3	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Describe the therapeutic significance of various topical dosage forms.	*describe*	L2 Comprehension / Understand
2	Evaluate the challenges in current topical applications of Ayurveda.	*evaluate*	L6 Evaluation / Evaluate
3	Design and develop novel topical dosage forms while preserving their effectiveness.	*design*	L5 Synthesis / Create
4	Integrate clinical knowledge to develop topical applications for specific areas, such as skin disorders, burns, wounds, and pain relief.	*integrate*	L5 Synthesis / Create
5	Access databases to develop strategies for topical applications in cosmeceuticals, with special regard to safety.	*access*	L? —

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Ophthalmic, Otic, and Buccal dosage forms and their clinical applications1.SOP for Ophthalmic, Otic and buccal dosage forms

4 obj · 10 h · Application / Guided response

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss classical and novel dosage forms for ophthalmic, otic, and buccal use.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe therapeutic significance and clinical applications of dosage forms for ophthalmic, otic, and buccal use. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Demonstrate various Netra Kalpa and their clinical applications.	*demonstrate*	Procedural	Practical 3h
Demonstrate the role of Ayurvedic ophthalmic preparations in various eye diseases.	*demonstrate*	Procedural	Experiential 5h

Unit 2 · Standardisation of Lepa and Malahara preparations and its application in various disorders

3 obj · 6 h · Application / Guided response

2.1 SOP for the preparation of various Lepas
2.2 Innovative approaches including Patches, Bandages, Bands
2.3 Clinical applications of Lepa and Malahara kalpa

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss classical topical dosage forms like Lepa, Malahara.	*discuss*	Conceptual	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe novel dosage forms for topical application, along with their regulatory guidelines. — teach at level 2, the verb *describe* is what will be asked	*describe*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Prepare a novel dosage form using one of the classical lepa Yoga.	*prepare*	Procedural	Practical 4h

Unit 3 · Standard Operating Procedure (SOP) for Upanaha, Dhupa, Dhumapana, and Avachurnana Kalpa and their therapeutic applications

4 obj · 12 h · Application / Origination

3.2 Anti-microbial activity of Dhupa
3.5 Clinical application of various Dhumapana yogas

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the therapeutic significance of Upanaha, Dhupa, Dhumapana. — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the preparation of a selected Upanaha which is useful in treating Shopha.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 3 · MECHANISM — 1			
Demonstrate the role of Malahara as a Topical application in Skin disease.	*demonstrate*	Procedural	Experiential 3h
DOING — LEVEL 5 · ORIGINATION — 1			
Demonstrate the role of lepa Yogas in managing various medical needs by innovative approaches.	*demonstrate*	Procedural	Experiential 5h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Origination on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 3 practical, 3 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Origination.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 31 · SEMESTER 6 · PAPER 4

Regulatory Acts and Rules applicable to Ayurvedic Drugs

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
3	6	15	16	75
90 notional hours	31 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	15 h	15	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	30 h	7	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	39 h	9	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 6 hours carry the modular assessment, worth 75 marks, bringing the module to its 90 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO



THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Chapter I of the Drugs and Cosmetics Act, 1940 and Rules 1945

3 obj · 6 h · Comprehension / Mechanism

1.1 Introduction to 'Drugs and Cosmetic Act, 1940; and 1945' and definitions

1.2 Prescribed Books on Ayurveda, Siddha, and Unani Drugs: The First Schedule

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 2			
Describe the Scope and Applicability of the Drugs and Cosmetics Act and Key Definitions.	*describe*	Conceptual °	Lecture 1h
Discuss the Legal importance of Drug Regulation and enlist prescribed books of ASU from the first Schedule of D&C Act.	*discuss*	Conceptual	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Analyse case law on the Factory Act.	*analyse*	Procedural	Experiential 4h

Unit 2 · Chapter II - Regulatory Bodies

3 obj · 9 h · Comprehension / Mechanism

2.1 The Drugs Technical Advisory Board (DTAB)
2.2 The Central Drugs Laboratory (CDL)
2.3 The Drugs Consultative Committee (DCC)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify key regulatory authorities and develop foundational knowledge of regulatory frameworks. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Record their observations regarding hypothetical issue through Mock composition exercise of demonstration of the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists.	*record*	Procedural	Practical 4h
DOING — LEVEL 3 · MECHANISM — 1			
Prepare a technical report on Adverse Drug Reactions (ADRs) for submission to DTAB.	*prepare*	Procedural	Experiential 4h

Unit 3 · Chapters III, IV, and IV-A - Regulation of Drugs and Cosmetics

8 obj · 18 h · Application / Mechanism

3.1 Chapter III: Standards of quality
3.2 Misbranded, adulterated, and spurious drugs and cosmetics
3.3 Prohibition of import of certain drugs or cosmetics
3.4 Manufacturing, Labelling, and Sale of Drugs and Cosmetics (Chapter IV-A)
3.5 Standards of quality for drugs and cosmetics (Sections 16 and 16A)
3.6 Prohibition of manufacture and sale of certain drugs and cosmetics (Sections 18 and 19)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Identify legal restrictions and prohibitions for misbranded and spurious drugs and. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 1h
Describe manufacturing, labelling, and sale provisions, including penalties and seizures (Chapter IV-A).	*describe*	Conceptual ◦	Lecture 1h
Justify the importance of standards of quality for drugs and cosmetics (Sections 16 and. — teach at level 6, the verb *justify* is what will be asked	*justify*	Conceptual	Lecture 1h
Discuss the prohibition of the manufacture and sale of certain drugs and cosmetics.	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Summarize the provisions of Chapter III regarding the import of drugs and cosmetics. — teach at level 2, the verb *summarize* is what will be asked	*summarize*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 2			
Prepare mock import license applications.	*prepare*	Procedural	Practical 4h
Demonstrate case scenarios of misbranded/spurious drugs and cosmetics.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 1			
Identify misbranded and spurious drugs/cosmetics.	*identify*	Procedural	Experiential 4h

Unit 4 · Schedules to Drugs and Cosmetics Rules related to ASU drugs and cosmetics

5 obj · 16 h · Application / Mechanism

4.1 Good Manufacturing Practices (GMP) for ASU Medicines

4.2 Prescription and Safety Requirements (Schedules H, G, X) and Pharmacological Information (Schedule Y)

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the provision of Schedule T and Schedule E (1).	*discuss*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Discuss the requirements of Schedules H, G, X (prescription and safety), and Schedule Y (pharmacological information). — teach at level 2, the verb *discuss* is what will be asked	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the formulation of a proprietary formula and apply for a license.	*demonstrate*	Procedural	Practical 5h
DOING — LEVEL 3 · MECHANISM — 2			
Apply for GMP certification.	*apply*	Procedural	Experiential 4h
Prepare a new formula and apply for a mock license.	*prepare*	Procedural	Experiential 5h

Unit 5 · Pharmaceutical observation of drugs and cosmetics ASU drugs

3 obj · 10 h · Comprehension / Mechanism

5.1 Shelf life and stability of ASU drugs: recent amendments

5.2 Use of Preservatives in ASU drugs

5.3 Packaging and labelling requirements for ASU drugs

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Describe Shelf life of ASU drugs: recent amendments, Preservatives, and. Discuss Packaging and labelling technique.	*describe*	Procedural ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 2			
Evaluate label compliance and identify shortcomings in given labels.	*evaluate*	Procedural	Experiential 5h
Design and develop a label for a hypothetical ASU product, ensuring compliance with regulatory requirements.	*design*	Procedural	Experiential 4h

Unit 6 · Miscellaneous Regulatory Acts related to drugs and cosmetics

9 obj · 25 h · Comprehension / Mechanism

6.1 Narcotic and Psychotropic Substances Act 1985
6.2 Factories Act, 1948
6.3 Industries Act, 1951
6.4 Biological Diversity Act 2002
6.5 Consumer Protection Act 1986
6.6 Drugs and Magic Remedies Act, 1954
6.7 Medicinal and Toilet Preparations Act, 1955
6.8 Opium Act, 1857
6.9 Pharmacy Act 1948
6.10 Recent Amendments in D & C Act concerning Ayurvedic Drug Industry

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 4			
Describe the key provisions of the Narcotic and Psychotropic Substances Act, 1985 Factories Act, 1948, and Industries Act, 1951.	*describe*	Conceptual ◦	Lecture 1h
Discuss the Biological Diversity Act 2002, Consumer Protection Act 1986, and Drugs and Magic Remedies Act, 1954.	*discuss*	Conceptual ◦	Lecture 1h
Discuss Medicinal and Toilet Preparations Act, 1955, Opium Act, 1857, and Pharmacy.	*discuss*	Conceptual ◦	Lecture 1h
Describe Recent Amendments in D & C Act concerning Ayurvedic Drug Industry.	*describe*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate Landmark Judgments under the NDPS Act.	*demonstrate*	Procedural	Practical 4h
Demonstrate Case Law on the Factory Act.	*demonstrate*	Procedural	Practical 4h
Demonstrate Case Study under the Drugs and Magic Remedies Act.	*demonstrate*	Procedural	Practical 4h
DOING — LEVEL 3 · MECHANISM — 2			
Analyse Landmark Judgments under the NDPS Act.	*analyse*	Procedural	Experiential 5h
Prepare Case Studies and Report related to Objectionable Advertisements.	*prepare*	Procedural	Experiential 4h

FOR THE TEACHER

- **This module tops out at Application on the thinking ladder and Mechanism on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 15 lecture, 7 practical, 9 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Mechanism.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

MODULE 32 · SEMESTER 6 · PAPER 4

Entrepreneurship and Start-Ups

Regulatory aspects and recent trends in Ayurvedic Pharmaceutics

CREDITS	UNITS	THINKING	DOING	FORMATIVE
1	4	6	8	25
30 notional hours	14 objectives	cognitive objectives	psychomotor objectives	modular assessment marks

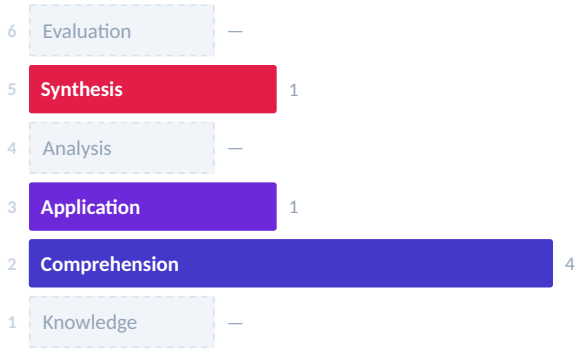
WHERE THE HOURS GO

KIND OF HOUR	HOURS	OBJECTIVES	WHAT IT MUST DELIVER IN THIS MODULE
Lecture	5 h	5	Put the material in front of students and make them retrieve it. On its own it cannot take them past level 2.
Practical	10 h	4	Move students from performing with help to performing unaided — the transition these hours exist for.
Experiential	13 h	5	Hand over an unfamiliar situation and require a judgement or a product. The modular assessment is drawn from here.

A further 2 hours carry the modular assessment, worth 25 marks, bringing the module to its 30 notional hours.

WHAT THIS MODULE ASKS STUDENTS TO DO

THINKING — 1956 / 2001



DOING — psychomotor



MODULE LEARNING OBJECTIVES

What the modular assessment tests, with the level of each verb.

#	OBJECTIVE	VERB	LEVEL
1	Define Entrepreneurship	*define*	L1 Knowledge / Remember
2	Identify Entrepreneurial Traits	*identify*	L1 Knowledge / Remember
3	Describe the Entrepreneurial Process	*describe*	L2 Comprehension / Understand
4	Enlist Different Types of Entrepreneurships	*enlist*	L1 Knowledge / Remember
5	Recognize the opportunities of entrepreneurship and start-ups	*recognize*	L1 Knowledge / Remember

THE UNITS, AND THEIR OBJECTIVES

Each unit below carries its own objectives, grouped by level. The kind of hour is printed against each: it is the curriculum's clearest statement of how a thing is meant to be learned.

Unit 1 · Entrepreneurship and Startup

3 obj · 4 h · Application / Mechanism

1.1 Introduction to Entrepreneurship and Start-ups
1.2 MSMED Act and its Process of Registration and Role of Department for Promotion of Industry and Internal Trade(DPIIT) and Bharat Startup Knowledge Access
1.3 Types of Enterprises and Role of Enterprises in National and Global Economy
1.4 Government Policies, Schemes, and Institutional Support for Entrepreneurship

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify the role of DPIIT (Department for Promotion of Industry and Internal Trade) BHASKAR (Bharat Startup Knowledge Access Registry) in the promotion of entrepreneurship and startups. — teach at level 1, the verb *identify* is what will be asked	*identify*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 3 · APPLICATION / APPLY — 1			
Describe entrepreneurship, the startup life cycle, the MSME Act, and the process of registration. — teach at level 2, the verb *describe* is what will be asked	*describe*	Procedural ◦	Lecture 1h
DOING — LEVEL 3 · MECHANISM — 1			
Create an elevator pitch for a startup idea.	*create*	Procedural	Experiential 2h

Unit 2 · Process of entrepreneurship development

3 obj · 8 h · Comprehension / Origination

2.1 Need and the process of entrepreneurship development
2.2 Dynamics of Entrepreneurial Motivation
2.3 Developing Entrepreneurial Competencies

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Discuss the dynamism of Entrepreneurial Motivation and key entrepreneurial competencies.	*discuss*	Conceptual ◦	Lecture 1h
DOING — LEVEL 2 · GUIDED RESPONSE — 1			
Demonstrate the process of development of a basic business model as per the template.	*demonstrate*	Procedural	Practical 3h
DOING — LEVEL 5 · ORIGINATION — 1			
Conduct and document market research for a start-up, including survey design and analysis.	*conduct*	Procedural	Experiential 4h

Unit 3 · Project Proposal for new Invention/startup

2 obj · 3 h · Comprehension / Adaptation

3.1 Challenges and the feasibility report of the Startups
3.2 Planning, resource mobilisation, and implementation of a start-up project

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Identify the challenges and feasibility report of the start-up. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Lecture 1h
DOING — LEVEL 4 · ADAPTATION — 1			
Carry out A Case Study-Based Role-Play, navigating a Startup's Early Days.	*carry*	Procedural	Experiential 2h

Unit 4 · Intellectual Property Rights and Patents

6 obj · 13 h · Synthesis / Mechanism

4.1 Introduction to IPR and Patents
4.2 Indian and international patent laws, proposed amendments as applicable to herbal/natural products and processes
4.3 Procedure for patent filing
4.4 Geographical Indication and Copyright

OBJECTIVE, AS WORDED BY NCISM	VERB	KNOWLEDGE	KIND OF HOUR
THINKING — LEVEL 2 · COMPREHENSION / UNDERSTAND — 1			
Define IPR, describe the purpose and importance of IPR, and the laws governing patent. — teach at level 1, the verb *define* is what will be asked	*define*	Conceptual ◦	Lecture 1h
THINKING — LEVEL 5 · SYNTHESIS / CREATE — 1			
Identify the patentability of a product. — teach at level 1, the verb *identify* is what will be asked	*identify*	Factual ◦	Experiential 2h
DOING — LEVEL 2 · GUIDED RESPONSE — 3			
Demonstrate the process of patent filing.	*demonstrate*	Procedural	Practical 3h
Demonstrate a case scenario of the new molecule patent.	*demonstrate*	Procedural	Practical 2h
Demonstrate a case scenario of patenting the formulation process.	*demonstrate*	Procedural	Practical 2h
DOING — LEVEL 3 · MECHANISM — 1			
Design an application to Patent the formulation process.	*design*	Procedural	Experiential 3h

FOR THE TEACHER

- **This module tops out at Synthesis on the thinking ladder and Origination on the doing ladder.** Set the modular assessment against those ceilings. An instrument that cannot reach them leaves the module untested at the level it claims; one that goes beyond them tests something the curriculum never taught.
- **Let the kind of hour decide the method.** The curriculum has already made that choice for every objective here: 5 lecture, 4 practical, 5 experiential. An objective set as experiential and delivered as a lecture has been demoted two levels, and nothing in the timetable will show it.
- **Most of the practical hours belong to one transition.** Getting a student from performing with help to performing unaided is slow, and it is what these hours exist for. Watch the middle of the group, not the strongest student.
- **Read the verb before planning.** Where an objective's verb sits at a different level from its code, the note beside it says which to teach to. The verb is what a student meets in the examination.

FOR THE STUDENT

- **Read the verb, and read the kind of hour.** Together they tell you what you will have to produce. You prepare for "discuss" and for "construct" in completely different ways.
- **Aim to work unaided.** Performing while someone guides you is level 2 of the doing ladder. Doing it alone and reliably is level 3, and that is the expectation.
- **Know why each step exists.** When a condition changes, only the reason tells you what to change with it — and this module reaches Origination.
- **Keep your record as you go.** A log written up at the end is a piece of writing; written as you work, it is evidence.

PART C

Reference

Nine appendices, all worked out from the curriculum's own objectives, so the counts here and the counts in Part B are the same numbers. C1 and C2 decode anything in the syllabus document. C3 compares every module and unit at a glance. C6 and C7 reproduce all three hundred and four practical and experiential activities with the level each serves, so a reference like "Experiential Learning 8.2" can be looked up without turning back to the syllabus.

1. **C1** The sixteen curriculum codes, decoded
2. **C2** The two schemes side by side
3. **C3** Level profile of every module and unit
4. **C4** Verb index — every verb the curriculum uses, and its level
5. **C5** Kinds of knowledge across the semester
6. **C6** All 141 practical training activities, mapped
7. **C7** All 163 experiential learning activities, mapped
8. **C8** Where the hours go, module by module
9. **C9** Sources and further reading

↑ Contents

PART C · APPENDIX C1

The sixteen curriculum codes, decoded

Every code that appears in Column 3E of Table 3, with the domain it belongs to, the level it sits at, and how many of the 577 objectives carry it. Keep this beside the syllabus document.

CODE	DOMAIN	LEVEL	NAME	TIMES USED
CK	Thinking	1	Knowledge / Remember	7
CC	Thinking	2	Comprehension / Understand	152
CAP	Thinking	3	Application / Apply	69
CAN	Thinking	4	Analysis / Analyse	32
CS	Thinking	5	Synthesis / Create	5
CE	Thinking	6	Evaluation / Evaluate	12
PSY-SET	Doing	1	Set	1
PSY-GUD	Doing	2	Guided response	194
PSY-MEC	Doing	3	Mechanism	65
PSY-ADT	Doing	4	Adaptation	30
PSY-ORG	Doing	5	Origination	4
AFT-REC	Going about it	1	Receiving	not used
AFT-RES	Going about it	2	Responding	not used
AFT-VAL	Going about it	3	Valuing	5
AFT-SET	Going about it	4	Organisation	1
AFT-CHR	Going about it	5	Characterisation	not used

Sixteen codes are defined; 13 are used here. Unlike the undergraduate curriculum, this course **does** use Synthesis and Origination, and every semester reaches Evaluation.

PART C · APPENDIX C2

The two schemes side by side

The six thinking levels in both schemes, with what a student actually does at each.

LEVEL	BLOOM 1956, AS PRINTED IN THE CURRICULUM	REVISED 2001	WHAT THE STUDENT DOES
Level 1	**Knowledge**	**Remember**	recalls it
Level 2	**Comprehension**	**Understand**	restates it in their own arrangement
Level 3	**Application**	**Apply**	uses it on something new
Level 4	**Analysis**	**Analyse**	takes it apart and shows how the parts relate
Level 5	**Synthesis**	**Evaluate** — the schemes differ here	makes something that was not given
Level 6	**Evaluation**	**Create** — the schemes differ here	judges it against stated criteria

The top two levels are the only place the schemes disagree. This book follows the 1956 numbering, because that is what the curriculum prints in Column 3E.

PART C · APPENDIX C3

Level profile of every module and unit

All thirty-two modules and one hundred and twenty-four units in one view. The fastest way to see where a semester's weight sits and to check that a modular assessment matches its ceiling.

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
SEMESTER 3						
PAPER 1 · MODULE 1 — CHRONOLOGICAL DEVELOPMENT AND TAXONOMY OF RASA DRAVYAS — 19 OBJ, 56 H, CEILING ANALYSIS / ADAPTATION						
1	1	Chronological Development and Overview of Rasashastra	5	9	Analysis	Adaptation
1	2	Taxonomy of Rasadravyas	5	13	Analysis	Guided response
1	3	Geological/mineralogical identification of Rasadravyas	6	22	Analysis	Mechanism
1	4	Overview of Sandigdha, Anupalabdha, Krutrima, and Pratinidhi Dravyas in the Context of Rasa Dravya	3	12	Comprehension	Guided response
PAPER 1 · MODULE 2 — PUTA AND SAMSKARA (TRANSFORMATIVE PROCESSES OF DRUG MANUFACTURING) — 21 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
2	1	Puta and Law of thermodynamics	6	18	Evaluation	Mechanism
2	2	Concept, understanding, application, SOPs and chemical changes of Jarana, Amritikarana and Lohitikarana	4	16	Analysis	Guided response
2	3	Concept, understanding, application, SOPs (Standard Operating Procedures) and chemical changes of Satva and Druti procedures	4	9	Application	Guided response
2	4	Samskara (Principles of Transformative processes of drug manufacturing)	2	3	Comprehension	Adaptation
2	5	Influence of Desha (place) and Kala(time and seasonal factors) of drug qualities	2	3	Comprehension	Mechanism
2	6	Kalaprakarsha- Processing with/Effect of specific time exposure and Bhajana (processing vessel)	3	7	Comprehension	Guided response
PAPER 2 · MODULE 9 — SCIENTIFIC FUNDAMENTALS OF AYURVEDIC PHARMACEUTICAL PREPARATIONS — 15 OBJ, 56 H, CEILING ANALYSIS / MECHANISM						
9	1	Classical Principles of drug processing and scientific basis of Panchavidha Kashaya Kalpana	3	6	Comprehension	Mechanism
9	2	Identity, Purity and Quality of Raw materials used in Ayurveda Pharmaceutics	3	12	Comprehension	Guided response
9	3	Pharmaceutical processing of drugs with its rationality	3	13	Analysis	Mechanism
9	4	Ayurvedic Classical Dosage Forms and their integration with modern scientific validation	6	25	Analysis	Guided response
PAPER 2 · MODULE 10 — PRINCIPLES OF PHARMACEUTICAL FORMULATION — 12 OBJ, 56 H, CEILING SYNTHESIS / MECHANISM						
10	1	Principles of Samskara (transformative processes of drug manufacturing) in relation to drug development	2	6	Application	Guided response
10	2	Rationality behind Composition of Formulations	2	6	Analysis	Guided response
10	3	Principles and designing of various dosage forms	2	8	—	Guided response
10	4	Principles of Prakshep dravyas (Pharmaceutical Additives)	3	14	Comprehension	Mechanism

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
10	5	Principles of Potency and Efficacy enhancement	3	22	Synthesis	Mechanism
PAPER 3 · MODULE 17 — GUIDELINES FOR ADMINISTRATION OF RASOUSHADHI IN CLINICAL PRACTICE — 22 OBJ, 56 H, CEILING EVALUATION / MECHANISM						
17	1	Concept of Rasa chikitsa and significance of Rasoushadhi	4	11	Application	Guided response
17	2	Concept of Kshetrikarana and its significance prior to administration of Rasoushadhi	4	11	Application	Guided response
17	3	Pathya -Apathya during Rasayoga Sevana	5	12	Application	Mechanism
17	4	Safety and toxicity of Rasoushadhi	5	14	Evaluation	Guided response
17	5	Heavy metal toxicity and chelation therapy	4	8	Comprehension	Mechanism
PAPER 3 · MODULE 18 — PRAYOGIKA BHAISHAJYA KALPANA(CLINICAL APPLICATION OF PHARMACEUTICAL PRODUCTS) — 24 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
18	1	Aushadha Sevana Kala (Time of administration of drug)	7	15	Application	Adaptation
18	2	Concept and scientific basis of Anupana	6	16	Evaluation	Mechanism
18	3	Matra (Posology)	5	12	Application	Guided response
18	4	Concept of drug repurposing	6	13	Application	Mechanism
PAPER 4 · MODULE 25 — ARTIFICIAL INTELLIGENCE IN DRUG DEVELOPMENT AND PHARMACOGENETICS; PHARMACOGENOMICS — 20 OBJ, 56 H, CEILING APPLICATION / ORIGINATION						
25	1	Role of Artificial intelligence in drug development	4	11	Application	Mechanism
25	2	Types of software used in Artificial intelligence	3	6	Application	Origination
25	3	3D printed drug delivery system	5	16	Application	Adaptation
25	4	Pharmacogenetics and Pharmacogenomics	4	11	Application	Guided response
25	5	Pharmacogenomics in drug discovery	4	12	Comprehension	Mechanism
PAPER 4 · MODULE 26 — PHARMACOLOGY — 16 OBJ, 56 H, CEILING SYNTHESIS / ADAPTATION						
26	1	Introduction to Pharmacology	3	12	Comprehension	Guided response
26	2	Pharmacokinetics, therapeutic index, and drug interactions	3	12	Application	Adaptation
26	3	Pharmacology of drugs that act on the peripheral nervous system	3	9	Synthesis	—
26	4	Pharmacology of drugs that act on the central nervous system I	3	9	Application	Mechanism
26	5	Pharmacology of drugs that act on the central nervous system II	4	14	Comprehension	Mechanism
SEMESTER 4						
PAPER 1 · MODULE 3 — PARADA (MERCURY) AND ITS PROCESSING: ANCIENT WISDOM AND SCIENTIFIC ADVANCES — 17 OBJ, 56 H, CEILING ANALYSIS / MECHANISM						
3	1	Parada: Ancient and Contemporary Perspectives on Its Shodhana and Underlying Principles	8	22	Comprehension	Mechanism
3	2	Parada Ashta Samskara and Ashtadasha Samskara	4	21	Comprehension	Guided response
3	3	Physico-chemical proprties and pharmacology of mercury and mercurial compounds	2	6	Analysis	Mechanism

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
3	4	Mercury toxicity, ADR and its management	3	7	Analysis	Guided response
PAPER 1 · MODULE 4 — MAHARASA DRAVYA, ITS PROCESSING AND UTILITY — 22 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
4	1	Maharasa Dravyas with special reference to Abhraka and Vaikranta	8	18	Analysis	Adaptation
4	2	Principles and Processing of Makshika and Vimala	3	11	Comprehension	Guided response
4	3	Principles and Processing of Shilajit and Sasyaka	3	10	Analysis	Guided response
4	4	Principles and Processing of Chapala and Kharpara, along with physico-chemical analysis, with research updates of all Maharasa Dravyas	4	9	Application	Mechanism
4	5	Utility of new metals and minerals	4	8	Evaluation	Guided response
PAPER 2 · MODULE 11 — MANA PARIBHASHA (METROLOGY IN AYURVEDIC PHARMACEUTICAL PREPARATION) — 16 OBJ, 56 H, CEILING ANALYSIS / ADAPTATION						
11	1	Mana Paribhasha (Classical system of measurement)	4	12	Analysis	Adaptation
11	2	Standard Units of Measurement	3	6	Knowledge	Adaptation
11	3	Volumetric Vs Gravimetric Methods	3	6	Application	Guided response
11	4	Rationality of Proportion of Ingredients in Ayurvedic formulations	3	16	Comprehension	Adaptation
11	5	Bulk production of Ayurvedic Formulations	3	16	Comprehension	Adaptation
PAPER 2 · MODULE 12 — PROCESS VALIDATION PROTOCOL — 10 OBJ, 56 H, CEILING EVALUATION / GUIDED RESPONSE						
12	1	Pharmaceutical Preparation process: Classical Dosage forms	6	41	Evaluation	Guided response
12	2	Pharmaceutical Preparation process: Recent advances of Dosage forms	3	10	Comprehension	Guided response
12	3	Novel Dosage and drug delivery Forms	1	5	—	Guided response
PAPER 3 · MODULE 19 — SOP AND CLINICAL APPLICATION OF CHATURVIDHA RASAYANA — 25 OBJ, 84 H, CEILING EVALUATION / ADAPTATION						
19	1	Standard Operating Procedure (SOP) and Clinical Application of Khalveeya Rasayana	5	17	Application	Adaptation
19	2	SOP and clinical application of Parpati Rasayana	6	15	Analysis	Adaptation
19	3	SOP and clinical application of Pottali Rasayana	7	25	Evaluation	Mechanism
19	4	SOP and clinical application of Kupipakwa Rasayana	7	27	Comprehension	Guided response
PAPER 3 · MODULE 20 — SOP AND CLINICAL APPLICATION OF AYASKRITI, LOUHA, AND MANDURA KALPA — 12 OBJ, 28 H, CEILING COMPREHENSION / ADAPTATION						
20	1	SOP and clinical application of Ayaskriti	3	8	Comprehension	Mechanism
20	2	SOP and clinical application of Louha kalpa	4	9	Comprehension	Adaptation
20	3	SOP and Clinical application of Mandura Kalpa	5	11	Comprehension	Mechanism
PAPER 4 · MODULE 27 — NETWORK PHARMACOLOGY & REVERSE PHARMACOLOGY — 16 OBJ, 56 H, CEILING ANALYSIS / MECHANISM						
27	1	Netwok Pharmacology	8	28	Analysis	Mechanism
27	2	Reverse Pharmacology	8	28	Analysis	Mechanism
PAPER 4 · MODULE 28 — PHARMACEUTICAL QUALITY ASSURANCE — 19 OBJ, 56 H, CEILING APPLICATION / ADAPTATION						
28	1	Quality Assurance and Quality Management concepts	6	19	Application	Guided response
28	2	Personnel Responsibilities	3	8	Comprehension	Adaptation

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
28	3	Containers, Packing materials	4	13	Comprehension	Mechanism
28	4	Good Laboratory Practices	3	8	Knowledge	Mechanism
28	5	Document maintenance in pharmaceutical industry	3	8	Application	Mechanism
SEMESTER 5						
PAPER 1 · MODULE 5 — CRITICAL APPRAISAL OF UPARASA DRAVYA AND SADHARANA RASA DRAVYA — 24 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
5	1	Principle and Processing of Uparasa Dravya	12	28	Application	Adaptation
5	2	Principle and Processing of Sadharana Rasa	12	28	Evaluation	Adaptation
PAPER 1 · MODULE 6 — CRITICAL APPRAISAL OF DHATU AND UPDHATU — 24 OBJ, 56 H, CEILING EVALUATION / ADAPTATION						
6	1	Principle and Processing of Dhatu / Metals	12	28	Analysis	Mechanism
6	2	Principle and Processing of Upadhatu	12	28	Evaluation	Adaptation
PAPER 2 · MODULE 13 — KRUTANNA (AYURVEDIC FOOD PREPARATIONS) — 15 OBJ, 56 H, CEILING ANALYSIS / ADAPTATION						
13	1	Classification of food substances and Formulations	3	12	Comprehension	Guided response
13	2	Customized food for treatment	6	24	Comprehension	Guided response
13	3	Ayurvedic Food Industry	3	10	Knowledge	Guided response
13	4	Food processing technology traditional and modern methods	3	10	Analysis	Adaptation
PAPER 2 · MODULE 14 — AYURVEDIC FORMULATIONS FOR HOLISTIC WELLBEING, VITALITY AND BODY CARE — 11 OBJ, 56 H, CEILING EVALUATION / MECHANISM						
14	1	Rasayana-Personalized Rejuvenation Formulations	5	24	Comprehension	Mechanism
14	2	Ayurvedic based cosmetic preparation	6	32	Evaluation	Mechanism
PAPER 3 · MODULE 21 — CLINICAL APPLICATION OF RASOUSHADHI ACCORDING TO SROTAS AND IN SELECTED CHALLENGING CLINICAL CONDITIONS — 13 OBJ, 28 H, CEILING APPLICATION / ADAPTATION						
21	1	Clinical application of Rasoushadhis specific to Srotas	4	8	Application	Adaptation
21	2	Clinical application of Rasoushadhi in emergency conditions	4	9	Comprehension	Adaptation
21	3	Rasoushadhi in specific, clinically challenging conditions	5	11	Application	Adaptation
PAPER 3 · MODULE 22 — YOGAS FROM DIFFERENT KALPANAS — 27 OBJ, 84 H, CEILING ANALYSIS / ADAPTATION						
22	1	SOP of Vati, Gutika, Guggulu and Varti Kalpas and their therapeutic applications	6	20	Application	Mechanism
22	2	SOP and Clinical applications of Avaleha and Khanda kalpana	5	16	Application	Mechanism
22	3	SOP of Sneha Kalpana and its clinical significance	5	17	Application	Adaptation
22	4	SOP and clinical applications of Arka and Dravakam in various disorders	5	14	Analysis	Mechanism
22	5	SOP and therapeutic applications of Sandhana Kalpana	6	17	Application	Mechanism
PAPER 4 · MODULE 29 — PHARMACY PRACTICE — 18 OBJ, 56 H, CEILING EVALUATION / ORIGINATION						
29	1	Hospital Pharmacy and Community Pharmacy	7	26	Comprehension	Mechanism
29	2	Therapeutic Drug Monitoring	7	17	Evaluation	Adaptation
29	3	Pharmacy and therapeutic committee	4	13	Comprehension	Origination

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
PAPER 4 · MODULE 30 — PHARMACEUTICAL MICROBIOLOGY — 18 OBJ, 56 H, CEILING APPLICATION / ADAPTATION						
30	1	Introduction to microbes with applications of sterilization	5	12	Application	Guided response
30	2	Sources of contamination	6	12	Application	Adaptation
30	3	Types of spoilage and use of antimicrobial agents	7	32	Application	Adaptation
SEMESTER 6						
PAPER 1 · MODULE 7 — SUDHA VARGA, SIKATA VARGA, VISHA VARGA DRAVYAS — 22 OBJ, 84 H, CEILING APPLICATION / GUIDED RESPONSE						
7	1	Characterization, Pharmaceutical Processing, and Clinical Applications of Sudha Varga Dravya	7	28	Application	Guided response
7	2	Characterization, pharmaceutical processing, and clinical applications of Sikata Varga Dravya	7	28	Application	Guided response
7	3	Pharmaceutical processing, phytopharmacology and clinical applications of Visha Varga dravya	8	28	Application	Guided response
PAPER 1 · MODULE 8 — APPRAISAL OF RATNA AND UPARATNA AND THERAPEUTIC SIGNIFICANCE OF PISHTI AND DRUTI KALPANA — 13 OBJ, 28 H, CEILING APPLICATION / MECHANISM						
8	1	Mineralogy, Crystal chemistry,and Pharmacotherapeutics of Ratna-Uparatna Varga	7	15	Comprehension	Mechanism
8	2	SOP and therapeutic applications of Pishti, Gandhaka Druti and Gandhaka Taila kalpana	6	13	Application	Guided response
PAPER 2 · MODULE 15 — DISEASE CONDITION SPECIFIC SELECTION OF FORMULATION — 12 OBJ, 56 H, CEILING ANALYSIS / MECHANISM						
15	1	Application of Yukti Pramana in Formulation Selection	2	2	Analysis	—
15	2	Selection of formulations for Speciality practice in Ayurveda	3	12	Analysis	Guided response
15	3	Routes of Drug Administration	2	12	Analysis	Mechanism
15	4	Designing formulations for Shodhana and Shamana Chikitsa	2	12	Analysis	Guided response
15	5	Innovation in Ayurvedic Formulations	3	18	Comprehension	Guided response
PAPER 2 · MODULE 16 — CURRENT TRENDS IN AYURVEDIC PHARMACEUTICAL INDUSTRY. — 12 OBJ, 56 H, CEILING APPLICATION / MECHANISM						
16	1	Key Competencies Required for Employment in Pharmacy	6	22	Comprehension	Mechanism
16	2	Internship/ Visits to GMP certified Pharmacies	3	16	Comprehension	—
16	3	Quality Assurance and Quality Control (QAQC) Laboratory Training	3	18	Application	Guided response
PAPER 3 · MODULE 23 — SOP AND CLINICAL APPLICATIONS OF CHURNA KALPANA(POWDER DOSAGE FORMS) — 26 OBJ, 84 H, CEILING EVALUATION / ADAPTATION						
23	1	SOP and clinical utility of Churna Kalpana	4	15	Application	Adaptation
23	2	Standardization, analytical profiling, and clinical application of Kshara	5	16	Application	Guided response
23	3	Lavana Kalpana with special reference to pharmaceutico-analytical aspects and therapeutic attributes	5	16	Application	Guided response
23	4	Satva Kalpana SOP and clinical significance	5	15	Evaluation	Guided response
23	5	Therapeutic attributes of Masi Kalpana	7	22	Synthesis	Adaptation

M	U	TITLE	OBJ	HRS	HIGHEST THINKING	HIGHEST DOING
PAPER 3 · MODULE 24 — THERAPEUTIC SIGNIFICANCE OF TOPICAL DOSAGE FORMS — 11 OBJ, 28 H, CEILING APPLICATION / ORIGINATION						
24	1	Ophthalmic, Otic, and Buccal dosage forms and their clinical applications1.SOP for Ophthalmic, Otic and buccal dosage forms	4	10	Application	Guided response
24	2	Standardisation of Lepa and Malahara preparations and its application in various disorders	3	6	Application	Guided response
24	3	Standard Operating Procedure (SOP) for Upanaha, Dhupa, Dhumapana, and Avachurnana Kalpa and their therapeutic applications	4	12	Application	Origination
PAPER 4 · MODULE 31 — REGULATORY ACTS AND RULES APPLICABLE TO AYURVEDIC DRUGS — 31 OBJ, 84 H, CEILING APPLICATION / MECHANISM						
31	1	Chapter I of the Drugs and Cosmetics Act, 1940 and Rules 1945	3	6	Comprehension	Mechanism
31	2	Chapter II - Regulatory Bodies	3	9	Comprehension	Mechanism
31	3	Chapters III, IV, and IV-A - Regulation of Drugs and Cosmetics	8	18	Application	Mechanism
31	4	Schedules to Drugs and Cosmetics Rules related to ASU drugs and cosmetics	5	16	Application	Mechanism
31	5	Pharmaceutical observation of drugs and cosmetics ASU drugs	3	10	Comprehension	Mechanism
31	6	Miscellaneous Regulatory Acts related to drugs and cosmetics	9	25	Comprehension	Mechanism
PAPER 4 · MODULE 32 — ENTREPRENEURSHIP AND START-UPS — 14 OBJ, 28 H, CEILING SYNTHESIS / ORIGINATION						
32	1	Entrepreneurship and Startup	3	4	Application	Mechanism
32	2	Process of entrepreneurship development	3	8	Comprehension	Origination
32	3	Project Proposal for new Invention/startup	2	3	Comprehension	Adaptation
32	4	Intellectual Property Rights and Patents	6	13	Synthesis	Mechanism

Ordered by semester and then paper, as a department teaches it, rather than by paper as Table 3 prints it. Both ceilings are shown because they move independently.

PART C · APPENDIX C4

Verb index — every verb the curriculum uses, and its level

The opening verb of all 577 objectives, sorted by the level that verb belongs to. The verb is what a student meets, which makes this the most practical table in the book.

LEVEL	NAME	VERBS THE CURRICULUM USES, WITH HOW OFTEN
Level 1	**Knowledge** Remember	identify 26, define 14, enlist 8, recognise 4, recognize 3, observe 2, record 2, recall 1, recite 1
Level 2	**Comprehension** Understand	discuss 110, describe 79, elaborate 8, illustrate 7, explain 2, summarize 2, present 1, classify 1
Level 3	**Application** Apply	demonstrate 137, perform 30, prepare 17, apply 9, conduct 4, carry 2
Level 4	**Analysis** Analyse	analyse 15, analyze 15, explore 5, review 4, compare 3, differentiate 2, examine 1, compile 1, investigate 1, select 1, categorize 1
Level 5	**Synthesis** Create	design 4, integrate 2, develop 2, construct 1, generate 1, create 1
Level 6	**Evaluation** Evaluate	evaluate 15, justify 7, interpret 5, appraise 5, assess 4, validate 1
Affective	**Attitude verbs**	debate 1, practice 1

Choose the level first, then take a verb from that row. Where an objective's verb sits at a different level from its code, Part B says which to pitch to.

PART C · APPENDIX C5

Kinds of knowledge across the course

The second axis of the 2001 scheme, counted across all 577 objectives.

KIND OF KNOWLEDGE	OBJECTIVES	HOW IT IS LEARNED
Factual	28	Rehearsal and self-testing. Only ten objectives here — this is not a memorisation course.
Conceptual	237	Explaining it aloud to somebody without notes.
Procedural	335	Carrying it out, more than once. **The dominant kind in this course**, and the reason reading cannot substitute for the practical hours.
Metacognitive	not used	Being made to look back at how you worked. Not asked for here.

An objective working on two kinds is counted under both, so the total exceeds 577. The kind is read from the wording of each objective; where the wording does not settle it, the label carries a small circle (°).

PART C · APPENDIX C6

All 141 practical training activities, mapped

Table 4 of the curriculum, with the level each activity is cited to serve. Practical training is where a student moves from performing with help to performing unaided.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 1 — CHRONOLOGICAL DEVELOPMENT AND TAXONOMY OF RASA DRAVYAS			
1.1	Demonstration of the application of the Mendeleev Periodic table in mapping Rasadravyas	Doing 2 · Guided response	The teacher will allot each student at least one category from the classification of Rasadravya or one drug from each category. Each student will map minerals/Metals to the Periodic Table and search the elemental properties of minerals/metals mentioned in Rasashastra. Students will locate these metals on the Mendeleev Periodic Table, noting their atomic numbers, groups, and periods. Each student will compare and analyze the traditional name with the Periodic table, along with other properties, etc.
1.2	Demonstration of different Rasadravyas mentioned in different classics and analysis of the classification of Rasa Dravyas	Doing 2 · Guided response	The teacher will demonstrate different Rasadravyas mentioned in different classics based on classical Grahya- Agrahya Lakshanas, compare them, and analyze. The teacher helps the students to identify similarities, differences, and the evolution of thought in Rasashastra and its dravyas.
1.3	Geological Mineralogical Identification of Rasadravyas	Doing 2 · Guided response	The teacher will demonstrate the identification of selected Rasadravya using geological/ mineralogical characters.
1.4	Demonstration of Krutrima and Pratinidhi Rasadravyas	Doing 2 · Guided response	The teacher will demonstrate the preparation of a Krutrima Rasadravya and discuss the steps involved, the rationale behind the process, and the therapeutic benefits of the resulting product. The teacher will encourage students to reflect on the challenges of identifying Krutima and Pratinidhi substances and the significance of these classifications in ensuring the integrity of Rasashastra formulations.
MODULE 2 — PUTA AND SAMSKARA (TRANSFORMATIVE PROCESSES OF DRUG MANUFACTURING)			
2.1	Demonstration and comparison of the temperature pattern and total duration of different	Doing 2 · Guided response	—
2.2	Demonstration of the art of Gandhaka Jarana: A Practical Exploration of the Rasa Jarana Process, Understanding its Role and Significance in Rasashastra	Doing 2 · Guided response	The teacher will demonstrate the Gandhaka Jarana procedure using a suitable method and explain the role and significance of Rasa Jarana. Students will then conduct the practical and present their findings at the end, highlighting their understanding of the process.
2.3	Demonstration of Amritikarana and Lohitikarana: Traditional Methods for Mica Processing in Rasashastra	Doing 2 · Guided response	The teacher will demonstrate the procedures of Abhraka Amritikarana and Abhraka Lohitikarana using a selected method. Students will observe, record their observations, and present their findings and insights at the end of the practical.
2.4	Demonstration and Observation of Satva Patana	Doing 2 · Guided response	The teacher will select any one ore sample used in Rasashastra and demonstrate the procedure of Satva Patana. The students will observe and present their findings at the end of the practical.
MODULE 3 — PARADA (MERCURY) AND ITS PROCESSING: ANCIENT WISDOM AND SCIENTIFIC ADVANCES			
3.1	Demonstration of Samanya and Vishesh Shodhana procedures of Parada	Doing 2 · Guided response	The teacher will demonstrate to each student the Samanya and Vishesha Shodhana of Parada by one reference. The teacher will help the students to observe the changes in the Parada before and after Shodhana, along with different stages of the process.
3.2	Parada Jarana, Parada Murchana and Bandhana processes	Doing 2 · Guided response	The teacher will demonstrate the Parada Jarana, Parada Murchana Bandhana by any one method. The students will note down the observations and record the findings

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
3.3	Demonstration of the Parada Samskara procedures- Swedana, Mardana, Murchana, and Utthapana	Doing 2 · Guided response	The teacher will demonstrate the selected procedures of Parada Samskara like, Swedana, Mardana, Murchana, and Utthapana. The observations, like changes in the Parada, total duration, temperature range, and yield, will be noted down by the students. Following this, the students will perform these Samskaras under the direct observation of the teacher, record their findings, and discuss.
MODULE 4 — MAHARASA DRAVYA, ITS PROCESSING AND UTILITY			
4.1	Analysis and Verification of Maharasa Dravya Samples. The students will identify and aut	Doing 2 · Guided response	—
4.2	Application of Shodhana, Marana, and Other Pharmaceutical Procedures for Abhraka and Vaikranta	Doing 2 · Guided response	The students will perform various procedures, such as Shodhana, Dhanyabhrakikarana, and Marana, related to Abhraka/Vaikranta under the teacher's supervision, following a selected method. Relevant observations and yield will be recorded, presented, and discussed in class. The discussion will include the topics like, basis of each type of sodhana/marana or each bhavana Dravya etc
4.3	Practical Application of Shodhana, Marana, and Other Pharmaceutical Procedures for Makshika and Vimala	Doing 2 · Guided response	The students will perform various procedures, such as Shodhana and Marana, related to Makshika and Vimala, using a selected method. Relevant observations and yield will be recorded, presented, and discussed in class.
4.4	Practical Application of Shodhana, Marana, and Other Pharmaceutical Procedures for Shilajatu and Sasyaka/ Tuttha	Doing 2 · Guided response	The students will perform various procedures, such as Shodhana and Marana, related to Shilajatu and Sasyaka/Tuttha, using a selected method. Relevant observations and yield will be recorded, presented, and discussed in class.
4.5	Practical Application of Shodhana, Marana, and Other Pharmaceutical Procedures for Chapala/Kharpara	Doing 2 · Guided response	The students will perform various procedures, such as Shodhana and Marana, related to Chapala/Kharpara, using a selected method. Relevant observations and yield will be recorded, presented, and discussed in class.
4.6	Physico-Chemical Characterization of Bhasma/Shodhita Products in Maharasa Dravya categor	Doing 2 · Guided response	—
MODULE 5 — CRITICAL APPRAISAL OF UPARASA DRAVYA AND SADHARANA RASA DRAVYA			
5.1	Authentication of Uparasa Dravyas	Doing 2 · Guided response	The teacher will select a drug from Uparasa category and will demonstrate the identification and authentication of the selected drug using classical Grahya Lakshana and mineralogical references with cross-verification through reference samples
5.2	Demonstration of Shodhana, Marana and other pharmaceutical procedures for Uparasa	Doing 2 · Guided response	The students will perform various procedures such as Shodhana, Marana, Pishti Nirmana etc related to Uparasa Dravyas under teacher's direct supervision, following a selected method. Relevant observations and yield will be recorded, presented, and discussed in class. The students may be allotted a selected Shodhana/Marana procedure of Uparasa Dravya mentioned in UG syllabus. The students will demonstrate the practical to UG students and discuss
5.3	Demonstration of Quality Control testing for the products of Uparasa Dravya	Doing 2 · Guided response	The teacher will select a processed drug from Uparasa category, such as a Shodhita Dravya, Bhasma or Pishti and will demonstrate the quality testing procedure using organoleptic, physico-chemical parameters and instrumental methods. The students will observe the procedures, record the findings, present the same and discuss
5.4	Identification and authentication of Sadharana Rasa Dravyas	Doing 2 · Guided response	The teacher will select a drug from Sadharana Rasa category and will demonstrate the identification and authentication of the selected drug using classical Grahya Lakshana, mineralogical characters and physico-chemical properties. The sample will be cross-verified with the reference samples.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
5.5	Demonstration of Shodhana, Marana and other pharmaceutical procedures for Sadharana Rasa	Doing 2 · Guided response	The students will perform various procedures such as Shodhana, Marana, Satva Patana etc related to Sadharana Rasa Dravyas under teacher's direct supervision, following a selected method. Relevant observations and yield will be recorded, presented, and discussed in class. The students may be allotted a selected Shodhana/Marana procedure of Sadharana Rasa Dravya mentioned in UG syllabus. The students will demonstrate the practical to UG students and discuss
5.6	Demonstration of Quality Control testing for the products of Sadharana Rasa Dravya	Doing 2 · Guided response	The teacher will select a processed drug from Sadharana Rasa category, such as a Shodhita Dravya, Bhasma and will demonstrate the quality testing procedure using organoleptic, physico-chemical parameters and instrumental methods. The students will observe the procedures, record the findings, present the same and discuss
MODULE 6 — CRITICAL APPRAISAL OF DHATU AND UPDHATU			
6.1	Demonstration of authentication of Dhatu Dravya	Doing 2 · Guided response	The teacher will select a Dhatu (metal) and demonstrate the identification and authentification of the selected Dhatu/metal using both classical and modern identification techniques using Grahya Lakshana and mineralogical methods. The students will not the observations and discuss
6.2	Pharmaceutical processing (Shodhana & Marana) of metals	Doing 2 · Guided response	The teacher will select a Dhatu(metal) and explain the pharmaceuticl process for that metal like, Shodhana and Marana. The students will perform these pharmaceutical procedures under the guidance of the teacher. Later, the students will demonstrate the relevant pharmaceutical process of a metal to the UG students as per the NCISM syllabus, describe and discuss under the direct sipervision of the teacher
6.3	Quality control of metallic preparations	Doing 2 · Guided response	The teacher will select a processed drug from Dhatu category, such as a Shodhita Dravya or Bhasma and will demonstrate the quality testing procedure using organoleptic, physico-chemical parameters and instrumental methods. The students will observe the procedures, record the findings, present the same and discuss
6.4	Identification of Upadhatu samples	Doing 2 · Guided response	The teacher will select a drug from Upadhatu category and will demonstrate the identification and authentication of the selected drug using classical Grahya Lakshana and mineralogical references with cross-verification through reference samples
6.5	Shodhana and Marana of Upadhatu	Doing 2 · Guided response	The students will perform various procedures such as Shodhana and Marana related to Upadhatu Dravyas under teacher's direct supervision, following a selected method. Relevant observations and yield will be recorded, presented, and discussed in class. The students may be allotted a selected Shodhana/Marana procedure of UpardhatuDravya mentioned in UG syllabus. The students will demonstrate the practical to UG students and discuss
6.6	Demonstrate the Quality control of sub-metallic formulations	Doing 2 · Guided response	The teacher will select a processed drug from Upadhatu category, such as a Shodhita Dravya, Bhasma or a formulation containing Upadhatu Bhasma and will demonstrate the quality testing procedure using organoleptic, physico-chemical parameters and instrumental methods. The students will observe the procedures, record the findings, present the same and discuss
MODULE 7 — SUDHA VARGA, SIKATA VARGA, VISHA VARGA DRAVYAS			
7.1	Demonstration of Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya	Doing 2 · Guided response	The teacher will select a drug from Sudha Varga and explain the Standard Operating Procedure (SOP) for Shodhana, Marana, and Pishti Nirmana based on a selected classical reference. Under direct supervision, students will then carry out Shodhana, Marana, and Pishtikarana of the selected drug, following the SOP. Students will record their observations and engage in a discussion.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
7.2	Demonstration of the quality testing procedures for Bhasma/Pishti of Sudha vargeeya Dravya	Doing 2 · Guided response	The teacher will select a sample of Bhasma/Pishti from Sudha Varga Dravya and demonstrate quality testing methods for the product, covering organoleptic, physicochemical parameters, and NPS tests. Students will observe the demonstration and record their observations.
7.3	Demonstration of Shodhana, marana and Pishti Nirmana of Sikata varga Dravya	Doing 2 · Guided response	The teacher will select a drug from Sikata Varga category and explain the Standard Operating Procedure (SOP) for Shodhana, Marana, and Pishti Nirmana based on a selected classical reference. Under direct supervision, students will then carry out Shodhana, Marana, and Pishtikarana of the selected drug, following the SOP. Students will record their observations and engage in a discussion.
7.4	Demonstration of the quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya	Doing 2 · Guided response	The teacher will select a sample of Bhasma/Pishti from Sikata Varga Dravya and demonstrate quality testing methods for the product, covering organoleptic, physicochemical parameters, and NPS tests. Students will observe the demonstration and record their observations.
7.5	Demonstration of Shodhana Process of Visha Dravya	Doing 2 · Guided response	The teacher will select a Visha Dravya and demonstrate one of its Shodhana processes, explaining the Standard Operating Procedure (SOP) involved. Students will then perform the Shodhana process under the teacher's direct supervision, adhering to the SOP.
7.6	Demonstration of the analytical testing process for Shodhita Visha dravya	Doing 2 · Guided response	The teacher will select a Visha Dravya and demonstrate the testing of the sample before and after Shodhana, using parameters such as organoleptic characters, physico-chemical parameters, qualitative phytochemical tests, and Thin Layer Chromatography (TLC).
MODULE 8 — APPRAISAL OF RATNA AND UPARATNA AND THERAPEUTIC SIGNIFICANCE OF PISHTI AND DRUTI KALPANA			
8.1	Mineralogical identification of precious and semiprecious stones under Ratna and Uparatna varga	Doing 2 · Guided response	The teacher will select a precious or semi-precious stone and demonstrate its identification using identification keys. The teacher will then discuss the physico-chemical properties and crystallography of the identified specimen.
8.2	Demonstration of SOP of Pishti Kalpana of Ratna-Uparatna	Doing 2 · Guided response	The teacher will select a drug from Ratna/Upartna Varga and instruct students to prepare its Pishti under their supervision and guidance. Students will then prepare the Pishti according to the Standard Operating Procedure (SOP) and present their observations.
8.3	Demonstration of different methods of Gandhaka Druti and Gandhaka taila	Doing 2 · Guided response	Under the direct supervision of the teacher, students will prepare Gandhaka Druti and Gandhaka Taila using various classical and modified methods as instructed. The students will then discuss their observations, note the physico-chemical characteristics, and explore the therapeutic applications of the prepared formulations.
8.4	Demonstration of the Pharmaceutical modification of Gandhaka Druti and Gandhaka Taila	Doing 2 · Guided response	The teacher will demonstrate the pharmaceutical modification of Gandhaka Druti/Taila into ointment/cream using suitable excipients. The students will observe the procedure and prepare the same under the guidance of the teacher.
MODULE 9 — SCIENTIFIC FUNDAMENTALS OF AYURVEDIC PHARMACEUTICAL PREPARATIONS			
9.1	Demonstration of unique Panchavidha Kashaya Kalpana	Doing 3 · Mechanism	The teacher will instruct the students to explore the classical references for unique Panchavidha Kashaya Kalpana like, Agnisiddha and Anagni Siddha Swarasa, Saptavidha Kashaya Kalpana, Sagni and Niragni Hima Kalpana etc along with the commentaries, and outline all steps of each preparation. The students will perform the fselected practical under the observation and guidance of the teacher. Each student will perform a different formulation, record their observations, and discuss

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
9.2	Demonstration of the application of QAQC parameters used for quality control of herbal drugs	Doing 2 · Guided response	The teacher will demonstrate the quality testing protocol of raw materials like organoleptic characters, estimation of foreign matter, bulk density, tapped density, extractive values, ash values, specific gravity, and refractive indexetc selecting an appropriate raw drug sample. The students will carry out the practical using different samples under the guidance of the teacher, record their findings, and compare them with the pharmacopoeial standards
9.3	Demonstration of formulations to understand uniqueness of proportion of the ingredients	Doing 3 · Mechanism	—
9.4	Assessment of Phytochemicals in the prepared dosage form by TLC The faculty will select	Doing 2 · Guided response	—
9.5	Demonstration of Kwatha Kalpana considering its process optimization	Doing 2 · Guided response	The faculty will choose a specific kwatha from a classical text for each student. Each student will demonstrate the methodical preparation of allotted kwatha. Everyone will observe the changes that occur by varying the proportions of water, temperature, and time. To explore various factors that affect the optimization of the process, the same kwatha preparation will be conducted multiple times to note the differences. Each student will be given drugs with different hardness and asked to prepare Kashaya using different ratio of water. The students will prepare kashaya samples, compare the time taken for the preparation, comparatively analyse them using organoleptic paramaters and basic physico-chemical parameters and present their findings.
MODULE 10 — PRINCIPLES OF PHARMACEUTICAL FORMULATION			
10.1	Demonstration of Samskara and alterations in the qualities observed during the preparati	Doing 2 · Guided response	—
10.2	Demonstration of formulations prepared with specific Aushadhi Gana and various classes(	Doing 2 · Guided response	—
10.3	Demonstration of Dosage Form	Doing 2 · Guided response	Design. Faculty will arrange a group discussion on the Principles of dosage form design in Modern Pharmaceutical Science 1) Pharmaceutical and Formulation considerations for dosage form design 2) Biopharmaceutical and Pharmacokinetic considerations for dosage form design Each student will develop and demonstrate any 2 formulations(for example- Mouthwash from Triphala decoction, Tooth paste from Bakula Pushpa churna, Ointment from Vrana Shodhana Taila, Cream from Kunkumadi taila, etc.)
10.4	Demonstration and analysis of the importance of preservatives, additives, excipients, fi	Doing 2 · Guided response	—
MODULE 11 — MANA PARIBHASHA (METROLOGY IN AYURVEDIC PHARMACEUTICAL PREPARATION)			
11.1	Application of Knowledge of mana Paribhasha in the preparation of various formulations	Doing 2 · Guided response	The students will design a chart of classical manaparibhasha applied to Ayurvedic Pharmaceutical preparation of Various dosage forms. They will study the following examples for evaluating the use of mana paribhasha in the preparation of the following classical formulations. Accordingly, each student will prepare a selected formulation from the list given below or any other relevant formulation under the direct observation of the teacher. 1.Bhagottara churna 2.Bahushal Guda 3.Yogaraja guggulu 4.Kutajavaleha 5.Kasisadya ghruta 6.Kutajarishta 7.Lauhadi Rasayana - critically appraise the measurements mentioned for Ayurveda 8. Pathya Ghruta

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
11.2	Unit Conversion for Pautava and Druvaya Mana	Doing 3 · Mechanism	Each student will study the Unit Converter for Pautava and Druvaya Mana Each student will select any 10 Dry samples of various sources and observe their mana based on unit conversion. A discussion will be held on the observations and application of this information.
11.3	Preparation of Formulations using the Volumetric Method of Measurement	Doing 2 · Guided response	Faculty will describe the differences between volumetric and gravimetric methods of measurement for various drugs, allowing students to explore the application of the volumetric method in preparing classical formulations. Preparation of Amruta Kwatha, Bala Kwatha, Kutaja Kwatha, Sharavari Kwatha, and Bhrungaraja Kwatha will be done using the volumetric method for measurement. Faculty will elaborate on the difference in the total amount of water required for the formulations when water is measured using the volumetric method. Verify the difference observed between the total amount of water required for the formulation when water is taken by the volumetric method.
11.4	Preparing formulations with a diverse range of raw materials. Faculty will elaborate on	Doing 3 · Mechanism	—
11.5	Demonstration of the machinery used for large-scale production in Ayurvedic pharmacies, with their mechanism	Doing 3 · Mechanism	The department will arrange a visit to a pharmacy where bulk production is carried out using advanced machinery. OR A video demonstration will be arranged to cover the assembly and operation of the equipment used for bulk production. Each student will-Observe and understand the working of these machines, along with understanding their working principle. Operate these machines in the teaching Pharmacy unit. Assess the modifications and impacts of bulk production on the final product. Explain the structure and operation of all these machines. Discuss with production unit personnel about the advantages and disadvantages of the machines used in Ayurvedic pharmacies.
MODULE 12 — PROCESS VALIDATION PROTOCOL			
12.1	Develop Process validation Protocol and demonstrate the preparation of formulations foll	Doing 2 · Guided response	—
12.2	Develop Process validation Protocol and demonstrate the preparation of formulations following SOP Each	Doing 2 · Guided response	Student will conduct practical sessions for undergraduate students(Minimum 3 for each PG student) to demonstrate, observe, assist, and perform the following practical as part of their NCISM curriculum. 1) Arka- Gomutra arka 2) Kshara-Palasha kshara 3) Lepa- Dashanga lepa 4) Malahara- Gairikadya malahara 5) Upanaha- Atasi upanaha For Netra Kalpana- Seka, Ashchyotana, Pindi, Bidal, Tarpana, Anjana, Putapaka- (Observe in Shalakya Netra department)
MODULE 13 — KRUTANNA (AYURVEDIC FOOD PREPARATIONS)			
13.1	Formulations from Classics prepared with aahara Dravya	Doing 2 · Guided response	The teacher will instruct each student to refer the classical texts for ingredients and proportions of all the formulations listed here. The students will prepare any 4 selected formulations following SOP. under the guidance of teacher. 1) Toya varga- Shadaga Paniya, Ushnodaka 2) Kshira varga- Shunthi Kshirapaka, Takrarishta, Tila navaneeta yoga, Panchagavya ghruta 3) Ikshu varga- Sharkara kalpana, Guda chatushtaya gutika 4) Madhu varga- Madhvasava, Madhu shukta 5) Taila Varga- masha Saindhava Taila, hingwadya Taila 6) Madya Varga - Yava Sura, Mardika, Varuni

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
13.2	Demonstration of therapeutic diet formulations	Doing 2 · Guided response	The teacher will instruct the students to refer the classical texts and explore disease specific diet formulations, understand their classical description and specific therapeutic utility. The students will prepare any 4 formulations following SOP under the supervision of the teacher. They may select the formulations from the list mentioned below or any other formulations as instructed by the teacher. 1) Jvara- - (Charaka Chi. 3/ 178 to 196) 2) Rakta pitta Tarpana- (Cha. Chi 4/33-39) 3) Aushadhi Siddha Yavagu - (Cha. Su. 2- summarize all 28 Yavagu) Prepare Gavedhuka Yavagu if material is available. 4) Amavata- Sapta mushtika Yusha- (Sharangadhara Ma. Kh. 2/164) 5) Chhardi - Draksha/ Kapittha Panaka (Ch.chi 20/36) 6) Atisara- Appraise formulations of Yavagu, Vilepi, Khada, Yusha, Rasaudana and their administration in different clinical conditions of atisara.(Ch.chi 19/23-41)
13.3	Demonstrate Protocol for the selected Aahara Kalpana-Each student will Demonstrate/ Obse	Doing 2 · Guided response	—
13.4	Demonstration of the preparation of novel Ayurvedic food products	Doing 2 · Guided response	The teacher will demonstrate selected novel Ayurvedic food products such as biscuits/cookies, candy/chocolate and granules and explain the SOP. The students will observe the procedure and prepare the same under the direct observation and guidance of the teacher. Based on the ingrdients the students will calculate the nutritional values of the products prepared
13.5	Demonstration of food processing techniques used in current Ayurvedic food Industries	Doing 2 · Guided response	The department will arrange a visit to the nearest food industry after getting prior permission. The students will observe the food processing technology, quality check, packaging etc. They will lcritically assess the potential effects of processing techniques on food quality. OR A visit will be arranged to the Ayurvedic pharmaceutical unit that prepares nutraceutical products, where the students will learn the entire process of manufacture, quality check, packaging etc. OR. Visit Ayurvedic food manufacturing units, ensuring that at least one is GMP-certified and FSSAI-approved, and observe various food processing OR A video demonstration of a food processing unit including various techniques, machinery, QC methods, packaging, research etc. The students will observe the video, record their findings, present them, and discuss. The students will visit the factory where freeze drying and vacuum drying, and other drying technologies are used, and understand the technique or a video demonstration of the same may be arranged. Students will review ready-to-cook (RTC) and ready-to-drink (RTD) Ayurvedic foods in the market.
MODULE 14 — AYURVEDIC FORMULATIONS FOR HOLISTIC WELLBEING, VITALITY AND BODY CARE			
14.1	Demonstration of Rasayana Formulations. Each student will prepare a selected Rasayana product under the supervision of the teacher, following the SOP. The teacher will allot any of the following products or any other Rasayana formulation. Chyavanprash Chaturvidha Triphala Rasayana Vrushya payasa yoga Apatyakara Ghruta Every year	Doing 3 · Mechanism	Faculty can allot different formulations selected from the classical texts.
14.2	Demonstration of Body care products. The teacher will demonstrate the preparation of a s	Doing 3 · Mechanism	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
14.3	Demonstration of the preparation of Modified Cosmeceuticals	Doing 2 · Guided response	The teacher will ask each student will search e-resources to assess the expectations of end users from the herbal cosmetics Industry. Based on the information collected, each student will design at least one their own innovative modified Cosmeceutical formulation under the guidance of the teacher developing an SOP. The students will note down the challenges faced, corrective actions done during the preparation, present their findings and discuss
MODULE 15 — DISEASE CONDITION SPECIFIC SELECTION OF FORMULATION			
15.1	Demonstration of formulations used in various branches of Ayurveda. Each student will prepare at least one formulation from each group under the direct observation of the teacher: 1.Kaumarbhrtya- Balachaturbhadra Rasa, Preenana modaka 2. Striroga- Pushyanuga Churna, PhalaGhruta 3.Garbhini- Garbhapal Rasa, Formulations from Masanumasiki Garbhini paricharya 4. Sutika- Bala taila, stanyashodhana Yusha 5. Shalyatantra- Kshara Sutra, Vranashodhan Taila, Murivenna 6. Shalakya Tantra- Bilwa taila / Irimedadi taila / Chandrodaya Varti 7.Kayachikitsa- Yavani shadava Churna, Vasaghruta 8. Panchakarma- Niruha Basti Samyojana, Pradhaman Nasya Yogas, Ghreya Virechana Yoga 9.Rasayana-Bhallataka Taila, Medhya Vachayoga 10.Vajikarana- Vrishya Payasa Yoga, Shatavari Ghruta	Doing 2 · Guided response	The teacher will explain and discuss the unique combination of ingredients, their proportions, SOP, endpoint, Matra, Anupana, and time for administration, probable mode of action, therapeutic efficacy, and relevant research articles
15.2	Survey the different dosage forms used by local health practitioners. The teacher will i	Doing 2 · Guided response	—
15.3	Demonstration of frequently used formulation for Shodhana and Shamana Chikitsa of common	Doing 2 · Guided response	—
MODULE 16 — CURRENT TRENDS IN AYURVEDIC PHARMACEUTICAL INDUSTRY.			
16.1	Market Status of Ayurvedic Pharmacy Industry	Doing 2 · Guided response	Students will-Conduct surveys, interviews, and focus groups with consumers, retailers, manufacturers, and healthcare professionals. Resourses Utilize market reports, industry publications, and academic research to gather data and insights.
16.2	Internship / Visits to GMP-certified Ayurvedic pharmacy Units(Minimum 5 units- Small/ Medium/ Large)	Affective · Valuing	The department will arrange a visit/training (internship) in a GMP-certified pharmacy. The students will observe the daily work patterns in all sections of the Pharmacy and get an on-hand training in various sections (based on the availability). They will also learn about inventory management and the documentation pattern of the Pharmacy. They will submit the report after the training session.
16.3	Perform any 5 QC practicals on formulation prepared by yourself. Faculty should demonstr	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 17 — GUIDELINES FOR ADMINISTRATION OF RASOUSHADHI IN CLINICAL PRACTICE			
17.1	Demonstration of the rational prescription for a selected disease	Doing 2 · Guided response	The teacher will create a case scenario of a clinical condition and demonstrate the clinical prescription of treatment modalities of Ayurveda, such as Shodhana, Shamana, Rasayana, and Vajikarana. Based on the demonstration, students will follow the same procedure in another disease, present the same, and discuss with a rationale.
17.2	Demonstration of the need and role of Kshetrikarana	Doing 2 · Guided response	The teacher will demonstrate the identification of the patients requiring kshetrikarana and explain why kshetrikarana is required in that case; Students will observe and record the agni- dosha- dushya and readiness for the administration of rasaoushadhi in that case. Based on the demonstration, each student will be given a different case scenario. Students will analyse the case and explore the requirement of Kshetrikarana in that case and design the protocol of Kshetrikarana.
17.3	Demonstration of common pathya and apathya Ahara during rasa yoga sevana	Doing 2 · Guided response	The teacher will provide a list of common Ayurvedic Pathya and Apathya Ahara based on classical guidelines. The teacher will assign each student a different list of Rasayoga. The students will prepare charts for the given rasa yogas and suggest appropriate pathya-apathya Ahara accordingly.
17.4	Demonstration of the common pathya-apathya Vihara during rasa yoga sevana	Doing 2 · Guided response	The teacher will provide a list of common Ayurvedic Pathya and Apathya Vihara based on classical guidelines, such as sleep, daily regimen, seasonal modifications etc. The teacher will assign each student a different list of Rasayoga. The students will prepare charts for the given rasa yogas and suggest appropriate pathya- apathya Vihara accordingly
17.5	Determination of signs and symptoms of rasa vikaras and their management	Doing 2 · Guided response	The teacher will demonstrate a case study of the patient experiencing side effects of Rasa Oushadhies. The students will identify and record the key signs and symptoms and analyse them. Accordingly, they will prepare at least one formulation for the management of the side effects.
17.6	Comparative assessment of Ashuddha and Shuddha Bhasma effects	Doing 2 · Guided response	The teacher will create two sets of case scenarios, one with properly prepared Bhasma and the other with improperly prepared Bhasma. Students will present variations in clinical outcomes and adverse reactions.
17.7	Identification of heavy metal toxicity	Doing 2 · Guided response	The teacher will give the clinical description of a case scenario with suspected heavy metal toxicity. The student will identify the heavy metal responsible for the toxic symptoms.
MODULE 18 — PRAYOGIKA BHAISHAJYA KALPANA(CLINICAL APPLICATION OF PHARMACEUTICAL PRODUCTS)			
18.1	Demonstration of chrono therapeutic alignment of Oushadha Sevana Kala with patient–disea	Doing 2 · Guided response	—
18.2	Demonstration of Modulation of dosage form action by chronotherapeutic administration in Ayurveda	Doing 4 · Adaptation	The teacher will select a dosage form and demonstrate the suitable Oushadha Sevana Kala for the selected dosage form based on classical guidelines of Oushadha Sevana Kala and Chronotherapeutics.
18.3	Demonstration of appropriate Anupana for the selected dosage form in different clinical	Doing 2 · Guided response	—
18.4	Demonstration of the calculation of paediatric dose using different formulae. The teache	Doing 2 · Guided response	—
18.5	Demonstration of the Patient-centric evaluation of Matra. The teacher will demonstrate m	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
18.6	Re-analysing classical ayurvedic drugs for novel indications	Doing 2 · Guided response	The teacher will select a classical Ayurvedic formulation and list out the ingredients and therapeutic indications of the formulation. With the available modern research articles on the pharmacological insights of the formulation's ingredients, the teacher will propose and justify a potential new therapeutic use for that formulation.
MODULE 19 — SOP AND CLINICAL APPLICATION OF CHATURVIDHA RASAYANA			
19.1	DOAP of preparation and SOP of Kharaleeya Rasayana. The teacher will select a Khalveeya	Doing 2 · Guided response	—
19.2	Demonstration of the method of preparation based on SOP of parpati kalpana. The teacher	Doing 2 · Guided response	—
19.3	Demonstration of the method of preparation and SOP of pottali kalpana. The teacher will	Doing 2 · Guided response	—
19.4	Demonstration of the method of preparation and SOP of Kupipakwa Rasayana. The teacher wi	Doing 2 · Guided response	—
MODULE 20 — SOP AND CLINICAL APPLICATION OF AYASKRITI, LOUHA, AND MANDURA KALPA			
20.1	Demonstration of the SOP and preparation of Ayaskriti. The teacher will describe the com	Doing 2 · Guided response	—
20.2	Demonstration of the SOP and preparation of Louha Kalpa. The teacher will select a Louha	Doing 2 · Guided response	—
20.3	Demonstration of the SOP and preparation of Mandura Kalpa.. The teacher will select a Ma	Doing 2 · Guided response	—
20.4	Clinical application of mandura kalpa. The teacher will demonstrate the clinical applica	Doing 2 · Guided response	—
MODULE 21 — CLINICAL APPLICATION OF RASOUSHADHI ACCORDING TO SROTAS AND IN SELECTED CHALLENGING CLINICAL CONDITIONS			
21.1	Demonstration of the clinical application of Rasoushadhi in selected Srotas, illustratin	Doing 2 · Guided response	—
21.2	Demonstration of the clinical applicability of Rasoushadhi in Medical Emergencies	Doing 2 · Guided response	The teacher will select a particular emergency condition and identify the applicable Rasa Oushadhi from Chaturvidha Rasayana. The teacher will then select a specific formulation and instruct the students to prepare it under their direct supervision. Students will record their observations and participate in a discussion. Following this, the teacher will demonstrate the rationality and probable mode of action of the selected formulation, drawing on classical properties, pharmacology of individual ingredients, and the method of preparation. Additionally, the teacher will analyze the findings in relation to published research works.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
21.3	Demonstration of role of Rasoushadhi in clinically challenging conditions	Doing 2 · Guided response	The teacher will select a relevant Rasoushadhi applicable in clinically challenging conditions, such as neurodevelopmental and neurodegenerative diseases. The teacher will then demonstrate the clinical application of the selected formulation, including the suitable Anupana (vehicle) and route of administration. Furthermore, the teacher will discuss the probable mechanism of action of the selected formulation, drawing on the classical properties and pharmacology of its ingredients. OR The teacher will demonstrate the therapeutic application of selected Rasoushadhies in managing specific, clinically challenging conditions within their respective outpatient departments (OPDs).
21.4	Demonstration of the role of Rasoushadhi as Rasayana in clinically challenging condition	Doing 2 · Guided response	—
MODULE 22 — YOGAS FROM DIFFERENT KALPANAS			
22.1	Demonstration of the preparation of Vati, Guggulu Kalpa, and their modified modern dosage forms	Doing 2 · Guided response	The teacher will select one formulation each from Vati and Guggulu categories. The students will prepare the selected formulations under the direct supervision of the teacher. Under the guidance of the teacher, the students will modify the selected formulation into tablet/capsule form using suitable excipients, present their observations, and discuss. Then the teacher will demonstrate the clinical utility of the prepared suppository based on pharmacological properties, classical descriptors, and the possible mode of action based on available published works.
22.2	Designing an S.O.P for the preparation of Ayurvedic suppository specifying its clinical utility based on Varti Kalpana	Doing 2 · Guided response	The Teacher will select a Varti formulation and instruct the students to prepare the formulation under the direct supervision of the teacher. The students will modify the Varti into modern dosage form suppository, using suitable excipient and mould under the guidance of the teacher. Then the Teacher will demonstrate the clinical utility of the prepared suppository based on pharmacological properties, classical descriptors, and the possible mode of action based on available published works.
22.3	Demonstration of the role of pharmaceutical additives and Samyak Paka of Avaleha	Doing 2 · Guided response	The teacher will assign each student to prepare an Avaleha, Paka, and Khanda Yoga and mark the critical end point based on Paka Lakshanas. The students will differentiate the Paka Lakshanas of Avaleha, Paka, and Khanda Kalpana and present their findings. The teacher will instruct the students to assess the authenticity and genuineness of the spices and condiments used in the preparation by comparing them with the pharmacopeial standards. Then the teacher will demonstrate how the pharmaceutical additives and leha paaka influence the clinical activity based on classical descriptors and using published research works.
22.4	Demonstration of S.O.P for a selected Sneha Kalpa which is widely used and determine its analytical profile	Doing 2 · Guided response	The teacher will assign each student a different Sneha Kalpa. The students will prepare the given formulation under the direct supervision of the teacher. The students will record the findings like, temperature changes, critical end points etc. The students will be instructed to analyze the prepared Sneha Kalpa using relevant quality control parameters and present their findings. Then the Teacher will demonstrate how the analytical profiling done matches with the pharmacopeial standards to ensure the quality of the Sneha Kalpa, which in turn will influence the clinical activity.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
22.5	Demonstration of S.O.P for the preparation of a selected Arka and Dravaka and create their analytical profile. The teacher will assign each student to prepare an arka and dravaka each. The students will prepare the given products following the SOP under the direct supervision of the teacher. The teacher will instruct the students to analyse the prepared products following relevant quality control parameters. The students will carry out the analysis accordingly under the guidance of the teacher and present their observations. Then the	Doing 2 · Guided response	Teacher will demonstrate how the pharmacopeial standards and classical descriptors help in determining the quality of the product, which in turn will influence the clinical activity.
22.6	Demonstration of S.O.P for preparing a selected sandhana Yoga and determine the analytical standards	Doing 2 · Guided response	The Teacher will select a Sandhana Yoga (Madhura and Amla) and assign the students to prepare the given formulation following SOP under the direct supervision of the teacher. The students will analyse the product after the completion of given formulation using relevant analytical parameters, including alcohol percentage. Then the teacher will demonstrate how the percentage of alcohol is significant in post-marketing surveillance. Note: It is expected that each batch will prepare a different Sandhana Kalpana, and the analysis will be carried out here only after the completion of fermentation.
MODULE 23 — SOP AND CLINICAL APPLICATIONS OF CHURNA KALPANA(POWDER DOSAGE FORMS)			
23.1	Demonstration of the SOP and quality analysis of a selected Churna yoga	Doing 2 · Guided response	The teacher will select different Churna Yogas according to the particle size (Sukshma churna, Yavakuta churna etc). The teacher will allot the selected Churna Kalpa to the students. The students will authenticate raw materials in the selected Yoga and will apply processing techniques like powdering, sieving using an appropriate mesh, and homogenous mixing of ingredients. Then the students will carry out the quality analysis of the prepared Churna following appropriate organoleptic, physico-chemical, particle size estimation, and other relevant parameters under the guidance of the teacher. The findings will be compared with the pharmacopoeial standards
23.2	Demonstration of the preparation of a selected Paneeya Kshara and Pratisaraneeya Kshara and their quality analysis	Doing 2 · Guided response	The teacher will assign students to collect and authenticate raw materials required for Kshara preparation. Students will apply processing techniques as per the provided SOP. Samples will be collected at the Paneeya Kshara and Pratisaraneeya Kshara stages. Under the teacher's guidance, collected samples will undergo quality testing using relevant parameters. The results will be compared with pharmacopeial standards to ensure compliance.
23.3	Demonstration of the preparation of a selected Lavana formulation following different methods and analytical profiling	Doing 2 · Guided response	The teacher will select a Lavana Kalpa and instruct students to prepare the product following the standard operating procedure (SOP). A Lavana Kalpa with multiple preparation methods (e.g., Sajala and Nirjala Narikela Lavana) will be chosen. Under the teacher's guidance, students will prepare the Lavana Kalpa, record observations, present, and discuss their findings. Additionally, students will analyze the prepared Lavana Kalpa using relevant quality parameters and compare the results with pharmacopeial standards. Alternatively, students may demonstrate the practical preparation of Lavana Kalpa to undergraduate students, as per the syllabus, under the teacher's supervision.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
23.4	Demonstration of the preparation of a selected Satva and performing its analytical profiling	Doing 2 · Guided response	The teacher will assign the students to collect and authenticate raw materials needed for the preparation of the selected Satva. Under the teacher's supervision, students will apply standard processing techniques for its preparation. The students will then perform various analytical tests as per pharmacopeial standards and create an analytical profile of the taken Satva.
23.5	Demonstration of the preparation of Masi and its analytical profiling	Doing 2 · Guided response	The teacher will select a Masi Kalpa prepared by different methods (Antardhuma and Bahirdhuma). Students will prepare Masi using these methods as per the SOP, guided by the teacher. Under the teacher's supervision, students will analyze the prepared Masi samples using relevant quality parameters, present their findings, and discuss.
23.6	Demonstration of the pharmaceutical modification of Masi Kalpana	Doing 2 · Guided response	The teacher will select a Masi Kalpa and demonstrate its modification into modern topical dosage forms, such as ointments, creams, and gels, using suitable bases and excipients. Students will observe the process and prepare the products under the teacher's supervision, then present their observations and discuss
MODULE 24 — THERAPEUTIC SIGNIFICANCE OF TOPICAL DOSAGE FORMS			
24.1	Demonstration of various Netra Kalpa and their clinical applications	Doing 2 · Guided response	The teacher will assign each student a unique Netra Kalpa to prepare under guidance. Students will then analyze its pharmacotherapeutic properties, identifying its effectiveness in eye diseases, and explain its probable mode of action based on pharmacological properties, classical descriptors, and research updates
24.2	Preparation of a novel dosage form using one of the classical lepa Yoga	Doing 2 · Guided response	The teacher will assign the students to authenticate raw materials in the selected Yoga. Then, the students will identify the pharmacotherapeutic properties of the selected Yoga, listing its ingredients, proportion, method of preparation, dose, mode of application, and explain its probable mode of action based on pharmacological properties, classical descriptors, and available research updates. The teacher will demonstrate the conversion of Lepa into a bandage using a suitable base. The students will prepare the same following the teacher's guidelines. Pharmaceutical modification of the classical Lepa into other novel forms like cream, ointment, gel, or liniment can also be selected for practical training.
24.3	Preparation of a selected Upanaha which is useful in treating Shopha	Doing 2 · Guided response	The teacher will assign the students to authenticate raw materials in the selected Yoga. Then, the students will identify the pharmacotherapeutic properties of the selected Yoga, which makes it effective as a bandage, and list the ingredients, proportion, method of preparation, dose, and mode of application. They will also explain the probable mode of action based on pharmacological properties, classical descriptors, and available research updates. Furthermore, students will prepare the formulation and create an analytical profile of the selected Yoga as per the Pharmacopeia standards to standardize the formulation.
MODULE 25 — ARTIFICIAL INTELLIGENCE IN DRUG DEVELOPMENT AND PHARMACOGENETICS; PHARMACOGENOMICS			
25.1	Demonstration of training in the basics of machine learning and its application	Doing 2 · Guided response	The teacher/experts in the field will give a Brief demo or guided tour of AI used cases (target identification, virtual screening)
25.2	Prediction of drug likeliness using machine learning	Doing 5 · Origination	Teacher/ experts will instruct/guide the scholars to load a dataset of small molecules (Ex. SMILES strings with molecular descriptors and bio-activity)
25.3	Demonstration of different 3D printers used in the drug delivery system	Doing 3 · Mechanism	The teacher/expert in the field will conduct a video demonstration of 3D printers used in the drug delivery system. Students will observe, record their findings, and present for discussion

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
25.4	Comparison of traditional dosage forms versus 3D printed ones regarding the structure and dissolution	Doing 4 · Adaptation	Teacher/experts in the field will conduct a video demonstration on the Comparison of traditional dosage forms versus 3D printed ones regarding the structure and dissolution
25.5	Demonstration of the prediction of Drug-Target Interaction	Doing 2 · Guided response	The teacher or expert in the field has to demonstrate Drug-Target Interaction Prediction by using tools like Swiss Target Prediction (Web-based), STITCH (Web-based, Binding DB (Database), Python + scikit-learn or Deep Purpose (for coding-based practical), etc.
MODULE 26 — PHARMACOLOGY			
26.1	Demonstration of pharmacological experiments using videos or simulation software	Doing 2 · Guided response	The teacher will demonstrate selected five pharmacological experiments like, study of different routes of drug administration in mice/rats, study of effect of hepatic microsomal enzyme inducers on the phenobarbitone sleeping time in mice, effect of drugs on ciliary motility of frog oesophagus, effect of drugs on rabbit eye, effects of skeletal muscle relaxants using rota-rod apparatus using videos or simulation software. Students will observe, discuss, and record the demonstration
26.2	Demonstrate the effect of drugs on animals by simulated experiments	Doing 4 · Adaptation	The teacher will demonstrate five selected pharmacological experiments like, effect of drugs on locomotor activity using actophotometer, anticonvulsant effect of drugs by MES and PTZ method, study of stereotype and anti-catatonic activity of drugs on rats/mice, study of anxiolytic activity of drugs using rats/mice and study of local anaesthetics by different methods. Students will observe, discuss and record the demonstration
MODULE 27 — NETWORK PHARMACOLOGY & REVERSE PHARMACOLOGY			
27.1	Demonstration of network pharmacology applications with the help of a video	Doing 2 · Guided response	The teacher will select a drug and demonstrate network pharmacology applications of the selected drug, with the help of a video demonstration, covering topics like polypharmacology, biological networks, computational approaches, drug repurposing, and systems pharmacology. The students will observe and record findings
27.2	Demonstration of reverse pharmacology applications with the help of a video	Doing 2 · Guided response	The teacher will select a drug and demonstrate reverse pharmacology applications of the selected drug, with the help of a video demonstration including the steps (1 hour for each step) like, Bioactivity Screening, Target Identification, Mechanism of Action Studies, Systems Biology, Clinical Observations, Computational Modelling, Animal Models, Safety and Toxicity Assessment, Ethnopharmacology and Phytochemical Analysis
MODULE 28 — PHARMACEUTICAL QUALITY ASSURANCE			
28.1	Demonstration of the process of mock GMP Inspection	Doing 2 · Guided response	The teacher will demonstrate the process of a mock GMP inspection by assigning different roles to the students, as QA officers and pharmacy personnel. The students will perform audits, check documentation, cleanliness, personnel hygiene, equipment logs etc
28.2	Demonstration of GMP Hierarchy and Responsibilities	Doing 2 · Guided response	The teacher will demonstrate GMP Hierarchy and assign Responsibilities, such as Quality Assurance (QA), Quality Control (QC), Production, Warehousing, Maintenance, Regulatory Affairs, Top Management, etc, to each student. The students will present their responsibilities, experiences, and challenges
28.3	Demonstration of Identification and Classification of Packaging Materials	Doing 2 · Guided response	The teacher will demonstrate a range of containers and packaging samples (e.g., glass bottles, blisters, plastic containers, aluminum foil, cartons) to the students, and the students will classify them as primary or secondary packaging and identify material type (HDPE, glass, PVC, etc.) and Record suitability for different dosage forms

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
28.4	Demonstration for SOP Writing of different equipment	Doing 2 · Guided response	The teacher will demonstrate SOP for an equipment or a lab process (e.g., pH measurement, reagent preparation), and the students will be asked to write a Standard Operating Procedure (SOP) of the demonstrated instrument
MODULE 29 — PHARMACY PRACTICE			
29.1	Drug Information and Case Analysis	Doing 2 · Guided response	The teacher will demonstrate a systematic approach to drug information queries using primary, secondary, and tertiary sources. The teacher will interpret lab reports to optimize drug therapy in real patient cases.
29.2	Prescription Handling and Dispensing	Doing 2 · Guided response	The teacher will demonstrate Prescription Handling, including legal compliance, checking for completeness, and identifying potential drug-drug interactions.
29.3	Data Interpretation and Pharmacokinetic Calculations	Doing 2 · Guided response	The teacher will select a formulation and demonstrate pharmacokinetic formulas, including volume of distribution, clearance, elimination constant, half-life, and calculation of loading and maintenance doses using nomograms and tailored regimens
29.4	Formulary Review and Management	Doing 2 · Guided response	The teacher will demonstrate selected case-based drug monographs, including discussion of cost, efficacy, and safety, with examples. The students will follow the guidelines and review different monographs and present.
MODULE 30 — PHARMACEUTICAL MICROBIOLOGY			
30.1	Demonstration of dry heat sterilization method	Doing 2 · Guided response	The teacher will select glassware contaminated with microbes based on the swab test and demonstrate the sterilization of the contaminated glassware using dry heat. The teacher will then demonstrate sterility testing of the glassware after dry heat sterilization. Additionally, the teacher will compare dry heat with moist heat (autoclaving) methods, discussing their applications and limitations. Students will observe the demonstration and then perform the dry heat sterilization process independently as part of experiential learning. OR A video demonstration of the dry heat sterilization method will be arranged. The students will observe the video, record the findings, present the same, and discuss
30.2	Demonstration of Aseptic Technique Practice for the transfer of microbes	Doing 2 · Guided response	The teacher will demonstrate proper aseptic techniques, including the use of sterile gloves, flame sterilization, and safe working practices near a Bunsen burner or in a laminar flow cabinet. The teacher will also demonstrate the aseptic transfer of microbes from one medium to another. OR A video demonstration of the aseptic technique practice in the microbiology lab will be arranged. The students will observe the video, record the findings, present the same, and discuss
30.3	Demonstration of Personnel Contamination Assessment	Doing 2 · Guided response	The teacher will demonstrate how personnel can be a source of contamination. Student volunteers will be asked to press their unwashed hands onto nutrient agar plates. The results will be compared with plates touched after hand washing or sanitizing. The plates will be incubated, and the differences in colony growth will be observed. The teacher will then explain the critical role of hand hygiene in contamination control. OR A video demonstration on personnel contamination assessment in a laboratory setting will be arranged, showcasing proper attire, hand hygiene, and awareness of potential contamination sources. Students will observe the video, record their findings, and participate in a class discussion.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
30.4	Demonstration of Different Types of Spoilage	Doing 2 · Guided response	The teacher will display spoiled food or pharmaceutical preparations. The teacher will then demonstrate the examination of food samples (e.g., bread, milk, fruit juice) or pharmaceutical preparations stored under different conditions (room temperature, refrigerated, exposed to air). Students will be instructed to note changes such as discoloration, mold growth, gas bubbles, off-odors, and texture changes. Over several days, students will record their observations and understand how microbial spoilage manifests physically and chemically.
MODULE 31 — REGULATORY ACTS AND RULES APPLICABLE TO AYURVEDIC DRUGS			
31.1	Recording their observations regarding hypothetical issue through Mock composition exercise of demonstration of the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists	Doing 2 · Guided response	The teacher will demonstrate the roles of DTAB (Drug Technical Advisory Board) composition, including representatives from DGHS, DCGI, pharmacologists, and others, and conduct a mock meeting to discuss a hypothetical issue, such as the approval of a new drug or banning a drug. The students will participate in the mock meeting, following the teacher's guidelines, and record their observations.
31.2	Mock Import License Application	Doing 2 · Guided response	The teacher will demonstrate filling a sample Form 10D for importing an ASU product, such as a herbal capsule, and show mock supporting documents to be attached, including the Free Sale Certificate and lab test report. The students will record their observations and engage in a class discussion.
31.3	Demonstration of case scenarios of Misbranded/spurious drugs and cosmetics	Doing 2 · Guided response	The teacher will demonstrate case scenarios related to misbranded or spurious drugs/cosmetics, explaining how to identify misbranded products based on misleading labelling and spurious products through fictitious company names and other deceptive tactics. The students will record their findings and observations.
31.4	Application for a license of a proprietary formula	Doing 2 · Guided response	The teacher will demonstrate how to formulate a treatment protocol for a selected disease, utilizing Schedule I and published literature. The teacher will also explain the process of applying for a license and showcase the necessary documents and study reports required for the application.
31.5	Demonstration of Landmark Judgments under the NDPS Act	Doing 2 · Guided response	The teacher will select 1-2 landmark judgments under the NDPS Act and demonstrate how courts have interpreted key provisions, such as Section 27 (consumption) and Section 21 (possession). The discussion will focus on crucial aspects, including quantity differentiation (small vs. commercial), presumption of guilt, and bail provisions under Section 37.
31.6	Case Law demonstration on the Factory Act	Doing 2 · Guided response	The teacher will present a notable case study related to labor law violations, focusing on issues such as worker safety, overtime pay disputes, or factory licensing. Students will document the case in a standard format and engage in a class discussion.
31.7	Case Study Demonstration under the Drugs and Magic Remedies Act	Doing 2 · Guided response	The teacher will demonstrate a case study illustrating the application of the Drugs and Magic Remedies Act.
MODULE 32 — ENTREPRENEURSHIP AND START-UPS			
32.1	Demonstration of the process of development of a basic business model as per the template	Doing 2 · Guided response	Students will design a new startup concept, develop a comprehensive business plan, create a prototype, and pitch their idea. They will discuss, brainstorm innovative ideas, and select one idea to develop further. Each student will prepare a written business plan, including market research, financial projections, and marketing strategies. They will also create a prototype or minimum viable product to demonstrate their idea. Students will then prepare a persuasive elevator pitch to present their startup idea to investors, and finally, they will present their pitches and prototypes to the class, receiving feedback and guidance

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
32.2	Demonstration of patent filing	Doing 2 · Guided response	The teacher will demonstrate filing a provisional patent in a real-life or simulated setting.
32.3	Demonstration of a scenario: The New Molecule Patent	Doing 2 · Guided response	The teacher will demonstrate a scenario where an R&D team in a mid-sized pharmaceutical company has discovered a promising new molecule or new chemical entity (NCE) for treating a rare disease condition. The demonstration will cover the process of patent filing, including timing, scope, strategy, disclosure of information, risks, and challenges.
32.4	Demonstration of a case scenario of patenting the formulation process	Doing 2 · Guided response	The teacher will demonstrate a scenario where an Ayurveda pharmaceutical company has developed a modified dosage form featuring a novel delivery mechanism, utilizing traditional ingredients. The demonstration will focus on patenting the formulation process, excluding the herbs themselves. Additionally, the teacher will explore utility model options (where applicable), and create an IP protection roadmap, encompassing design patents, trademarks, and copyrights for packaging and branding.

141 activities. The level shown is that of the objective in Table 3 which cites the activity, so it tells you what the activity is being asked to certify.

PART C · APPENDIX C7

All 163 experiential learning activities, mapped

Table 5 of the curriculum. These are the largest block of hours in the semester and where the upper levels are built; the modular assessment is drawn from them.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 1 — CHRONOLOGICAL DEVELOPMENT AND TAXONOMY OF RASA DRAVYAS			
1.1	Preparing a chronological tree of Rasashastra and its integration with other medical traditions	Doing 4 · Adaptation	Each student will be asked to create a chronological timeline highlighting key texts and developments in Rasashastra. They will compare and contrast excerpts from classical texts to understand the evolution of alchemical practices. Students will investigate the influence of Rasashastra on contemporary Ayurvedic practices and its integration with other medical traditions. Each student will present the findings and discuss
1.2	Integration of the selected metals and minerals with the periodic table elements	Thinking 4 · Analysis	The teacher will allot each student at least one category from the classification of Rasadravya or one drug from each category. Each student will map minerals/Metals to the Periodic Table and search the elemental properties of minerals/metals mentioned in Rasashastra. Students will locate these metals on the Mendeleev Periodic Table, noting their atomic numbers, groups, and periods. Each student will compare and analyze the traditional name with the Periodic table, along with other properties, etc.
1.3	Recognise Rasadravya based on their Graahyatva-Agraahyatva	Doing 3 · Mechanism	The teacher will allot each student at least one category from the classification of Rasadravya or one drug from each category. The student will identify the Rasadravya based on classical characters and find out the ideal characters for acceptance (Grahya)
1.4	Identification of Rasadravya using geological/mineralogical characters	Doing 3 · Mechanism	The teacher will allot each student at least one category from the classification of Rasadravya or one drug from each category. The student will identify the Rasadravya based on geological/mineralogical characteristics
1.5	Exploring Sandigdha and Anuplabhdha Rasadravyas and propose suitable alternatives	Doing 1 · Set	Each student will collect different samples of Sandigdha Rasadravyas and identify them with labelling. Student will also explore present scenarios where certain Rasadravyas are absent or unavailable, with proposed alternative substances or modifications to compensate for the absence, drawing from their knowledge of Rasashastra principles
MODULE 2 — PUTA AND SAMSKARA (TRANSFORMATIVE PROCESSES OF DRUG MANUFACTURING)			
2.1	Hands-on Comparison of Traditional and Modern Puta Methods: Observing and Analyzing Temperature Patterns and Duration	Doing 3 · Mechanism	Each student will go through the different laws of thermodynamics and relate with traditional Puta and understand the concept of heatiing and energy. Student will analyse and monitor the temperature at various points within the Puta and Sharava Samputa. Then compare the observed data with theoretical predictions based on thermodynamic principles.
2.2	Exploring Thermal Dynamics in Traditional Puta Systems: An Experiential Investigation of Fuel Variations, Temperature Patterns, and Thermodynamic Principles	Doing 3 · Mechanism	The teacher will assign a unique Puta to each student, asking them to utilize different fuel materials and observe the resulting temperature patterns. Students will present their findings using graphical representations. Additionally, students will explore the laws of thermodynamics and relate them to traditional Puta, gaining insight into heating and energy concepts. They will also analyze and monitor temperature fluctuations at various points within the Puta and Sharava Samputa.
2.3	Perform an experiential Investigation of Amritikarana and Lohitikarana Procedures, Explo	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
2.4	Performing and analyzing the Satva Patana procedure. The teacher will allot each student	Doing 2 · Guided response	—
2.5	Applcation of Time-Tested Principles of Classical Drug Manufacturing: by integrating Toy	Doing 4 · Adaptation	—
2.6	Exploring the Interplay of Desha and Kala on Pharmaceutical Quality: A Hands-on Investigation of Geographical and Temporal Influences on Drug Manufacturing, Analyzing Processes and Findings to Enhance Quality, Safety, and Efficacy	Doing 3 · Mechanism	Each student will research and conduct practicals investigating the impact of Desha (geographical location and manufacturing site) and Kala (processing time, seasonal variations, and herb maturity) on pharmaceutical quality. Following hands-on experience, students will analyze the processes and findings, correlating them with manufacturing principles to enhance quality, safety, and efficacy, and present their conclusions in a class presentation.
2.7	Develop Mastery of Classical Drug Manufacturing: Integrating Time-Tested Principles of Vasana and Bhavana to Optimize Pharmaceutical Quality, Safety, and Efficacy	Doing 2 · Guided response	Each student will search for and perform practicals related to Vasana (infusion of aroma and effect of proximity), and Bhavana (effect of levigation/trituration). After the procedure, they will analyze each process and findings, correlating them with different principles of manufacturing to enhance quality, safety, and efficacy, and present their findings in a class presentation.
2.8	Demonstration of the effect of Kalaprakarsha and Bhajana in Pharmaceutical Processing, Analyzing Processes and Findings to Optimize Quality, and stability	Doing 2 · Guided response	Each student will conduct practical on Kalaprakarsha (investigating the effect of specific time exposure) and Bhajana (examining the influence of processing vessels). Following hands-on experience, students will analyze each process, correlate findings with manufacturing principles, and present their insights in a class presentation to enhance understanding of quality, stability and efficacy.
MODULE 3 — PARADA (MERCURY) AND ITS PROCESSING: ANCIENT WISDOM AND SCIENTIFIC ADVANCES			
3.1	Samanya Shodhana of Parada	Doing 2 · Guided response	The teacher will allot each student to perform the Samanya Shodhana of Parada following references other than the one that is not taught during practical hours. The students will record the observations and present their findings, and discuss
3.2	Vishesha Shodhana procedures of Parada	Doing 3 · Mechanism	The teacher will allot each student to perform the Vishesha Shodhana of Parada following references other than the one that is not taught during practical hours. The students will record the observations and present their findings, and discuss
3.3	Performance of the Parada Ashta Samskara procedures- Patana, Bodhana, Niyamana, and Deepana	Doing 2 · Guided response	Each student or group will practically explore and perform the Samskaras of Parada like, Patana, Bodhana, Niyamana, and Deepana. The observations, like changes in the Parada, total duration, temperature range, and yield, will be noted down and presented in class.
3.4	Comparison and analysis between the Pharmacodynamics and pharmacokinetics of mercury and mercurial products	Doing 3 · Mechanism	The teacher will allot each student different mercurial products/ compounds and ask them to explore and compare the pharmacodynamic and pharmacokinetic behaviour of allotted products/compounds using both classical Ayurvedic theory and modern biomedical models like Rasa Panchaka and modern concepts of pharmacodynamics and pharmacokinetics. The students will also explore available published data. They will present their findings and discuss.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
3.5	Analysis of mercury toxicity and ADR incidences	Doing 2 · Guided response	The teacher will ask each student to search and record any case of mercury toxicity among OPD/IPD patients using mercurial products, explore the probable toxicokinetics, and suggest its management using relevant classical antidotes and modern therapeutics. OR The teacher will allot each student a case scenario of mercurial toxicity. The student will be asked to explore the toxico-kinetics and suggest its management using relevant classical antidotes and modern therapeutics.
MODULE 4 — MAHARASA DRAVYA, ITS PROCESSING AND UTILITY			
4.1	Pharmacotherapeutic attributes of Abhraka Bhasma and Vaikranta Bhasma	Doing 3 · Mechanism	The teacher will assign each student a specific topic related to the pharmacological, toxicokinetic, and clinical attributes and applications of Abhraka and Vaikranta. Students will search and gather information from classical texts, published scientific literature, and case studies. They will then present their findings and engage in a class discussion
4.2	Critical Analysis of Abhraka and Vaikranta: Pharmacology and Clinical Applications	Doing 4 · Adaptation	The teacher will assign each student a specific topic related to the pharmacological, toxicokinetic, and clinical attributes and applications of Abhraka and Vaikranta. Students will search and gather information from classical texts, published scientific literature, and case studies. They will then present their findings and engage in a class discussion
4.3	Explore and perform the Makshika Bhasma and Vimala Bhasma	Doing 2 · Guided response	Each student will prepare Makshika/Vimala Bhasma using at least one method. The student will record their observations and findings, and then present their results in class. The teacher will provide feedback and recent updates on the subject
4.4	Exploration and performance of Shodhana and other pharmaceutical procedures of Shilajit and Sasyaka/ Tuttha	Doing 2 · Guided response	Each student will perform at least one method of Shodhana/Marana for Shilajit and Sasyaka/Tuttha. The student will record their observations and findings, and then present their results in class. The teacher will provide feedback and recent updates on the subject.
4.5	Explore and perform the pharmaceutical procedure, like Shodhana, Marana for Chapala/Khar	Doing 3 · Mechanism	—
4.6	Exploring New Metallic/Mineral Resources that align with Maharasa Dravyas Principles. Ea	Doing 2 · Guided response	—
MODULE 5 — CRITICAL APPRAISAL OF UPARASA DRAVYA AND SADHARANA RASA DRAVYA			
5.1	Identify and perform an authentication process of Uparasa Dravyas	Doing 4 · Adaptation	The teacher will allot each student with a different drug from Uparasa category. The students will identify and authenticate the given drug based on classical Grahya Lakshana and mineralogical references with cross-verification through reference samples, The students will present their findings and discuss The teacher will allot each student with different drug from Uparasa category. The teacher will instruct each student to search the relevant database regarding the mineralogical characters, physico-chemical properties and crystallography(wherever applicable) of the given samples. The students will explore, note down the findings, present the same and discuss.
5.2	Performing Quality Control tests for the products of Uparasa Dravya	Doing 2 · Guided response	The teacher will allot each student with different processed drug from Uparasa category, such as a Shodhita Dravya, Bhasma or Pishti. The students will test the given sample following organoleptic, physico-chemical parameters and instrumental methods. The students will record the findings, present the same and discuss

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
5.3	Preparation and presentation of formulations of Uparasa Dravya	Doing 2 · Guided response	The teacher will allot each student with a different formulation of Uparasa Dravyas such as, Gandhaka Rasayana, Kasisa Bhasma etc. The students will explore the reference, Standard Manufacturing Process and will prepare the given formulation accordingly. The students will present their findings in the class and discuss the steps, challenges, and outcomes
5.4	Critical Review of toxicity and ADRs of Haratala and Manahshila	Doing 3 · Mechanism	The teacher will instruct each student to explore real or hypothetical case reports involving ADRs or Toxicity of the products containing Haratala and Manahshila. The students will carry out literature review from classics and modern toxicology texts, critically discuss the findings and suggest preventive and remedial measures
5.5	Identify and authenticate Sadharana Rasa Dravyas	Doing 2 · Guided response	The teacher will allot each student with a different drug from Sadharana rasa category. The students will identify and authenticate the given drug based on classical Grahya Lakshana and mineralogical references with cross-verification through reference samples, The students will present their findings and discuss The teacher will allot each student with different drug from Sadharana Rasa. The teacher will instruct each student to search the relevant database regarding the mineralogical characters, physico-chemical properties and crystallography(wherever applicable) of the given samples. The students will explore, note down the findings, present the same and discuss.
5.6	Performing Quality Control tests for the products of Sadharana Rasa Dravya	Doing 4 · Adaptation	The teacher will allot each student with different processed drug from Sadharana Rasa category, such as a Shodhita Dravya, Bhasma or Pishti. The students will test the given sample following organoleptic, physico-chemical parameters and instrumental methods. The students will record the findings, present the same and discuss
5.7	Preparation and presentation of formulations of Sadharana Rasa Dravya	Doing 4 · Adaptation	The teacher will allot each student with a different formulation of Sadharana Rasa Dravyas. The students will explore the reference, Standard Manufacturing Process and will prepare the given formulation accordingly. The students will present their findings in the class and discuss the steps, challenges, and outcomes
5.8	Field Survey: Manufacturer/clinicians using Sadharana Dravyas	Doing 2 · Guided response	The teacher will instruct each student to explore the Sadharana Rasa Dravya used by manufacturers/clinicians/traditional healers. The students will interact, and collect the information regarding Sadharana Rasa Dravyas. They will compile the data and present
MODULE 6 — CRITICAL APPRAISAL OF DHATU AND UPDHATU			
6.1	Identification & documentation of classical Dhatus	Doing 3 · Mechanism	The teacher will allot each student with a different drug from Dhatu category. The students will identify and authenticate the given drug based on classical Grahya Lakshana and mineralogical references with cross-verification through reference samples, The students will present their findings and discuss
6.2	Quality assessment of Dhatu Bhasma	Doing 2 · Guided response	The teacher will allot each student with different Dhatu Bhasma (Loha Bhasma, tamra Bhasma etc). The students will test the given sample following classical Bhasma pareeksha, organoleptic, physico-chemical parameters and instrumental methods. The students will record the findings, present the same and discuss
6.3	Formulation preparation using Dhatu Bhasma	Doing 2 · Guided response	The teacher will allot each student with a different formulation containing Dhatu Bhasma(e.g.Navayasa Louha). The students will explore the reference, Standard Manufacturing Process and will prepare the given formulation accordingly. The students will present their findings in the class and discuss the steps, challenges, and outcomes

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
6.4	Critical appraisal of ADRs due to metals	Doing 2 · Guided response	The teacher will instruct each student to explore real or hypothetical case reports involving ADRs or Toxicity of the products containing metals. The students will carry out literature review from classics and modern toxicology texts, critically discuss the findings and suggest preventive and remedial measures
6.5	Identification of Upadhatu samples	Doing 2 · Guided response	The teacher will allot each student with a different drug from Upadhatu category. The students will identify and authenticate the given drug based on classical Grahya Lakshana and mineralogical references with cross-verification through reference samples, The students will present their findings and discuss
6.6	Performing Quality Control tests for the products of Upadhatu	Doing 2 · Guided response	The teacher will allot each student with different processed drug from Upardhatu category, such as a Shodhita Dravya or Bhasma. The students will test the given sample following organoleptic, physico-chemical parameters and instrumental methods. The students will record the findings, present the same and discuss
6.7	Preparation of Upadhatu formulations	Doing 4 · Adaptation	The teacher will allot each student with a different formulation of Upadhatu Dravyas. The students will explore the reference, Standard Manufacturing Process and will prepare the given formulation accordingly. The students will present their findings in the class and discuss the steps, challenges, and outcomes
6.8	Field survey: Manufacturers or Vaidyas using Dhatu/Upadhatu	Doing 2 · Guided response	The teacher will instruct each student to explore the formulations of Dhatu/Upadhatu used by manufacturers/clinicians. The students will interact, and collect the information regarding Dhatu/Upadhatu Dravyas. They will carry out the Case documentation on sourcing, processing, and therapeutic applications
MODULE 7 — SUDHA VARGA, SIKATA VARGA, VISHA VARGA DRAVYAS			
7.1	Shodhana, marana, and Pishti Nirmana of Sudha varga Dravya	Doing 2 · Guided response	The teacher will assign a different Sudha Varga drug to each student and instruct them to carry out Shodhana, Marana, and Pishtikarana of the allotted drug. Students will conduct the practical following the Standard Operating Procedure (SOP), record their observations, present their findings, and engage in a discussion.
7.2	Quality testing procedures for Bhasma/Pishti of Sudha Vargeeya Dravya	Doing 2 · Guided response	The teacher will assign each student a unique sample of Bhasma/ Pishti from Sudha Varga Dravya. Students will conduct quality testing of the assigned product, evaluating organoleptic, physicochemical parameters, and NPS tests. They will observe, record, and present their findings, followed by a class discussion.
7.3	Shodhana, marana and Pishti Nirmana of Sikata varga Dravya	Doing 2 · Guided response	The teacher will assign a different drug from Sikata Varga category to each student. Students will then carry out Shodhana, Marana, and Pishtikarana of their allotted drug, following the Standard Operating Procedure (SOP). They will record their observations, present their findings, and engage in a class discussion
7.4	Quality testing procedures for Bhasma/Pishti of Sikata vargeeya Dravya	Doing 2 · Guided response	The teacher will assign each student a unique sample of Bhasma/ Pishti from Sikata Varga Dravya. Students will then conduct quality testing of their assigned product, evaluating organoleptic, physicochemical parameters, and NPS tests. They will observe, record, and present their findings, followed by a class discussion.
7.5	Performing Shodhana of Visha Dravyas as per different classics	Doing 2 · Guided response	The teacher will assign a different Visha Dravya to each student. Students will then carry out Shodhana of their assigned drug, following the Standard Operating Procedure (SOP). They will present their observations and engage in a class discussion. Alternatively, the teacher may assign the same drug to all students and instruct them to perform Shodhana using different classical methods.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
7.6	Performing analytical testing process for Shodhita Visha dravya	Doing 2 · Guided response	The teacher will assign each student a different Visha Dravya, providing both unprocessed (Ashuddha) and processed (Shodhita) samples. Students will then conduct comparative testing of the given samples, evaluating parameters such as organoleptic characters, physico-chemical properties, qualitative phytochemical tests, and Thin Layer Chromatography (TLC).
MODULE 8 — APPRAISAL OF RATNA AND UPARATNA AND THERAPEUTIC SIGNIFICANCE OF PISHTI AND DRUTI KALPANA			
8.1	Identification of the precious and semiprecious stones under Ratna and Uparatna varga using keys for mineralogical identification	Doing 2 · Guided response	The teacher will randomly assign a precious or semi-precious stone to each student. Students will then identify their assigned stone using identification keys, present their findings, and discuss the physico-chemical properties and crystallography of the identified specimen.
8.2	Identification of substitutes for Ratnavarga and Uparatna varga drugs	Doing 3 · Mechanism	The teacher will assign each student a different precious or semi-precious stone and instruct them to identify at least one substitute from the museum and demonstrate the method of identification of the selected substitute. The students will present their findings and discuss the physico-chemical properties, crystallography, and pharmacology of the selected specimen.
8.3	Preparation of the Pishti of Selected Ratna or Uparatna	Doing 2 · Guided response	The teacher will assign each student a different drug from Ratna/ Upartna Varga and instruct them to prepare the Pishti using various media for trituration. Students will then prepare the Pishti according to the Standard Operating Procedure (SOP) and present their observations.
8.4	Preparation of Gandhaka Druti using different media	Doing 2 · Guided response	The teacher will instruct each student to prepare Gandhaka Druti using various media (Tila Taila, Ghrita, Sarshapa taila) and different proportions. Students will then prepare the products, compare their physico-chemical properties, present their observations, and discuss
MODULE 9 — SCIENTIFIC FUNDAMENTALS OF AYURVEDIC PHARMACEUTICAL PREPARATIONS			
9.1	Preparation of unique formulation from the category of Panchavidha Kashaya Kalpana	Doing 2 · Guided response	Each student will enlist unique examples of formulations (minimum 10) of Panchavidha Kashaya kalpana from classical texts, and will enlist the steps of their preparation. All students will discuss to evaluate all dimensions of the formulations, viz. ingredients, water proportion, time required, therapeutic efficacy, samskaras performed, and the difference between various classical paribhasha. Each student will prepare at least one formulation from each group, record the observations, present the same, and discuss. Each student will be allotted with different formulation and the students will explore preparing the given formulation with different methods with due importance to heat, proportion etc and prepare the formulations, note their observations, present the same with scientific rationale.
9.2	Standardization of raw material used in Bhaishajya Kalpana. Each student will refer to a	Doing 2 · Guided response	—
9.3	Interpretation of the significance of understanding the proportion of ingredients of formulations quoted in the classics	Thinking 4 · Analysis	Each student will compile 6 examples of classical formulations of different dosage forms on the basis of the uniqueness of ingredients and their proportions. The faculty will discuss and explain its significance. Each student will demonstrate preparation of one formulation, present the findings and discuss the rationality behind the proportion of the ingredients
9.4	Performing the extraction of phytochemicals in Ayurvedic formulations. All students will	Doing 2 · Guided response	—
9.5	Review of Research Updates in the field of Ayurvedic pharmaceutics along with modern sci	Affective · Valuing	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 10 — PRINCIPLES OF PHARMACEUTICAL FORMULATION			
10.1	Illustration of various Prakshepa Dravyas and their proportions stated in the classics. All students will summarize the list of Prakshepa dravyas in the library session and assess the identity, purity, quality, and guna karma of the following Prakshepa dravyas-1.Madhu 2. Sita/ sharkara 3.Guda 4.Kshara 5.Lavana 6.Ghruta 7.Taila 8.Churna- Jiraka, Shunthi,Hingu, Pippali, Trikatu, Chaturjata 9.Kalka 10.Drava- Jala, Ksheera, Madya, Mutra 11.Guggulu 12. Shilajit	Doing 3 · Mechanism	Faculty will allot 2 or 3 prak?epa dravya for studying, and each student will submit an essay based on the findings. Each student will prepare at least 2 Formulations from the following (or any other) to understand the importance of Prakshepa Dravya-1.Brahmi Svarasa 2.Vasa kvatha with different combinations 3.Draksha Phanta 4.Rasona Kalka 5.Narikela Khandapaka 6.Khadirarishta
10.2	Classical techniques of Shaktyutkarsha/ Potency enhancement	Doing 3 · Mechanism	The faculty will plan a workshop where all the students will demonstrate the following or any relevant practical-1) Ghana Nirmana-Kutaja Ghana 2) Bhavana- Shatabhavita Amalaki Rasayana 3) Punaravartita Paka-Anu Taila Nirmana 4) Sura Nirmana-Rasona Sura(Chakradatta, Bhaishjya Ratnavali- Aamavata) 5) Addition of Yogavahi Dravya- addition of Pippali/Kajjali/ Rasasindura Students will compile classical examples of formulations that utilize shaktyutkarsha techniques. A brainstorming session will be conducted following the faculty's concluding remarks.
10.3	Evaluation of Methods used to improve potency and efficacy in modern pharmaceutics A bra	Thinking 5 · Synthesis	—
MODULE 11 — MANA PARIBHASHA (METROLOGY IN AYURVEDIC PHARMACEUTICAL PREPARATION)			
11.1	Mana Paribhasaha from classics and its importance for preparing classical formulations	Doing 4 · Adaptation	—
11.2	Preparation of Classical Formulations by applying knowledge of converted units Group act	Doing 4 · Adaptation	—
11.3	Application of classical knowledge of Mana paribhsha with understanding its effect on pr	Doing 2 · Guided response	—
11.4	Significance of proportion of ingredients . Brainstorming sessions will be conducted wit	Doing 4 · Adaptation	—
11.5	Assessment of advanced tools and techniques used currently for manufacturing Ayurvedic f	Doing 4 · Adaptation	—
MODULE 12 — PROCESS VALIDATION PROTOCOL			
12.1	Studying the documentation related to Standard Operating Procedures (SOPs) used in pharm	Doing 2 · Guided response	—
12.2	Construct a protocol for unique classical formulations and prepare the selected product	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
12.3	Preparation of	Doing 2 · Guided response	Prepare different novel modern dosage forms The faculty will organize a workshop focused on practical training in modern dosage forms. OR will arrange an educational visit to a nearby GMP-certified drug pharmacy or a well-regarded pharmacy college. This visit will provide an opportunity to observe and assist in the preparation of modern dosage forms. OR will incorporate video recordings to train participants on the Standard Operating Procedures (SOP) for preparing modern dosage forms. This should include, Demonstrate SOP for-1. Solid dosage forms - Powders, Pills, Capsules, Tablets, Capsules, Lozenges. 2. Liquid dosage forms - Emulsions, Suspensions, Lotions, Liniments, Syrups, Nasal Drops, Gargle. 3. Semi-solid dosage forms - Ointments, Suppositories, Gel, Paste, Cream. Plasters, Suppositories. 4. Gaseous dosage forms- Aerosols, nasal Spray, Inhalers.
12.4	Overview of Novel Dosage Forms. After conducting a thorough search by all students in the library and on mobile platforms for relevant articles, the	Doing 2 · Guided response	Faculty will arrange a departmental seminar on the topic: "Overview of Novel Dosage Forms." The discussion will focus on the ongoing evolution of healthcare and how the development of advanced formulations is crucial for delivering safe and effective treatments to patients.
MODULE 13 — KRUTANNA (AYURVEDIC FOOD PREPARATIONS)			
13.1	Critical appraisal of unique formulations prepared from Aahara dravya	Doing 2 · Guided response	The teacher will instruct students to study the Ahara Varga from classical texts in depth, accompanied by commentary, for better understanding. Each student will search the formulations from the E-samhita portal or manually from the printed Samhitas. Each student will prepare any 4 (one from each group mentioned below) selected formulations following SOP. 1) Shuka dhanya varga- Vatya manda, Tandulodaka 2) Shimbi Dhanya Varga- Masura Ghruta, Mudgadi Ghruta (Yogaratnakara Striroga) 3) Mamsa Varga- Mayura Ghruta(Ch. chi 26/156-158)Tittiri Putapaka(Sushruta. Uttara 40/84,85), Mushika taila (Sushruta/ Kshudraroga/ Gudabhramsha chikitsa (Su.chi. 20/61-63)Pishita odana(Su. U.43/20) 4) Shaka Varga- Patoladi Ghruta, Surana Vataka, Mulaka Taila, Kushmandavaleha 5) Pushpa varga- Shriphala kusuma vati, Lavangarishta(Gada-nigraha) , Palash Pushpasava. 6) Phala Varga- Dadimavaleha, Draksharishta, Kharjuradi Mantha, Bilvavaleha, Kapitthashtaka Churna. 7) Lavana Varga- Hingu sauvarchala ghruta, Vallabha Ghruta(Yogaratnakara, Hrudroga) Bhaskar Lavana Churna. 8) Spices- Pancha kola Phanta, Eladi Guti, Hingvashtaka Chrna, Haridra Khanda, Jirakarishta
13.2	Disease specific Ayurvedic Ahara	Doing 2 · Guided response	Faculty will convey to the students the plan of the studyas follows-1. Select three Adhyayas from Charaka Samhita Chikitsa Sthana and three Adhyayas from Sushruta Samhita Uttarsthana. 2. Distinguish and list the Aahara dravyas (food substances) and dosage forms used to treat the specified diseases. 3. Interpret the rationale behind the combinations of the ingredients used in these treatments. 4. Explain the correlation between the stages of the disease (Rogavastha) and the recommended therapeutic diet. 5. Appraise the classical descriptions of the vyadhi-pratyanika (disease-specific) and Lakshana-pratyanika (symptom-specific) therapeutic diet recommendations. 6. Demonstrate any four preparations from your compilation of dietary yogas based on the Charaka and Sushruta Samhita. OR The teacher will instruct the students to visitthe hospital and review the prescribed diet for the diseases the students have studied. The students will note down the dietary formulations and present their findings with the method of preparation and rationality of prescription.
13.3	Personalised prescription for Aushadhi Siddha Ahara. The teacher will instruct the stude	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
13.4	Exploratory Survey about the Ayurvedic Food Industry and Ayurvedic Food products availab	Doing 2 · Guided response	—
13.5	Analysis of Functional Food and Modified Food Formulations in Ayurvedic Food Industry. T	Doing 4 · Adaptation	—
MODULE 14 — AYURVEDIC FORMULATIONS FOR HOLISTIC WELLBEING, VITALITY AND BODY CARE			
14.1	Role of Ayurvedic Rasayana Vajikarana Formulations	Doing 2 · Guided response	The students will compile classical formulations having Rasayana and Vajikarana action, classify them based on various criteria, Viz. Dhatu Karya, Avayava Karya, Vyadhikshamattva Vardhana etc. They will present their findings in front of the peers. The contents of presentation should include, the rationality behind the selection and combination of Ingredients and the proportion of ingredients in the selected formulation, justification of the role of Rasayana based various mechanisms of action and the role of Vajikarana formulations in Suprajanana. The students will refer relavant research articles published in peer reviewed journals
14.2	Role of Ayurvedic Formulations in the wellness of specific body parts	Doing 2 · Guided response	The teacher will allot each student with different target specific Rasayana Dravya such as, Hridya, Varnya, Tvachya etc. Each student will explore the principle of designing classical Formulations and their specific target and compile a list of formulations having a specific target action. The faculty will distribute 2 formulations to each student from the following list(or any relevant formulations) 1.Varnya - Kunkumadi Taila, Kanaka Taila(Charaka samhita) 2.Tvachya- Eladi taila, Chandan bala Lakshadi taila, Nalpamaradi taila (sahasrayoga) 3.Kanthya- Khadiradi Vati, Kanthasudharaka vati 4.Keshya - Bhrungaraja Taila 5.Hrudya- Prabhakara Vati 6.Medhya- Brahmi Vati
14.3	Market survey for Ayurvedic Body care products. Students will be instructed to conduct a	Doing 2 · Guided response	—
14.4	Designing and preparing novel Ayurvedic Cosmeceuticals. The teacher will allot each stud	Doing 2 · Guided response	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 15 — DISEASE CONDITION SPECIFIC SELECTION OF FORMULATION			
15.1	Understanding the routes of drug administration is crucial, as it highlights the importance of well-designed dosage forms that improve patient outcomes and therapeutic effectiveness. The teacher will instruct each student to conduct the following UG curriculum teaching class and revise knowledge about the route of drug administration.They will discuss the advantages and disadvantages of each aushadhi prayoga marga and probable mode of action after administration of the following dosage forms-1. Mukha (Oral Cavity): Vati, Gutika, Churna, Asava, Arishta, Kashaya, Avaleha, Khanda, Sneha (Ghrita/Taila), 2. Nasa(Nasal Route)- Dosage form used: Churna, taila, swarasa, arka 3. Karna (Through Ear)- Taila, Ghrita 4. Akshi (Through Eyes) - Ghrita, Taila, 5. Twak (Through Skin)- Lepa, Alepa, Pralepa, Malahara,upanaha,pradeha, abhyanga, udvartana • Shirodhara - Takra • Abhyanga- Sahacharadi Taila • Asthti Bhagna- Murivenna Taila • Vrana- Jatyadi Taila • Smashru – Shankha Bhasma • Kesha Ghanata- Bringaraja Taila • Akala Palita – Hasthi Danta Masi • Indralupta – Icchabhedi Rasa • Lomashatana – Lomashatana Lepa 6. Guda (Anal Route)- Dosage forms - Vartis, taila, ghrita, kalka, churna, kashaya 7. Mutra marga (Through urethra)- Uttara Basti with dosage forms- Taila, ghrita 8. Yoni marga (Through vagina)- Yoni Dharana, Yoni Dhavana, Yoni Pichu, Yoni Dhoopana	Doing 3 · Mechanism	Each student will prepare a suitable dosage form from the list mentioned below, present their findings, and discuss: 1.Nasya Kalpana- Navana, Avapidaka, Dhuma, Pratimarsha, Pradhamana 2.Mukha Kalpana- Kavala, Gandusha,P ratisaran,manjan 3.Basti kalpana- Yapana, Anuvasana, Karshana, Sneha, Bruhana, Vajikara, Rasayana, Matra 4.Netra Kalpana- Seka,Aschyotana,Pinda,Bidaal,Tarpana,putapaka, 5.Nabhipurana- Bindu Taila (Sha. Sam. Ma. Kh. 9/61-66) 6.Other bahya Kalpana- Lepa,Dhara,Bandha,Dhupana,Agnikarma,Kshara karma,Avagaha,Pinda, Drava sveda, Taap sveda,Vrana karma,jalauka, Paadabhyanga Evaluate the role and responsibility of effective dosage form design for the aforementioned Ayurvedic treatment modalities. Refer to relevant articles to enhance your understanding and organize a departmental presentation on this topic.
15.2	Hands-on training focusing on innovative formulations	Doing 2 · Guided response	The faculty will instruct each student to prepare different innovative formulations such as transdermal patches, nebulizers, and eye masks. They will discuss the necessity and current trends in innovative formulations within the field of Ayurvedic pharmaceutics, review modern pharmaceutical literature, consult experts, or watch videos to understand the SOP for transdermal patches, nebulizers, eyemasks, and other innovative formulations. Each student will prepare the allotted innovative product, present the observations, and discuss the challenges faced
15.3	Learning about the R&D department of Ayurvedic Pharmacy. The students will be posted to	Doing 2 · Guided response	—
MODULE 16 — CURRENT TRENDS IN AYURVEDIC PHARMACEUTICAL INDUSTRY.			
16.1	Assessment of the real-life experience of the market for the growing popularity of Ayurv	Doing 3 · Mechanism	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
16.2	Internship in the Ayurvedic Pharmacy. Each student should-Understand current trends in t	Affective · Organisation	—
16.3	QAQC Laboratory Training	Affective · Valuing	The students will visit the QAQC laboratory and observe and understand the working pattern of the Laboratory. The students will interact with the concerned experts and obtain updated information about Leading Drug research laboratories / CCRAS drug research units, regulatory and certification bodies for QAQC of Ayurvedic Formulations. The students will be able to justify the need and importance of QAQC in the field of Ayurvedic Pharmaceutics. The students will also observe the Calibration of instruments used for analysis. They will prepare a report of their experience, present the same, and discuss.
MODULE 17 — GUIDELINES FOR ADMINISTRATION OF RASOUSHADHI IN CLINICAL PRACTICE			
17.1	Exploring the Rasa Chikitsa through field visits. The teacher will ask the students to e	Doing 2 · Guided response	—
17.2	Exploring the procedure and benefits of kshetrikarana	Doing 2 · Guided response	The teacher will instruct each student to visit Panchakarma unit to observe the process of Kshetrikarana. They will record their observations and present different steps of Kshetrikarana and discuss the benefits
17.3	Outcome of compliance and non-compliance with the pathya apathya guidelines	Doing 3 · Mechanism	The students will observe case records of the patients undergoing Rasayoga sevana and document the benefits of compliance with prescribed Pathya-Apathya measures. Students will be assigned case studies where deviation from Pathya-Apathya resulted in suboptimal Rasayoga effect. Students will analyse the cases, identify key factors, and present how the patient outcome was influenced by non-adherence to Pathya-Apathya
17.4	Evaluation of prescription errors containing Rasoushadhis	Thinking 6 · Evaluation	The teacher will create sample prescriptions of Rasoushadhi with prescription errors. The students will analyse them for potential errors such as overdose, improper combinations, etc, and they will present their findings and discuss.
17.5	Analysis of chelation therapy in heavy metal toxicity	Doing 3 · Mechanism	Students will present a case scenario of heavy metal toxicity with a description of clinical signs and symptoms, and suggest treatment strategies with suitable chelating agents. They will also explore classical methods of detoxification and compare them with contemporary chelation practices.
MODULE 18 — PRAYOGIKA BHAISHAJYA KALPANA(CLINICAL APPLICATION OF PHARMACEUTICAL PRODUCTS)			
18.1	Exploring chrono therapeutic alignment of Oushadha Sevana Kala with patient–disease dyna	Doing 2 · Guided response	—
18.2	Exploring the modulation of the action of the dosage form by chronotherapeutic administr	Doing 2 · Guided response	—
18.3	Enlisting appropriate Anupanas for various formulations in different clinical conditions	Doing 3 · Mechanism	—
18.4	Optimising drug response through synergistic Anupana integration. Each student will anal	Affective · Valuing	—
18.5	Exploring the role of Anupana in the modulation of drug tolerance and safety. The studen	Doing 3 · Mechanism	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
18.6	Exploring posology principles from an ayurvedic and modern pharmacology perspective. The	Doing 2 · Guided response	—
18.7	Exploring novel indications for classical Ayurvedic formulations	Doing 3 · Mechanism	Each student will be allotted five formulations, and students will evaluate the new therapeutic use of that formulation. With the available modern research articles on the pharmacological insights of the formulation's ingredients, the students will propose and justify a potential new therapeutic use for that formulation
18.8	Exploring the Reverse pharmacological approach for drug repurposing in Ayurveda	Doing 3 · Mechanism	The students will explore various scientific articles for successful clinical case studies of classical ayurvedic formulations for new indications (COVID, lifestyle disorders, etc). Students will create a pharmacological profile of the ingredients of the formulation from the ayurvedic text and the modern context. The students will link the clinical success of the formulation to the pharmacological profile of the ingredients of the formulation.
MODULE 19 — SOP AND CLINICAL APPLICATION OF CHATURVIDHA RASAYANA			
19.1	Validation of utility of kharaleeya rasayana to clinical case practice. Each student wil	Doing 4 · Adaptation	—
19.2	Demonstration of the impact of Bhavana – A bioactivation tool in Kharaleeya Rasayana. Ea	Doing 2 · Guided response	—
19.3	The pharmacodynamic rationale and clinical application of parpati kalpana. Each student	Doing 4 · Adaptation	—
19.4	Preparation of Pottali following SOP The teacher will assign each student a different po	Doing 3 · Mechanism	—
19.5	Analysis of the pharmacological profile of the ingredients, bhavana dravya, and the phar	Doing 3 · Mechanism	—
19.6	Preparation of selected Kupipakwa Rasayana	Doing 2 · Guided response	The teacher will assign one Sagandha and one Nirgandha Kupipakwa Rasayana for the group of students (other than the ones prepared in practical). The students will prepare the allotted formulation following SOP, record their findings, present the same and discuss
19.7	Demonstration of the scinetific rationality for the selection of specific Kupipakwa Rasa	Doing 2 · Guided response	—
MODULE 20 — SOP AND CLINICAL APPLICATION OF AYASKRITI, LOUHA, AND MANDURA KALPA			
20.1	Decoding the role of Ayaskriti. as a multi-target formulation. The students will list ou	Doing 3 · Mechanism	—
20.2	Application of Louha Kalpa by integrating through textual insights with real word cases	Doing 4 · Adaptation	—

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
20.3	Gomutra as a medium of pharmaceutical processing and as a bio enhancer – "Re-evaluating its role in mandura kalpa." The teacher will instruct each student to carry out pharmaceutical processes like Shodhana where Gomutra is used. The students will carry out the practical, record their observations, and present them and discuss	Doing 3 · Mechanism	The students will list out mandura kalpas containing gomutra. They will critically evaluate how go mutra functions as a bio enhancer in the preparation of the mandura kalpa, enhancing therapeutic potency, absorption, and clinical efficacy from both ayurvedic and modern pharmacological perspectives. The students will also refer to the scientific articles which substantiate the role of gomutra in attaining therapeutic benefits.
MODULE 21 — CLINICAL APPLICATION OF RASOUSHADHI ACCORDING TO SROTAS AND IN SELECTED CHALLENGING CLINICAL CONDITIONS			
21.1	Therapeutic application of Rasoushadhi in different Srotas Rasoushadhi applicable to the	Doing 4 · Adaptation	—
21.2	Exploring the clinical utility of Rasoushadhi in Medical Emergencies	Doing 4 · Adaptation	The teacher will select certain emergency conditions and instruct the students to explore and list the relevant Rasoushadhi from Chaturvidha Rasayana. Each student will then be assigned a different formulation from the listed Rasoushadhi to prepare, record their observations, and present their findings. Following this, students will delve into the rationality and probable mode of action of their allotted formulations, drawing on the classical properties and pharmacology of the ingredients. Additionally, they will analyze the findings of relevant scientific articles published on their assigned formulations.
21.3	Exploring the role of Rasoushadhi in clinically challenging conditions such as Cancer. T	Doing 2 · Guided response	—
21.4	The role of Rasoushadhi in clinically challenging conditions such as neurodevelopmental	Doing 4 · Adaptation	—
MODULE 22 — YOGAS FROM DIFFERENT KALPANAS			
22.1	Analysis of the role of Gugulu Kalpana in various clinical conditions by selecting a specific Gulggulu Yoga. The	Doing 3 · Mechanism	Teacher will select different gugulu preparations and ask the student to select one Guggulu formulation and assign them to list the therapeutic actions of that formulation. Also, the student will analyse how the formulation is effective in those clinical conditions by analysing the possible mode of action, discussing the pharmacological activity of each ingredient, method of preparation, and published research works. OR The teacher will allot each student a different Vati/ Guggulu Kalpa and instruct them to explore the prescriptions containing those formulations in OPD/IPD files of the Hospital. The students will note the dose, Anupana, duration of administration, clinical indication, and therapeutic outcome, present their observations, and discuss
22.2	Analysis of the role of selected leha yoga in different clinical conditions. The	Doing 3 · Mechanism	Teacher will assign each student a particular Avaleha/Paka/Khanda Yoga, and the student will analyse how the selected Avaleha Yoga works in different clinical conditions and demonstrate the mode of action based on classical descriptors, pharmacology of individual ingredients, and published research works. OR The teacher will instruct each student to explore the prescription containing Avaleha/Paka/Khanda. The students will note down the dose, Anupana, duration, clinical indication, and therapeutic outcome. They will present their findings and discuss

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
22.3	Analysis of different stages of Sneha Paka	Doing 4 · Adaptation	The teacher will assign each student a different Sneha Kalpa and will ask them to carry out Sneha Paka. The students will carry out the Sneha Paka till Dagdha Paka and collect the samples at different stages of Sneha Paka (Ama, Mridu, Madhyama, Khara, and Dagdha). The students will comparatively analyse the samples of different Paka stages and present their findings. Based on the analytical parameters, therapeutic significance of the samples will be discussed.
22.4	Illustration of the clinical application of selected Arka/Dravaka Kalpa	Doing 3 · Mechanism	The Teacher will assign different Arka/Dravaka formulations to each student. The students will explore the clinical indications of the allotted dosage form and discuss the dose, clinical application, and mechanism of action based on classical and pharmacological descriptors of the ingredients. They will refer relevant publications (case studies/case reports), present the same, and discuss.
22.5	Comparative analysis of Asava Arishta Samples having different percentage and types of sugar	Doing 3 · Mechanism	The teacher will allot each student a different Asava/Arishta sample containing different proportions of sweetening substances and different types of sweeteners like Sharkara, Sita, Guda, madhu or their combinations. The students will analyse the given samples for alcohol percentage, total, reducing and non reducing sugars, present their findings and discuss
22.6	Record the dose, duration, indications, and clinical outcome of the various formulation	Doing 2 · Guided response	—
MODULE 23 — SOP AND CLINICAL APPLICATIONS OF CHURNA KALPANA(POWDER DOSAGE FORMS)			
23.1	Evaluating the role of a selected Churna in a specific clinical condition	Doing 4 · Adaptation	The teacher will allot each student a Churna yoga. The students will explore the reference, note down the classical indications. The students will anayse how each component ingredient helps in the management of said clinical condition. Also, the students will plot the possible mode of action by analysing the pharmacology, classical descriptors, dose, anupana, time of administration, Pharmacotherapeutic properties of ingredients, and their role in the mentioned disease condition and research updates. OR The teacher will allot each student a Churna yoga and instruct them to explore the prescriptions containing those formulations in OPD/IPD files of the Hospital. The students will record the dose, duration, indications, and clinical outcome, present their observations and discuss.
23.2	Clinical applications of Kshara and Ksharasutra	Doing 2 · Guided response	The teacher will allot each student a Kshara and Ksharasutra yoga. Students will explore the reference, note clinical indications, and analyze how Kshara Sutra manages conditions like Arsha and Bhagandara. They will also examine the mechanism of action of the allotted Kshara in specific disease conditions. Furthermore, students will plot the possible mode of action by analyzing pharmacology, classical descriptors, dose, anupana, administration time, pharmacotherapeutic properties of ingredients for Kshara Kalpa, and the mechanism of action of Ksharasutra in conditions like Arsha, Bhagandara, incorporating research updates. OR The teacher will allot each student to explore a prescription containing Kshara (oral or Pratisarana) and a Kshara Sutra yoga in OPD/IPD files of the Hospital. Students will record the dose, duration, indications, and clinical outcome, present their observations, and discuss.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
23.3	Demonstration of role of Lavana Kalpana in various clinical conditions	Doing 2 · Guided response	The teacher will allot each student a different yoga from Lavana Kalpana. Students will explore the reference, note clinical indications, and analyze how each component ingredient contributes to managing the specified disease. They will plot the possible mode of action by analyzing pharmacology, classical descriptors, dose, anupana, administration time, pharmacotherapeutic properties of ingredients, and their role in specific diseases, incorporating research updates. OR The teacher will instruct the students to explore a prescription containing Lavana Kalpa, from OPD/IPD files of the Hospital. The students will record the dose, duration, indications, and clinical outcome, present their observations, and discuss.
23.4	Role of a selected satva in a specified disease condition	Doing 2 · Guided response	The teacher will allot each student a yoga from Satva Kalpana. Students will explore their clinical indications and plot the possible mode of action by analyzing pharmacology, classical descriptors, dose, anupana, administration time, pharmacotherapeutic properties of ingredients, and their role in specific diseases, incorporating research updates. OR The teacher will instruct the students to explore prescriptions containing Guduchi Satva in various clinical conditions. Students will explore such prescriptions in OPD/IPD files of the Hospital, record the dose, duration, indications, and clinical outcome, present their observations, and discuss.
23.5	Role of Malaharas as a Topical application in Skin disease, The teacher will allot each	Doing 4 · Adaptation	—
MODULE 24 — THERAPEUTIC SIGNIFICANCE OF TOPICAL DOSAGE FORMS			
24.1	Exploring the role of Ayurvedic ophthalmic preparations in various eye diseases	Doing 2 · Guided response	The teacher will allot each student a different Netra Kalpa indicated for various eye diseases. Students will analyze how each component ingredient contributes to managing the indicated eye diseases. They will plot the possible mode of action by analyzing the pharmacology, classical descriptors, mode and time of administration, pharmacotherapeutic properties of ingredients, and their role in specific eye diseases, incorporating research updates. OR The teacher will instruct the students to search for prescriptions in OPD/IPD files for different marketed eye drops. Students will record the product name, ingredients, dose, duration, indications, and clinical outcome, then present their observations and discuss.
24.2	Demonstration of the role of lepa Yogas in managing various medical needs by innovative approaches	Doing 5 · Origination	The teacher will allot each student a selected Lepa yoga, indicated for external application in various diseases. Students will analyze how each component ingredient alleviates symptoms of the specific disease condition. They will plot the possible mode of action by analyzing pharmacology, classical descriptors, mode and medium of application, pharmacotherapeutic properties of ingredients, and discuss research updates.
24.3	The role of Malahara as a Topical application in Skin disease. The teacher will allot ea	Doing 3 · Mechanism	—
MODULE 25 — ARTIFICIAL INTELLIGENCE IN DRUG DEVELOPMENT AND PHARMACOGENETICS; PHARMACOGENOMICS			
25.1	Performing De Novo Drug	Doing 3 · Mechanism	Design Using Generative Models Each scholar is given with different assignment and is made to experience De Novo Drug Design Using Generative Models using appropriate tools and methods. Their findings are presented to peers and discussed
25.2	Analysis of ADMET properties using AI tools	Doing 4 · Adaptation	Each scholar is given with different drug molecule and is made to experience using pre-trained models or public web tools (Ex, Swiss ADME, pkCSM) to predict pharmacokinetic profile and compare predictions to real-world outcomes from literature

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
25.3	Identification of potential adverse effects of drug candidates using a dataset and tools	Doing 2 · Guided response	Each scholar is given with different natural drug molecule and is made to experience Side Effect Prediction following a suitable method, dataset, and tools. The outcome will be presented and discussed
25.4	Extraction of potential drug-disease relationship by using through Literature /data Mining	Doing 3 · Mechanism	Each scholar is asked to carry out Literature Mining for Drug Repurposing using appropriate tools and methods. They are asked to use NLP to extract potential drug-disease relationships, present their findings, and discuss
25.5	Interpretion of Pharmacokinetics and Toxicology Modelling using appropriate AI tools and methods	Doing 3 · Mechanism	Each scholar is asked to use appropriate tools and methods to experience Pharmacokinetics and Toxicology Modelling. Predicted ADMET properties are presented and discussed
MODULE 26 — PHARMACOLOGY			
26.1	Analysis of the mechanism of action of drugs acting on the Cardiovascular system through a case study scenario	Thinking 5 · Synthesis	The teacher will create a case scenario of cardiovascular disease, like a 60-year-old man who presents with chest pain and is diagnosed with stable angina. His blood pressure is 160/95 mm Hg. Now, discuss the mechanism of action and rationale for using nitrates, beta-blockers, and calcium channel blockers. Students will explore the role of antihypertensive agents (e.g., ACE inhibitors, diuretics) and recognize potential drug interactions and side effects.
26.2	Analysis of the mechanism of action of drugs acting on the Respiratory system through a case	Doing 3 · Mechanism	The teacher will create a case scenario of respiratory disease, such as a 28-year-old woman has a history of asthma with recurrent wheezing, and dyspnoea. She is currently experiencing an exacerbation. Now, differentiate between short-acting and long-acting bronchodilators, and explain the role of corticosteroids in asthma management. The teacher may create a case scenario of diseases pertaining to different systems for every batch.
26.3	Analyse Adverse Drug Reaction in a polypharmacy patient through a case scenario	Doing 3 · Mechanism	The teacher will create a case scenario of Adverse Drug Reaction in a polypharmacy patient, supposing a 70-year-old patient who is on digoxin, furosemide, and amlodipine. He complains of nausea, confusion, and visual disturbances. Students will be instructed to focus on Digoxin toxicity, Electrolyte imbalances (e.g., hypokalaemia from furosemide), Pharmacokinetics in elderly patients, and will try to find out the points like factors that predispose the elderly to ADRs, management of digoxin toxicity, and lab monitoring
26.4	Discussion on Emergency Pharmacology through a case scenario of anaphylaxis management	Doing 3 · Mechanism	The teacher will create a case scenario of an emergency condition about different systems, like the cardiovascular system, respiratory system, etc. Students will be asked to find out a first-line treatment/ management for the said anaphylaxis condition
MODULE 27 — NETWORK PHARMACOLOGY & REVERSE PHARMACOLOGY			
27.1	Practice of Network Pharmacology	Doing 3 · Mechanism	The teacher will give each student selected hands-on activities and approaches in network pharmacology that can involve a variety of hands-on activities and approaches that help deepen understanding of drug interactions, mechanisms of action, and the overall pharmacological landscape. The activities can be, simulation exercises OR, field trips, OR, interdisciplinary Workshops OR, Online Courses and MOOCs. The student will perform/attend one or more activities spanning not less than 10 hours. After learning, students will present their experience and discuss
27.2	A mini-project on network pharmacology	Doing 3 · Mechanism	The teacher will allot each student a different herb. Students will be asked to select a major pharmacological activity of the given herb and follow a systems biology approach to validate reverse pharmacology. After the completion of the project, students will present the outcomes and discuss

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
27.3	Presentation of observations of analysis of Reverse Pharmacology in selected cases. The	Doing 3 · Mechanism	—
27.4	Interpret Reverse pharmacology with a systems biology approach. The teacher will allot e	Doing 3 · Mechanism	—
MODULE 28 — PHARMACEUTICAL QUALITY ASSURANCE			
28.1	Identification of the Root cause, affected personnel roles, and suggest corrective and preventive measures through Case Study Discussion on GMP Violation	Doing 2 · Guided response	The teacher will create a case scenario of GMP Violation due to Poor Personnel Management. The students will be provided with a real or fictional GMP warning letter or case study involving personnel errors. The students will identify the Root cause, affected personnel roles, and suggest corrective and preventive measures
28.2	Documentation of a personal record of pharmacy personnel	Doing 4 · Adaptation	The teacher will ask the students to create a mock record of an employee of the pharmacy, and it will be cross-checked by another student for errors
28.3	Inspection of Packaging Materials to check their compliance with quality specifications	Doing 3 · Mechanism	The teacher will give mock purchase orders, specification sheets, and provide some packaging material samples (some compliant, some defective). The students will inspect the materials, match them against specifications (dimensions, clarity, print quality), and approve or reject with reasons
28.4	Calibration of Analytical Instruments	Doing 3 · Mechanism	The students will perform the calibration of different analytical instruments, pH meter, an Analytical balance, UV-Vis spectrophotometer
28.5	QA Documentation audit	Doing 3 · Mechanism	The teacher will allot each student mock documents like Batch Manufacturing Records (BMR), deviations, Change control forms, etc. The students will audit the given documents for errorsapprove/reject documentation and justify their decisions
MODULE 29 — PHARMACY PRACTICE			
29.1	Identification of Adverse drug reaction in selected case and .documentmentation cases of Adverse Drug Reaction	Doing 2 · Guided response	The students will be posted in the hospital. They will document a minimum of 10 patient cases concerning anupana, sahapana, pathya-apathya, kala maryada, and ADRs, if any. They may also document the cases retrieved from Hospital records.
29.2	Experience about Basic Health Screening at community screening	Doing 3 · Mechanism	Students will perform hands-on tests using community pharmacy equipment in a community setting, on a minimum of 10 patients, including measurements of blood pressure using a sphygmomanometer, capillary blood glucose using a glucometer, lung function using a peak flow meter and incentive spirometer, pulse oximetry to measure SPO?, and BMI assessment
29.3	Discuss on Multidisciplinary Integration for TDM(Therapeutic Drug Monitoring) decision making	Doing 4 · Adaptation	Students will interact with clinicians and laboratory personnel, communicating Therapeutic Drug Monitoring (TDM) recommendations, discussing laboratory results, and planning monitoring strategies. Additionally, students will conduct case-based ward rounds, incorporating lab values, patient status, and polypharmacy considerations into TDM decision-making.
29.4	Development of Standard Therapeutic Guidelines	Doing 5 · Origination	Students will draft protocols for common conditions, such as hypertension, based on evidence-based medicine. These protocols will undergo peer review by a committee, which will evaluate and refine the guidelines before their formal adoption.

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
MODULE 30 — PHARMACEUTICAL MICROBIOLOGY			
30.1	Conduction of Dry heat sterilization practice	Doing 2 · Guided response	Each student will be assigned an article contaminated with microbes. The students will then sterilize the articles using the dry heat method in a hot air oven. After sterilization, students will test for sterility by inoculating the nutrient broth and observing for growth. The recorded findings will be presented and discussed.
30.2	Simulation Activity for Aseptic Practices	Doing 4 · Adaptation	The students will simulate activities such as preparing sterile media, gowning, and handling materials in an aseptic area. Peers will observe and provide feedback on potential contamination risks. The students will then discuss how behaviour and protocols can reduce contamination risks.
30.3	Real-life case studies on spoilage outbreaks and preservative failures	Doing 3 · Mechanism	The teacher will assign each student a spoiled food item, a pharmaceutical preparation, or a scenario. Students will then identify the type of spoilage, determine the probable cause of the spoilage, and suggest ways in which contamination could have been prevented.
30.4	Observing and detecting Spoilage of Pharmaceutical Product in Real Time	Doing 4 · Adaptation	The teacher will provide each student with two samples of the same dosage form (one fresh and one old) of a pharmaceutical product, such as Panchavidha Kashaya Kalpana, and their suitable Upakalpana. Students will then assess both samples for spoilage using organoleptic and basic physicochemical parameters. They will record their observations on the comparative stability of the fresh and old samples. By applying theoretical knowledge to practical scenarios, students will highlight the importance of microbial limit tests in quality control of pharmaceutical products.
MODULE 31 — REGULATORY ACTS AND RULES APPLICABLE TO AYURVEDIC DRUGS			
31.1	Case Law analysis on the Factory Act	Doing 3 · Mechanism	The teacher will instruct each student to explore a case study related to Worker safety violations/Overtime pay disputes/Factory licensing issues. The students will present the case in a standard format.
31.2	Technical Report to DTAB: Reporting Adverse Drug Reactions (ADRs)	Doing 3 · Mechanism	The teacher will assign students to prepare a mock technical report on the adverse effects of a new cosmetic product, adhering to the technical and regulatory guidelines outlined in the Drugs and Cosmetics (D&C) Act. Students will present their reports, which will be reviewed and verified by the teacher for accuracy and compliance.
31.3	Identification of Misbranded/ Spurious Drugs and Cosmetics	Doing 3 · Mechanism	Students will be tasked with collecting and compiling real-life case studies on misbranded or spurious drugs and cosmetics from published literature or electronic resources. They will analyze each case to identify and document the key parameters that classify a product as misbranded or spurious. Finally, students will present their findings and engage in a class discussion to share insights and perspectives.
31.4	Application for GMP certification	Doing 3 · Mechanism	The students will be instructed to compile all the mock forms required to get GMP certification and try to fill out the forms. The teacher will discuss - How to get the licence for single herb/herbo-mineral compound/extract, and how to apply for a Loan licence. Students will record the findings, discuss with the teacher
31.5	Development and Licensing of Proprietary Herbal/ Ayurvedic Formulations	Doing 3 · Mechanism	The teacher will assign each student a different disease and instruct them to formulate a herbal/ Ayurvedic formula for the given condition. Students will then develop the necessary documents to support a license application for their formulated product. The teacher will review and discuss the submissions, providing conclusions and feedback

REF	ACTIVITY	LEVEL IT SERVES	WHAT THE STUDENT DOES
31.6	Label Compliance Check	Doing 3 · Mechanism	The teacher will distribute various Ayurvedic, Siddha, and Unani (ASU) drug labels (authentic or printed) among the students. Each student will assess their assigned label for compliance with the requirements outlined in Rule 161, documenting their findings. The students will then discuss their observations with the teacher
31.7	Product Label Designing as per D&C Act	Doing 3 · Mechanism	The teacher will assign each student a different dosage form and instruct them to design a product label in compliance with the Drugs and Cosmetics (D&C) Act. Students will create their labels, present them to the class, and engage in a discussion with the teacher to receive feedback and guidance.
31.8	Analysis of Landmark Judgments under the NDPS Act	Doing 3 · Mechanism	The teacher will instruct each student to explore a landmark judgment under the NDPS Act, evaluating court interpretations of key provisions (Sections 27, 21, etc.) and focusing on quantity differentiation, presumption of guilt, and bail provisions under Section 37.
31.9	Objectionable Advertisements: Case Studies and Report	Doing 3 · Mechanism	The students will be instructed to explore the cases coming under the Drug and Magic Remedies (Objectionable Advertisements) Act, 1954. Each student will explore a case, present the observations, and report the same in standard format to the respective pharmacovigilance cell.
MODULE 32 — ENTREPRENEURSHIP AND START-UPS			
32.1	Creating an elevator pitch for a startup idea	Doing 3 · Mechanism	Students will design a new startup concept, develop a comprehensive business plan, create a prototype, and pitch their idea. They will discuss, brainstorm innovative ideas, and select one idea to develop further. Each student will prepare a written business plan, including market research, financial projections, and marketing strategies. They will also create a prototype or minimum viable product to demonstrate their idea. Students will then prepare a persuasive elevator pitch to present their startup idea to investors, and finally, they will present their pitches and prototypes to the class, receiving feedback and guidance.
32.2	Conducting market research for a start-up, including survey design and analysis	Doing 5 · Origination	Each student will conduct a market survey for a startup, then design a startup proposal based on the survey findings, including logical analysis and recommendations. Students will record and present their proposals, demonstrating their understanding of market research and its application in startup development
32.3	Navigating a Startup's Early Days: A Case Study-Based Role-Play	Doing 4 · Adaptation	The teacher will assign each student a case study of a startup in its early days. Provide each student with a set of role-play scenarios related to the startup's early days, including pitching to investors, managing conflicts between co-founders, handling customer complaints, and deciding on a marketing strategy. After the role-play, each student will be asked to reflect on their experiences and discuss questions such as what challenges they faced, how they overcame these challenges, and what they learned from the experience. Finally, ask each student to present their findings and insights to the class
32.4	Designing of an application to Patent the formulation process	Doing 3 · Mechanism	The teacher will allot each student a different scenario related to pharmaceutical modification. The student will design and develop a patent application for the formulation process
32.5	Identification of the patentability of a product	Thinking 5 · Synthesis	The teacher will allot each student a drug/product containing the ingredient mentioned in Ayurvedic classics. Students will be asked to check the patentability of the product, present their findings, and discuss

163 activities. The level shown is that of the objective in Table 3 which cites the activity, so it tells you what the activity is being asked to certify.

PART C · APPENDIX C8

Where the hours go, module by module

The lecture, practical and experiential hours of every module, computed from Table 3 and reconciled against each module's notional hours. Read alongside Chapter 6.

M	MODULE	CR	LECTURE	PRACTICAL	EXPERIENTIAL	ASSESS	TOTAL
SEMESTER 3							
1	Chronological Development and Taxonomy of Rasa Dravyas	2	10 h	20 h	26 h	4 h	60 h
2	Puta and Samskara (Transformative processes of drug manufacturing)	2	10 h	20 h	26 h	4 h	60 h
9	Scientific fundamentals of Ayurvedic Pharmaceutical preparations	2	10 h	20 h	26 h	4 h	60 h
10	Principles of pharmaceutical formulation	2	10 h	20 h	26 h	4 h	60 h
17	Guidelines for administration of Rasoushadhi in clinical Practice	2	10 h	20 h	26 h	4 h	60 h
18	Prayogika Bhaishajya Kalpana(Clinical application of Pharmaceutical products)	2	10 h	20 h	26 h	4 h	60 h
25	Artificial intelligence in drug development and Pharmacogenetics; Pharmacogenomics	2	10 h	20 h	26 h	4 h	60 h
26	Pharmacology	2	10 h	20 h	26 h	4 h	60 h
SEMESTER 4							
3	Parada (Mercury) and Its Processing: Ancient wisdom and Scientific advances	2	10 h	20 h	26 h	4 h	60 h
4	Maharasa Dravya, its processing and utility	2	10 h	20 h	26 h	4 h	60 h
11	Mana Paribhasha (Metrology in Ayurvedic Pharmaceutical preparation)	2	10 h	20 h	26 h	4 h	60 h
12	Process validation Protocol	2	10 h	20 h	26 h	4 h	60 h
19	SOP and clinical application of Chaturvidha Rasayana	3	15 h	30 h	39 h	6 h	90 h
20	SOP and Clinical application of Ayaskriti, Louha, and Mandura Kalpa	1	5 h	10 h	13 h	2 h	30 h
27	Network pharmacology & Reverse Pharmacology	2	10 h	20 h	26 h	4 h	60 h
28	Pharmaceutical Quality Assurance	2	10 h	20 h	26 h	4 h	60 h
SEMESTER 5							
5	Critical appraisal of Uparasa Dravya and Sadharana Rasa Dravya	2	10 h	20 h	26 h	4 h	60 h
6	Critical appraisal of Dhatu and Updhatu	2	10 h	20 h	26 h	4 h	60 h
13	Krutanna (Ayurvedic Food Preparations)	2	10 h	20 h	26 h	4 h	60 h
14	Ayurvedic Formulations for Holistic Wellbeing, Vitality and Body Care	2	10 h	20 h	26 h	4 h	60 h
21	Clinical application of Rasoushadhi according to Srotas and in selected challenging clinical conditions	1	5 h	10 h	13 h	2 h	30 h

M	MODULE	CR	LECTURE	PRACTICAL	EXPERIENTIAL	ASSESS	TOTAL
22	Yogas from different Kalpanas	3	15 h	30 h	39 h	6 h	90 h
29	Pharmacy Practice	2	10 h	20 h	26 h	4 h	60 h
30	Pharmaceutical Microbiology	2	10 h	20 h	26 h	4 h	60 h
SEMESTER 6							
7	Sudha Varga, Sikata Varga, Visha Varga Dravyas	3	15 h	30 h	39 h	6 h	90 h
8	Appraisal of Ratna and Uparatna and therapeutic significance of Pishti and Druti Kalpana	1	5 h	10 h	13 h	2 h	30 h
15	Disease condition specific selection of formulation	2	10 h	20 h	26 h	4 h	60 h
16	Current trends in Ayurvedic Pharmaceutical Industry.	2	10 h	20 h	26 h	4 h	60 h
23	SOP and clinical applications of Churna Kalpana(Powder dosage forms)	3	15 h	30 h	39 h	6 h	90 h
24	Therapeutic significance of Topical dosage forms	1	5 h	10 h	13 h	2 h	30 h
31	Regulatory Acts and Rules applicable to Ayurvedic Drugs	3	15 h	30 h	39 h	6 h	90 h
32	Entrepreneurship and Start-Ups	1	5 h	10 h	13 h	2 h	30 h

Course totals: lecture **320 h**, practical **640 h**, experiential **832 h**, modular assessment **128 h** — **1920 notional hours** over 64 credits. The split is fixed by the credit framework at one part lecture to two parts practical to three parts experiential, so two thirds of this course happens with the teacher not talking.

PART C · APPENDIX C9

Sources and further reading

The primary sources for the taxonomies, the curriculum this book is built on, and a note on what was and was not altered.

THE TAXONOMIES

- Bloom, B. S. (ed.), *Taxonomy of Educational Objectives. Handbook I: Cognitive Domain.* New York: Longmans, Green, 1956. The original, and the scheme Column 3E uses.
- Anderson, L. W. and Krathwohl, D. R. (eds.), *A Taxonomy for Learning, Teaching and Assessing.* New York: Longman, 2001. The revision: verb names, the swapped top two levels, and the knowledge axis.
- Krathwohl, D. R., Bloom, B. S. and Masia, B. B., *Taxonomy of Educational Objectives, Handbook II: Affective Domain.* New York: David McKay, 1964.
- Simpson, E. J., *The Classification of Educational Objectives in the Psychomotor Domain.* Washington: Gryphon House, 1972; and Dave, R. H., in R. J. Armstrong (ed.), *Developing and Writing Behavioral Objectives*, 1970.
- Krathwohl, D. R., 'A Revision of Bloom's Taxonomy: An Overview', *Theory into Practice*, 41(4), 2002.

THE CURRICULUM

- National Commission for Indian System of Medicine, Board of Ayurveda. *Curriculum for MD/MS Ayurveda, Semesters III to VI — Rasashastra and Bhaishajya Kalpana (AYPG-RB).* New Delhi: NCISM. Applicable from the 2024–25 batch. Tables 2, 3, 4 and 5 are the source of every objective, code, hour, mode and activity in this book; Table 6 is the source of the assessment figures in Chapter 8.

A NOTE ON HOW THIS BOOK WAS MADE

Every objective in Part B is reproduced in the curriculum's own words. Plain-English spelling errors have been corrected and sentence punctuation lost to the source document's line wrapping has been restored. No Sanskrit or technical term has been altered, and no subject content has been added, removed or reworded.

Extracting this curriculum needed several repairs peculiar to its layout: domain codes split across line breaks and interleaved with the level beside them, hours joined to the end of an objective's own full stop, activity references torn apart mid-word, syllabus points glued to the unit titles they follow, and the activity description tables which sit inside Table 3 and had to be separated from the objectives around them. Table 3 is also printed paper-first rather than semester-first, and has been reordered here.

The result reconciles against the curriculum's own credit framework at three levels: 577 objectives accounting for 1792 teaching hours, plus 128 hours of modular assessment, giving the **1920 notional hours** of 64 credits; each of the four semesters independently totalling 480 hours; and 304 activities matching the 141 practical and 163 experiential entries of Tables 4 and 5, with every reference in Table 3 resolving to one of them.

The counts and tables in Part C are generated from the same parsed data as Part B, so the two cannot disagree. Where a judgement had to be made — chiefly in reading the kind of knowledge from an objective's wording — it is made from the verb and its grammatical object, and marked with a small circle (°) where the wording supports a reading without settling it.